

Lower Russian River Bathymetric Analysis

Revision 2

Methods, Equipment and Results

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1. INTRODUCTION

1.1 PROJECT DESCRIPTION

As part of the implementation of the Lower Russian River Biological Opinion, The Sonoma County Water Agency contracted Environmental Data Solutions (EDS) to design and conducted an extensive field data collection program within the Lower Russian River Estuary.

The goal of the field effort was to produce a 2 (two) foot interval contour map of the river channel and surrounding topography. EDS collected high-resolution bathymetric data, sidescan imagery data, bottom sediment sample data, water surface elevation time series data (WSE) and topographic data.

The purpose of this field data collection report is to outline methods, procedures, locations and results of this field effort.

1.2 PROJECT FOOTPRINT

The project area is bounded by the river mouth downstream in Jenner and by Austin Creek upstream and across from the Cassini Ranch Family Campground in Duncan's Mills, Ca. The 7.2 mile study reach also included the Willow Creek watershed on river left, 0.40 miles upstream from the Highway 1 Bridge (Appendix A - Figure 1).

1.3 HYDROLOGIC MONITORING

As part of an effort to be better understand areas of inundation resulting from river mouth closure, five (5) WSE recorders were installed throughout the project reach (Figure 2). These data describe the characteristics of diurnal tidal amplitude changes and phase shifts as well provide calibration data for future hydrologic modeling. In addition to describing diurnal tidal data, the time series also captured two mouth closure events and one moderate storm.

1.4 BATHYMETRIC SURVEYING

The bulk of the field effort focused on collecting high resolution bathymetric data throughout the survey reach and into Willow Creek. EDS utilized interferometric sidescan sonar (*Teledyne Benthos C3D*) which collects bathymetric data and sidescan imagery concurrently.

In areas that were too shallow for the C3D, a shallow-water singlebeam "sweep system" was utilized to collect bathymetric data. This configuration was successful in collecting data in shallow areas such as Willow Creek and on the large cobble bars in the main stem.

1.5 TOPOGRAPHIC SURVEYING

As part of the original Request for Proposal and scope of work, topographic data (using land-based survey equipment) were to be collected along the entire reach to the mean high-water mark (during mouth closure) in order to supplement the final bathymetric contour map. The EDS field crew collected

topographic data in certain areas before the decision was made to collect data using Light Detection and Ranging (LiDAR).

The Sonoma County Water Agency (SCWA) contracted *Delta Geomatics Inc.* (Santa Rosa, CA) to collect LiDAR data from the river centerline out to the floodplain on either side of the river. The LiDAR and bathymetric datasets were then used to generate the contour maps.

As a result of the extremely thick willow canopy and shallow creek bathymetry, land-based survey equipment was necessary to collect channel bathymetric data throughout the lower Willow Creek reach.

1.6 SIDESCAN SONAR IMAGERY

As part of the sediment mapping effort, sidescan imagery was collected throughout the project reach. Sidescan sonar data was collected concurrently with the bathymetric data using the *Benthos C3D*. Sidescan sonar is a category of sonar used to create images of the river bed and features contained therein. Sidescan emits fan-shaped pulses through the water column to the river bed across a wide angle perpendicular to the survey vessel path. Coupled with positioning data (from a Real-Time Kinematic GPS), sidescan imagery produces high-resolution, geo-rectified TIFFS of the riverbed.

Over 290 grab samples were collected and analyzed in order to aid in the calibration of surface textures evident in the sidescan imagery. A resulting sediment-type map was then produced which shows bedrock outcrops, large patches of sub-aquatic vegetation (SAV), snags and other debris on the river bed.

1.7 CONTOUR MAP GENERATION

The final product from the bathymetric, LiDAR and topographic survey efforts is a complete set of two (2) foot interval contour maps covering the entire survey reach as well as the entire Willow Creek watershed. A one (1) meter grid was used to aid on the generation of a Digital Elevation (DEM) and contour map generated in AutoCAD 2009 (using Civil 3D).

2. METHODS

2.1 SURVEY CONTROL NETWORK ESTABLISHMENT

A Static GPS survey was performed on three control point monuments that were located throughout the survey reach. The survey control network also included occupying two found control points: 1) National Geodetic Survey (NGS) High-Accuracy Reference Network (HARN) station PID JT9623, and 2) Central Coast Height Modernization Project (CENCHM2007) point DUNC.

Static GPS surveys allow various systematic errors to be resolved when high accuracy vertical and horizontal positioning is required. Static procedures are used to produce baselines between stationary GPS units and Continuously Operating Reference Stations (CORS) by recording data over an extended period of time during which the satellite geometry changes. Resulting Receiver Independent Exchange (RINEX) files are then uploaded to the Online User Position Service (OPUS) computers operated by the NGS for final baseline solutions. The static surveys were planned and executed by James Kulpa and Thomas Tucker LS#4460.

These control points were used as the basis for all hydrographic and topographic survey data collection. A *Leica System 1200* GPS base station was established on one of the three bench marks during each hydrographic and topographic data collection day. Before and after each data collection episode (hydrographic and topographic), a GPS rover was used to check-into an established control point to ensure proper calibration of the resultant data stream emanating from the rover unit.

2.1.1 GPS Network Design

On October 2, 2008, Thomas Tucker (LS 4460) established three primary control points using static GPS data processed using OPUS-derived baseline solutions to constrain the network. *Trimble 4000 SSI* dual frequency GPS receivers were utilized to establish three control points: 1) capped rebar at the Jenner visitor center – **BM 1**, 2) capped rebar at the Route 1 / Highway 116 interchange – **BM 2** and 3) a found survey disc in a well monument on Moscow Road – **BM 3** (Figure 3).

The three control points were occupied for a second time; Table 1 lists meta-data for each static session and Appendix B contains the original OPUS data processing sheets as well as field observation logs from the static GPS sessions.

Table 1 – Hydrographic Survey Control Points – OPUS Processing Meta-Data

Point Id	Static Survey Date	Start Data Collection (UTC)	Stop	CORS Base Stations Used (PID)	Observations Used	Geoid Model	RMS (m)	Notes
BM-1	10/2/2008	18:11	19:44:3	OPUS: DH7229 DK6396 DG8210 OPUS RS: DH7229 DK6396 DG8210 DI1085 AM7021 DH3879 AF 9690 DH9024 DG8213	OPUS: 3756/3895 OPUS RS: 11016/11475:	Geoid 03	0.01M	At the visitors parking lot
BM-1X	2/20/2009	19:06	21:52	OPUS: DH7229 DK6396 DG8210 OPUS RS: DH7229 DK6396 DG8210 DI1085 DH3882 DH9021 DG8213	OPUS: 7306/7320 OPUS RS: 20079/21501	Geoid 03	0.009M	This is an eccentric to BM1 – a truck was parked on BM1. After a differential level tie between BM1-X and BM1, I checked into BM1 by 0.008m.
BM-2	10/2/2008	18:32	19:40	OPUS: DH7229 DK6396 DG8210 OPUS RS: DH7229 DK6396 DG8210 DI1085 DH3882 DH9021 DG8213	OPUS: 2355/2646 OPUS RS: 6363/7920	Geoid 03	0.011M	Intersection of 116 and Route 1
BM-2	12/30/2008	19:36	21:32	OPUS RS: DH7229 DK6396 DG8210 AM7021 DH3879 DH9024 DH9021 DH5855 AF9690	OPUS RS: 6363/7920	Geoid 03	0.013M	Data processed using OPUS-RS (no OPUS-S processing)
BM-3	10/2/2008	20:22	21:43	OPUS: DH7229 DK6396 DG8210 OPUS RS: DH7229 DK6396 DG8210 DI1085 AM7021 DH3879 DH9024 DG8213	OPUS: 2314/2729 OPUS RS: 5049/7542	Geoid 03	0.012	This point is also CSRC point DCAN
BM-3	3/16/2011	23:53	02:33	OPUS: DH7229 DK6396 DG8210	OPUS: 7317/7788	Geoid 03	0.012	Data processed using OPUS-S (no OPUS-RS processing)

Table 1 - Continued								
Point Id	Static Survey Date	Start Data Collection (UTC)	Stop	CORS Base Stations Used (PID)	Observations Used	Geoid Model	RMS (m)	Notes
JT9623	2/20/2009	19:28	21:30	OPUS: DH7229 DK6396 DG8210 OPUS RS: DH7229 DK6402 DG8210 DI1085 AM7021 DH3879 DH9021 DH3882	OPUS: 5328/5391 OPUS RS: 12546/15426	Geoid 03	0.01	

2.1.2 Datums

The NAD83 datum (2009.00 epoch) was used for horizontal value and were referenced to the California Coordinate System (CCS 83), Zone 2 for the entire survey reach. U.S. Survey Feet, NAVD 88 (OPUS-derived using GEOID 03). The GEOID 03 model was loaded into the bathymetric survey data acquisition program, as well as all RTK-GPS units.

2.1.3 Quality Assurance / Quality Control

The first QA/QC procedure was simply to choose an optimum location for each control point. Each location was scrutinized for line of sight to the horizon (unobstructed 360° horizon) and location relative to each survey reach. Standardized field sheets were used to collect site meta-data and GPS data logging information (Appendix B).

All control points were occupied twice with each resultant RINEX file processed using both OPUS and OPUS-RS processing routines. The vertical difference and accuracies between the two types of processing for each point were compared and are shown in Section 3.1.

Two Central Coast Height Modernization (California Spatial Reference Center (CSRC) - CENCHM2007) points were occupied as well: 1) National Geodetic Survey (NGS) designation – PID JT9623 and 2) CSRC point DCAN (which is also EDS BM-3). Results of these occupations are discussed in Section 3.1.

Once the points were established and verified, RTK-GPS was used to take rover check shoots on each point. An inverse function was then used to check the reported XYZ against the surveyed XYZ.

2.1.3.1 *Precise Point Positioning*

As an additional quality assurance check, a Precise Point Positioning (PPP) analysis, using *TerraPOS* software (www.frontiergeomatics.com), was conducted on each of the EDS control points raw data file. PPP is a processing method that provides centimeter positioning accuracy. PPP takes advantage of high-resolution carrier phase observations and uses very precise satellite orbit and clock information instead of

using broadcast ephemeris information. This is possible due to The International GPS Service (IGS) providing GPS orbit products to the scientific community with increased precision and timeliness.

PPP also models and corrects error effects resulting from:

- 1) Satellite antenna offset and phase center variations
- 2) Satellite yaw
- 3) Ionospheric delays
- 4) Tropospheric delays
- 5) Receiver antenna offset and phase center variations
- 6) Solid earth tides
- 7) Rotational deformation due to polar motion

2.2 TIDAL MONITORING

In order to collect continuous WSE data throughout the survey reach, EDS located and installed five (5) bottom-moored tide gages (Figure 2). Water level time series data were collected using a submersible, non-vented pressure / temperature sensor designed and manufactured by *Coastal Leasing Inc.* of Cambridge, Massachusetts (*MacroTide* series). The gage has a pressure transducer port and a temperature sensor on the top of the unit. The sensors are self-contained in that the data logger is built directly into the unit.

The data loggers were programmed to turn on and excite each sensor every six minutes and take readings at 2 hertz for 30 seconds. These 30-second “bursts” were then averaged and stored into memory with a date and time stamp (Pacific Standard Time).

Non-vented pressure transducers measure absolute pressure (water column pressure + atmospheric pressure) by converting the deformation of a pressure diaphragm to a millivolt reading. The millivolt reading is then converted to an absolute pressure reading reported in PSIA (Pounds Per Square Inch Absolute).

When the data loggers are downloaded, a comma-delimited file containing a temperature and pressure, reading with an associated time and date stamp is produced. Data post-processing includes the subtraction of concurrent atmospheric pressure values from each absolute pressure reading. A separate, atmospheric pressure time series was acquired using a pressure transducer mounted above the Mean Higher-High Water elevation.

The atmospheric data were subtracted from each PSIA reading during the conversion from PSIA to depth. Depth of water above the pressure port was calculated using:

$$D=(P-A)*K$$

Where:

D = depth, in meters

P = total pressure, from Macro Tide, in PSIA

A = atmospheric pressure

K = pressure-to-depth conversion factor, which is a function of temperature and conductivity

The conversion from pressure to depth can be derived from the density of water as follows:

$$K = 0.703242 \text{ m/psi} * 1/\text{Rho (T) (relative V as a function of temp.)} * 1/\text{Rho (S) (relative volume as a function of salinity)}$$

2.2.1 Deployment

Channel Bottom-Mounted Moorings

The tide gage was placed within a bottom-mounted mooring that consisted of a one-foot long, six inch diameter PVC, horizontally-bolted to the top of a 100 pound cement and lead block. The sensor was placed within the protective PVC housing. The mooring block was tapered toward the top and wide at the bottom to ensure stability. To keep the mooring stable and from sinking into the sediment, four, three-foot long thin wood “feet” were attached to the bottom of the mooring block (Figure 4).

Each mooring had a 40 Ft. drag chain attached that was then deployed in a known compass direction, and where possible, attached to an old pile or log.

2.2.2 Tide Gage Survey

To ensure all of the WSE time series are in a relative datum (Ft. NAVD 88), surveying of the actual water surface (concurrent with a scheduled sensor reading) before and after each instrument reconnaissance was integral to this data collection effort.

Before the tide gages were deployed, a local temporary tide gage benchmark (TBM) was established in the vicinity of each gage in order to facilitate the reporting of the tide time series in U.S. survey feet, NAVD 88. Using a digital level (*Leica Sprinter*), a level-line loop was conducted between each tidal TBM and a project primary benchmark. Each loop was closed-out to the primary mark (J. Kulpa and T. Tucker) and all level loop closure errors were less than 0.05 ft.

After the mooring was initially deployed, the distance between the water surface and the TBM, was surveyed using a digital survey level (and recorded with time). This protocol was also repeated prior to and after each recovery of the instrument platform.

To covert from feet of water above the pressure transducer (units after PSIA conversion) to the project datum (Ft. NAVD 88), conversions between NAVD and depths were derived from the survey data. Each instrument deployment “data block” (data collected between each deployment and recovery) had a separate conversion factor since the instrument was moved and relocated to a slightly different vertical position during each site visit.

2.2.3 Tide Monitoring QA / QC

Instrument deployments and recoveries involved sensor checks on the following constituents:

- An open-air pressure reading was taken every time the sensor was recovered to ensure accuracy against atmospheric pressure reading taken by the atmospheric gage.
- The conversion (before instrument recovery) between sensor reading and corresponding measured water surface elevations in the project datum was checked against the conversion derived from the previous post-deployment readings. Post deployment and pre-recovery conversions need to be the same; differences in conversions are indicative of either the sensor physically moving or experiencing transducer drift during that particular deployment.
- All conversion offsets between raw tide gage data and corresponding WSE (in NAVD 88) were within 0.15 feet.
- Instruments inspected for physical damage or corrosion.
- Ensuring O-rings are not cracked and are properly lubricated.
- Checking battery power and remaining free memory.
- Cleaning the instrument of any bio fouling or sediment accumulation.

2.3 TIDAL DATUM ANALYSIS

A tidal datum analysis was performed in order calculate mean high water and mean lower water at each tide gage location. At each station, the high and low tides were selected out from the dataset by comparing each water level data record to all other records in a 3 hour window before and after it. If the record was the highest elevation, it was selected as a “high tide”. If the record was the lowest elevation, it was selected as a “low tide” (Figures 5 and 6). Once the high and low tides were queried from the dataset, they were plotted and all erroneous data points during times when the mouth of the river mouth was closed were removed.

MHW was calculated at each station as the average of all of the high tides. MLW was calculated at each station as the average of all the low tides.

Please note that the MHW and MLW elevations determined in this effort **do not** constitute true tidal datums. The Method of Corresponding Tides, which relates the tidal elevations at a local monitoring station to those found at a nearby reference monitoring station, was not used. The MHW and MLW values given in this report will only indicate what the local average high and low tide values were at each station during the observation period. A true tidal datum reckoning can be performed to determine MHW for the overall survey area if required.

2.4 BATHYMETRIC SURVEY

EDS collected bathymetric data while the mouth was closed during the first half of October, 2009.

An interferometric bathymetric system is comprised of five components integrated into one robust, survey solution. A 17.0 Ft. Boston Whaler survey vessel was outfitted with a *Teledyne Benthos C3D*

Bathymetric Sidescan shallow water multibeam system attached to an over-the-side mount (Figure 7). The *C3D* system contains a series of transducers arrayed within a tow fish, a topside high-speed processing unit connected to a *Intel / Windows* – based CPU and associated software (*Hypack Version 2008* for navigation, and *CARIS Hips Version 6.1* for acquisition and patch testing).

The *C3D* system is integrated with a heading / heave / pitch / roll sensor (motion reference unit or MRU) and a Real-Time Kinematic GPS (RTK-GPS) unit for position and ellipsoid heights.

2.4.1 Survey Vessel

Bathymetric-based field operations were performed from Environmental Data Solution's 17-ft fiberglass Boston Whaler survey vessel, based in San Rafael, CA. This vessel has a large pilothouse for survey electronics, a 5KW generator, a hydraulic winch, and large open deck suitable for the deployment of hydrographic equipment.

The survey hardware package (sidescan transducer, MRU, GPS antennas) was mounted on the same over-the-side mount, which was designed and engineered specifically for the Russian River operations (Figure 7). Having the MRU on the same lever arm as the sonar equipment results in a very "tight" association between motion corrections and returned sonar data.

The field crew consisted of: James Kulpa, Hydrographer (EDS); Donny Brouillette, Hydrographer (EDS) and Darren Gewant, Survey Technician (EDS).

2.4.2 Interferometric Sidescan Sonar

The *C3D* is an advanced bathymetric sidescan sonar with extremely high resolution and dynamically focused beams. It is very well suited for detailed riverbed mapping and inspection in water depths from 1.5 meters up to approximately 150 meters in the ocean. The spacing between soundings as well as the acoustic footprints are set nearly constant over the swath in order to provide a uniform and high detection mapping performance. Dynamic focusing of all receive beams optimizes the system performance and resolution for short range applications.

SARA CAATI (Small Aperture Range Angle and Computed Angle of Arrival Transient Imaging) is a patented technique licensed to *Teledyne Benthos* by Simon Frazier University. It is a method for estimating the backscatter arrival spectrum and assigning a distinct position and elevation to each arrival data point. SARA CAATI uses an angle-of-arrival estimation similar to interferometry. However, unlike interferometry, the *C3D* solves the problem of concurrent arrivals from multiple angles.

The *C3D* produces a swath of sonar approximately 3.5 to 4 times the water depth and collects approximately 400 soundings per square meter. Sound velocity was calculated using an *Odom Digibar Pro* sound velocity meter.

2.4.3 Motion Reference Unit

Unlike a singlebeam echosounder, depths from a multibeam unit are calculated from signal returns at extremely steep grazing angles from nadir. It is these outer beams that are most influenced by vessel motion and attitude. To compensate for the motion of the vessel, an accelerometer based motion reference unit is installed on the vessel (in this case, it was installed above the transducer) to provide real-time heave, pitch, roll and heading measurements to the processing unit of the *C3D*.

EDS utilized a *Coda Octopus F-180* MRU which provides highly accurate and reliable motion and position data updated at a maximum rate of 70 Hz. The *F-180* consists of a single electronic box housing the internal electronic motion sensors coupled with two GPS antennas which aid in heave and heading calculations (Figure 7).

The *F-180* was mounted in line with the *C3D* transducer on the same mount. This is an extremely robust way to mount a MRU – opposed to mounting the unit offset from the axis of the sidescan transducer and thus requiring the implementation of a virtual lever arm.

2.4.4 Bathymetric Survey Control

The most common problem in accurately measuring the seafloor with any sonar-based system is the calculation of the tidal elevation offset, or in the case of a closed Russian River, a rising stage due to mouth closure. Commonly a tide staff or electronic gauge is deployed in one location near the survey site and is used to calculate the tides for the entire survey area. However, it is widely understood that non-linear tidal phenomena, phase lags and tidal gradients can drastically influence the tidal elevation spatially throughout a channel and therefore the use of a single point measurement is often unreliable.

To avoid these potential tidal elevation errors which can translate into significant departures from the true bottom depth, we used geodetic Global Positioning Systems (GPS) with **real-time kinematic (RTK) baseline processing that is integrated with the sidescan and inertial navigation instruments**. The motion and Geoid 03 compensated positions and orthometric elevations of the RTK-GPS data stream are tagged with each sonar ping. In effect, the RTK-GPS mounted on the hydrographic survey vessel acts as a roving tide gauge collecting the most accurate tidal measurements throughout the survey area.

The survey team utilized a *Leica System 1200* RTK base station and rover unit (mounted directly over the MRU and sidescan transducer) coupled with a *Pacific Crest* UHF radio modem. The RTK base station was set over control point BM-1 with the rover unit programmed to collect data in California State Plane Coordinates, Zone 2 (NAD 83 Datum) and Ft. NAVD.

In order to correct raw sounding data for WSE height changes, transducer ellipsoid heights were collected at a rate of 10Hz and stored into a *Hypack* proprietary file called an *RTK tides file*. A geoid separation value was then applied to this file in order to transform the fathometer elevations from ellipsoidal heights to orthometric heights (Ft. NAVD 88).

2.4.5 Speed of Sound Measurements

Fathometers calculate water depth by using algorithms based on the speed of sound through the water column. The survey crew utilized an *Odom Digibar Pro* speed of sound probe to measure sound velocity multiple times during each survey day.

Mounted near the end of the sound velocity probe is a high frequency "sing-around" transducer and its associated reflector. This precisely spaced pair is used to measure the velocity of sound in water by transmitting and receiving a signal across a known separation distance.

2.4.6 Bathymetric Data Reduction

After field data acquisition is complete, the data was initially processed and edited in the field to ensure data quality and coverage. The next step was applying the sound velocity profiles to all sounding data. RTK vertical control is integrated in the data acquisition process and applied to all sounding data in post-processing when producing the final set of soundings.

Each individual sounding line must also be reviewed for both position and depth accuracy. In-depth office processing of logged bathymetric swath data and individual sensor data is conducted using *Caris HIPS* hydrographic survey data processing software. Final post-processing (contouring) was done using AutoCAD [™] 2009.

2.4.7 Bathymetric Survey QA/QC

Sidescan swath sonar systems combine a complex array of instruments, consisting of the transducer, motion sensor, gyrocompass, and geodetic GPS system. Standards developed by the International Hydrographic Organization (IHO), USACE Standards for Hydrographic Surveys, and the NOS Hydrographic Surveys Specifications and Deliverables for shallow water (<30 m) hydrography (IHO 1987; USACE 2003; NOS 2003) are used as the protocol for calibration. Proper alignment of these instruments with one another and with the vessel's reference frame is critical to achieve the high-accuracy required in the SOW.

Calculation of the horizontal and vertical offsets between each of the instruments is followed by a series of open-water measurements known as patch tests. The patch test is performed to calculate several residual biases influenced by the dynamics of the survey vessel and the alignment of the instruments. Results of the patch test are used to calculate a pitch, roll and heading offset and positioning time delay or navigation latency. Additional calibration measures are performed in the field including comparison of nadir depths with a lead line and frequent sound velocity profiles (SVP).

2.4.7.1 *Channel Bottom Topographic Shots*

The most robust protocol for checking bathymetric data accuracy is the use of topographic survey gear in order to survey channel bottom shots. The EDS field crew shot numerous channel shots using the RTK-

GPS gear. Figure 8 shows areas where RTK-GPS was used to survey points on the channel bottom: The black polygon represents bathymetric data coverage, and the green shapes indicate RTK-GPS shots. Table X shows RTK-GPS-derived channel bottom elevations listed against surrounding bathymetric points (the final bathymetric dataset was exported to a 1 meter grid).

Table 2 – RTK-GPS Channel Spot Elevations Compared to Surrounding Bathymetric Data

Check Area (Referring to Figure 8)	RTK-GPS Elevation	Surrounding Bathymetric Elevations			
Area 1	2.2	2.5	2.9	2.9	2.4
	1.4	1.4	1.7	1.7	1.9
	2.2	2.1	2.2		
	3.2	3.3	3.3	3.0	2.9
	1.6	1.6	1.6	1.6	1.5
	2.5	2.6	3.1	3.3	2.7
	1.9	1.9	2.1	2.1	1.8
	2.4	2.4	2.8	2.9	2.5
	3.5	4.1	4.5	4.2	4.0
	1.6	1.5	1.4	1.6	1.7
	1.4	1.8	2.4	2.3	1.7
	4.1	4.2	4.1	4.3	4.3
	2.9	3.2	3.2	3.2	3.3
	2.9	3.3	3.2	3.2	3.6
	2.7	1.6	3.6	2.4	3.6
	2.5	2.9	3.0	3.1	3.1
	2.5	2.5	2.1	3.2	2.0
	1.1	1.3	1.1	1.1	1.5
	0.9	1.0	1.0	1.1	1.1
	1.3	1.6	1.7	1.7	2.1
	3.9	4.0	3.9	3.9	4.1
Area 2	4.8	4.7	4.4	5.1	5.3
	4.9	3.9	5.8	5.1	5.2
	2.2	1.9	1.5	1.5	1.4
	4.0	3.6	2.9	2.8	2.2
	3.8	4.2	3.0	3.2	4.3
	3.4	3.5	3.2	3.6	3.6
Area 3	0.9	1.3	1.4	1.4	1.5
	1.2	1.6	1.3	1.2	1.5
	3.0	3.9	2.7	3.5	4.9
	4.3	5.1	-4.7	-7.5	5.7
	4.3	4.3	4.2	4.0	4.2
	2.7	3.8	2.5	3.6	4.9
	3.1	3.0	3.5	3.6	4.3
	-0.6	-0.7	-0.9	-0.6	-0.8

Table 2 - Continued					
Check Area (Referring to Figure 8)	RTK-GPS Elevation	Surrounding Bathymetric Elevations			
Area 3	-1.0	-1.1	-1.0	-0.9	-0.7
	1.6	1.7	1.5	1.8	1.9
	1.1	1.4	1.4	1.5	1.5
	1.4	1.6	1.6	1.6	1.8
	2.4	2.1	2.3	2.2	2.5
	1.8	1.7	1.5	1.6	1.2
	2.0	2.0	2.0	2.2	1.9
Area 4	3.1	3.3	3.2	3.0	3.0
	3.2	3.4	3.5	3.2	3.4
	4.0	4.0	4.7	4.3	4.4

2.4.7.2 Survey Planning

The generation of pre-determined survey vessel navigation lines is essential if near 100% bottom coverage is the goal. Since the sonar swath width is directly related to water column depths, it is important that the navigation lines in a line plan are spaced such that the ends of the sonar swath on each survey line overlap the ends of adjoining sonar swaths.

This overlapping of survey swaths is not only essential for coverage, it is a key QA / QC tool in that bottom elevations can be compared from two different survey swaths run in opposite directions and at different times.

The initial survey plan estimated an average water depth along the project reach of 8 Ft.; which translates into a sonar swath width of approximately 32 - 40 ft. The average width of the project reach is approximately 300 ft. and taking into account sonar swath overlap, the navigation lines were spaced at 30 ft intervals resulting in approximately 10 longitudinal navigation lines for an average reach. Some areas of the project reach are very shallow requiring lines spaced at much smaller intervals.

2.4.7.3 Patch Tests

To ensure data quality, the installation and subsequent calibration of the integrated sidescan system is dependent on a proper system alignment. EDS uses *Caris HIPS* hydrographic data processing software which provides a facility for accurately representing system alignment. The HIPS Vessel File (HVF) allows for the senior surveyor or Hydrographer to easily define the survey sensor configuration for accurate processing of virtually any survey platform. Sensor offsets and errors are time tagged which allows for the HVF to be utilized for the life of the survey vessel and evolve as changes are made to the survey platform. The HIPS Vessel Editor also provides a 3D view of the vessel and survey sensors for visualization of the sensor offsets.

Sidescan calibrations must be designed carefully and individually in consideration of systems, vessel, location, environmental conditions and survey requirements. The calibration procedure should determine

or verify system offsets and calibration correctors (residual system biases) for draft (static and dynamic), horizontal position control (RTK-GPS), navigation timing error, heading, roll, and pitch. These procedures are used to measure static navigation timing error, transducer pitch offset, transducer roll offset, and transducer azimuth offset (yaw). Patch tests are performed after initial mobilization, and periodically thereafter if sensors are modified during the survey, to quantify any residual biases from the initial system alignment. During this calibration series, four separate tests are performed to determine residual alignment biases for:

- Position Time Delay (Latency)
- Pitch Offset
- Roll Offset
- Yaw/Azimuthal Offset

Roll Offset Measurement

A roll calibration consists of running a set of two inverse survey lines to calculate sonar and motion reference unit (MRU) alignment relative to vertical. Any misalignment can cause large depth and position errors at the outer beams (Figure 9).

Pitch Offset Measurement

Pitch calibration tests consist of running a set of two identical survey lines run inverse to each other at varying speeds over a sloping seafloor or over a known object to detect sonar and MRU alignment relative to vertical. Any misalignment can cause depth and position errors across the swath (Figure 9).

Yaw Offset Measurement

A Yaw calibration test is run using a set of adjacent survey lines run at the same speed over a known object on the seafloor in order to calculate sonar and MRU alignment relative to vertical. A Yaw offset can cause depth and position errors across the swath (Figure 10)

GPS Latency Measurement

A latency test consisting of a set of two identical survey lines run over a sloping area at varying speeds to calculate the delay between position fix and transmission. Any offset detected will cause positional errors if not corrected. The error is independent of the sidescan system (Figure 10).

The results of each patch test can be seen in the patch test report located in Appendix C

2.5 SINGLEBEAM BATHYMETRIC SURVEY EQUIPMENT

The final C3D bathymetric survey dataset contained gaps as a result of extremely shallow areas (depth < 4 - 5 feet, see figures 13 - 15). In order to collect bathymetric data within these gaps, the field crew utilized a shallow water jet boat outfitted with shallow-water single beam sonar transducers (Figure 11).

The survey crew utilized a 17' Jon boat manufactured by *Lowe Inc.* as the primary singlebeam survey vessel. The vessel was equipped with a 50 horsepower (hp) *Johnson* outboard jet with automatic tilt. The vessel is compliant with all U.S. Coast Guard safety regulations and was inspected by the U.S. Coast

Guard prior to the survey in January, 2009. A graph displaying the results from a squat correction test conducted for the vessel in February 2008 is aboard the vessel (per U.S. Army Corps of Engineers' specifications).

The hydrographic surveys utilized Class 1 methods and accuracies as outlined in the Army Corps of Engineers' January 2002 *Hydrographic Surveying Manual* (EM 1110-2-1003). Singlebeam bathymetric data were collected using four *Omhex Sonarmite* fathometers arranged in a swath system across the beam of the boat mounted approximately six feet apart resulting in a 24 Ft. swath width.

2.5.1 Heading and Roll Sensor

One of the most important components of the survey system is the heading and roll sensor. Due to the large distances between the outer transducers, and the vessel centerline, a data string containing individual transducer heading and roll data is necessary (updated at a rate of 20 Hz) to ensure accurately placed data.

As the survey vessel rolls from side to side, the outer transducers exaggerate this movement by moving up and down. In order to correct for this movement, data from the roll sensor was recorded in the *Hypack* raw file and applied to the sounding data during post-processing.

The survey crew utilized a *Crescent VS100* DGPS heading and roll sensor which provides accurate heading and roll data at high update rates. The unit consists of a high performance GPS engine and two (2) multi path-resistant antennas for GPS signal processing. The *Crescent VS100* uses moving base station Real-Time Kinematic (RTK) technology to achieve very precise heading data.

Moving base station RTK technology uses two antennas; one antenna is designated as the primary GPS, and the other as the secondary GPS. Knowing the fixed distance between the primary and secondary antennas allows the *Crescent VS100* to constrain its search volume when computing GPS satellite positioning data.

The *Crescent VS100* contains an integrated electronic tilt sensor and gyro. The tilt sensor constrains the RTK heading solution beyond the volume associated with just a fixed antenna separation. It is able to do so since the *Crescent VS100* knows the approximate inclination of the secondary antenna with respect to the primary antenna.

2.5.2 Survey Software and Data Hub

Survey planning, calibration, data collection and post-processing were accomplished using *Hypack Max* (Version 9.0) survey software running on an *Intel Pentium / Windows XP* platform. *Hypack* was used to plan the survey utilizing ortho-rectified aerial photos, planned line files and configuration files that describe each survey component and calibration parameters.

The *Hypack* configuration file describes each transducer in terms of physical (horizontal and vertical) position on the vessel and relative position to the other survey components (RTK, heading / roll sensor

and survey vessel centerline). The configuration file also contains the software drivers for each component and manages the data stream into the appropriate raw data files.

Survey data was integrated into an on-board network controlled by a central serial to network hub manufactured by *Quatech Inc (Model EME-100D)*. The high speed hub accepts 8 nine (9)-pin serial cables and manages the data through-put to the survey computer via a network cable connection. The network hub manages data strings from the heading and roll sensor, the RTK-GPS and each of the four fathometers. *Hypack* then accepts these strings and creates a raw data file for each transducer. A *Hypack* raw file contains the following parameters:

- 1) Time
- 2) Latitude and longitude of each depth data point
- 3) Northing and Easting for each depth data point
- 4) Ellipsoid height
- 5) Position quality
- 6) Raw depth
- 7) Depth corrected for transducer draft
- 8) Vessel speed
- 9) Vessel squat
- 10) Roll
- 11) Vertical offset of each transducer based on a virtual lever arm and the roll reading
- 12) Heading

When the survey vessel steams to the beginning of a new planned line, logging commences automatically with each logging episode resulting in a new raw file.

2.5.3 Singlebeam Survey QA/QC

There are two standard procedures used to ensure fathometer data accuracy: 1) correct local speed of sound calculations and 2) fathometer barcheck calibrations. Fathometers calculate water depth by using algorithms based on the speed of sound through the water column. Since the speed of sound is affected by water temperature and salinity, these values were recorded each day before the bathymetric surveys began and entered into *Hypack* (utilizing the *Odom Digi-Bar Pro*).

The second protocol is a barcheck calibration which was performed on each fathometer before and after each survey day and during the course of the day. This procedure consisted of checking actual water depth at each transducer using a stadia rod and comparing this measurement to the reported depth from the fathometer. Transducer drafts (depth of transducer below the water surface) were measured and recorded while the vessel was underway and stationary.

All barcheck measurements agreed with fathometer output to within less than 0.10 feet every survey day for every transducer. Channel bottom shots taken with the RTK GPS also provided an additional check against the final bathymetric data: All RTK shots were imported into *Hypack* and checked against bottom elevations derived from the bathymetric survey during the final step of data processing.

2.6 SIDESCAN SONAR IMAGERY RE-ACQUISITION

Due to the shallow nature of the project reach and the presence of a salt water lens, about 40% of the original sidescan data was below accuracy standards. The resulting data gap necessitated the need to re-acquire sidescan imagery using a more specialized instrument.

EDS utilized *Teledyne Benthos SIS-1624* dual frequency (200 kHz and 450 kHz) sidescan sonar for the imagery re-acquisition. The *SIS-1624* is a professional grade sidescan sonar survey system that includes a topside acquisition computer and tow cable. Due to the shallow nature of the survey reach, the tow fish was attached to the bow of the survey vessel instead of being towed behind the vessel (traditional sidescan setup).

The range setting on the *SIS-1624* was set to 25 meters (the range on each side of the vessel) in order to maximize sonar imagery resolution. The *SIS-1624* has variable ranges settings ranging from 25 – 100 meters. While setting the range to 25 meters maximizes the image resolution, this requires more survey vessel passes to achieve 100% bottom coverage.

Planned side scan survey lines were set at 25 meters in order to achieve 50% overlap on each survey swath. This protocol ensures that all sides of an object on the river bed are “illuminated” by the side scan swath.

2.6.1 Sidescan Data Processing

Sidescan imagery was acquired and processed using the software package *Triton Isis v7.1*, with final image mosaicing completed in *Triton Map v3.1*. Acquisition of sidescan images resulted in 87 files each processed individually using the following guidelines:

1. We applied the amplitude option for bottom tracking and slant range correction, using the point and click method to ensure a clean removal of the water column where needed.
2. Time varying gain was set during acquisition, but was adjusted where necessary following bottom tracking and slant range correction to maximize image quality.
3. A coverage map was generated using the course made good option and the default 9 point boxcar filter. Navigation was smoothed, and outside points eliminated when necessary. Few GPS-RTK points were removed as survey was never permitted when out of GPS lock, or when a weak signal was present.
4. Resolution was set at 0.05m, to maximize final image quality.
5. Mosaicing in *Triton Map v3.1* divided the survey into 12 regions, where a small number of select images were mosaiced relative to a specific area to reduce file size and improve the mosaicing image.
6. Each completed mosaic was exported as a geo-referenced TIF file set to the projection identified in step 4.

2.6.2 Sediment Sampling

Using a ponar sampler, 300 sediment grab samples were collected throughout the project reach. The analysis of these samples aided in the interpretation of the side scan imagery and resulting sediment-type map.

Sediment sample locations were chosen via a random sampling along regularly spaced transects or by analyzing preliminary sidescan imagery and navigating to pre-determined areas of interest. Using *Hypack* survey navigation software, a target file containing the northing and easting of a desired sampling location was loaded into the survey computer. The sampling vessel then navigated to this location, dropped anchor, and a sediment sample was brought to the deck using the ponar sampler.

A variety of meta data was recorded for each sample: 1) position, 2) photo documentation and a qualitative description of the sample. The water was then drained from the sample and the sample was then placed in a sample bag and logged.

2.6.3 Sediment Sample Analysis

The sediment samples were then sent to the University of California at Berkeley's fluvial sediment analysis lab lead by Dr. Leonard Sklar and run by Stuart Foster (Building 256, Lark Drive, RFS, Richmond, CA).

Depending upon sample and water content, each was either air dried or dried in an oven at 60 degrees Celsius. The samples that were dried in the oven were first removed from their bags, and all other samples were cut open and allowed to dry within the bags. Some samples formed large aggregate masses when drying and were broken up using a mortar and pestle being careful not to destroy original grain sizes.

Once the samples were completely dried they were transferred to a stack of sieves labeled 8mm, No.10, No.30, No.70, and No. 230 from top to bottom. At the bottom of the stack there was a collection pan. The stack of sieves is then placed in a *RotoTap* for five to fifteen minutes depending upon the size and type of the sample. The stack of sieves is then removed from the *RotoTap* and each individual sieve is carefully emptied. A brass wire brush was then used to remove grains which may be stuck within the mesh of the sieves. After collecting the entire contents from each sieve they are weighed and recorded. Large rocks are measured by hand to see if they are greater than 64 mm and fall in the cobble category. It is also noted if there is a noticeable amount of organic matter in the samples. All large organic matter is removed before sieving.

The sediment samples were divided into eight classes consisting of: Bedrock, Clean Gravel, Poor Gravel, Silty Gravel, Clean Sand, Poor Sand, Silty Sand, and Mud. The sample locations were then plotted in AutoCAD, separated and color coded by layer. These were used as the main reference for delineating the sediment boundaries on the sidescan imagery mosaic. Table 3 outlines the criteria by which the sediment samples were categorized.

Table 3 - Sediment Sample Classification Criteria

<u>Sediment Classification</u>	<u>Sieve Size Criteria</u>
Cobble	>64mm
Gravel	>8mm
Clean Gravel	>2mm
Coarse Sand	>0.595mm
Medium Sand	>0.21mm
Fine Sand	>63µm
Mud	<63µm

A separate layer delineating sub-aquatic vegetation was then created based on vegetation samples taken as well as GPS waypoints collected around sub-aquatic vegetation patches.

2.7 TOPOGRAPHIC SURVEY EQUIPMENT

In order to collect survey data above the waterline, and in the Willow Creek Channel and floodplain, the EDS field crew utilized two types of land-based survey equipment: 1) RTK-GPS and an 2) Electronic Theodolite (total station).

2.7.1 Real Time Kinematic GPS

RTK surveying requires dual frequency LI/L2 GPS receivers: One receiver is set over a known point (base) and the rover receiver collects XYZ data as well as point meta-data as the field surveyor travels along a pre-determined survey transect. The base station receiver relays satellite correction information to the rover via a multi frequency *Pacific Crest PDL* radio modem.

The survey team utilized a *Leica System 1200* RTK base station and rover setup. The RTK base station was set over control point BM 1 at the visitor center with the rover unit programmed to collect data in California State Plane Coordinates, Zone 2 (NAD 83 Datum) and Ft. NAVD 88.

2.7.2 Total Station Surveys

In areas of dense vegetation, a total station was used to collect topographic point data. The total station was setup over a temporary benchmark (TBM) set using the RTK-GPS and oriented (back sighted) to a second TBM. Total station raw data (prism distances, azimuth angles and vertical angles) were collected by a *Carlson Survey Explorer* data collector which calculated and recorded northings, eastings, elevations and codes.

Before and after the total station control points were set, the RTK rover checked into one of the three project control points before and after each total station BM survey.

2.7.3 Topographic Surveys QA/QC

There are many environmental and operator-based influences that can affect the accuracy of RTK-GPS and the resultant baseline solutions. Although RTK-GPS is an emerging tool among hydrographers, little attention has been given to an accuracy standard for this methodology—especially in the field of coastal mapping and monitoring. In an effort to limit operator error and to quantify daily environmental error, we have developed an internal standards protocol for RTK error estimation based on thresholds developed by the California Department of Transportation and the US Army Corps of Engineers (USACE) Topographic Accuracy Standards (CALTRANS 2002; USACE 1994).

The first step in our protocol is to determine an appropriate land-based GPS station that will provide the most accurate corrections and range to the outer limits of the survey area. We chose to set a mark on at the intersection of Highways 1 and 116 (BM 2).

The second component of our RTK-GPS protocol is to perform a detailed GPS site calibration prior to the collection of any hydrographic survey data. The site calibration is used to determine the base station quality relative to the local network of NGS and NOS survey control and to analyze any potential spatial separations between the local geoid heights (GEOID 03) and ellipsoidal values (WGS-84) that may influence the resulting orthometric elevations. The calibration entails selecting the control to be used for the RTK-GPS base station receiver and radio broadcast system and then checking at least two known geodetic benchmarks of exceptional horizontal and vertical quality within and even outside the survey boundaries. The benchmarks are occupied in “site calibration mode” over 300 epochs or approximately 3 to 5 minutes.

2.8 CONTOUR MAP GENERATION

Topographic features were surveyed using data from the LIDAR aerial survey, while hydrographic features were surveyed with multibeam bathymetric soundings. The two surveys were combined to create one, cohesive map of the entire reach.

2.8.1 LiDAR Data

As stated in Section 1.6, filtered LiDAR data was supplied to EDS by *Delta Geomatics Inc* (Appendix D contains meta-data reports supplied to EDS by *Delta Geomatics*). These data were then combined with the bathymetric and other topographic data collected by EDS in order to create the contour maps.

The *Delta Geomatics* field crew used NGS JT9623 as the basis for the LiDAR surveys. *Delta Geomatics* used an Ellipsoid height reported at 31.218 meters (207.40 US Ft. NAVD 88 Vertcon) for this point (Appendix D). However, this point was adjusted in 2007 (as part of the Central Coast Height Modernization Project undertaken by the California Spatial Reference Center) and assigned a new Ellipsoid height of 31.059 M, a difference of 0.159 meters (0.554 feet). The OPUS-derived (EDS) elevation for this point is 206.30 US Ft. NAVD 88; a difference of -1.05 Ft. between what Delta used and what EDS used. This difference was then applied to the LiDAR data.

LiDAR Ground-Truthing

The LiDAR data was ground-truthed by EDS surveyors on January 15, 2009. A topographic grid was established and surveyed (using RTK GPS) upstream of the Willow Creek channel, river left (Figures 12 and 12a). Areas within this grid included bare earth on the cobble bar, low vegetation on the floodplain, and a tall vegetation canopy within the forested area.

RTK GPS data were then compared with LiDAR data using AutoCAD Civil 3D (2009) resulting in an elevation difference map of the area (see Figure 12). The colors represent the elevational difference between the supplied LiDAR data and the RTK data. The red areas are where the LiDAR data is 1 to 7 feet below the rover data, the green areas are where the LiDAR data is 0 to 1 feet below the rover data. The blue areas are where the LiDAR data is 0 to 1 feet above the rover data and the purple areas are where the LiDAR data is 1 to 13 feet above the rover data. The red and purple areas represent areas of extremely dense and tall vegetation.

2.8.2 AutoCAD Work

Due to the large quantity of data collected, it was impractical to combine all data into a single surface model. Therefore, 5,000 x 5,000 foot grids were setup, and a surface model was generated for each grid area. Adjacent grid data was temporarily combined in order to generate accurate contours at the grid border. Boundaries were drawn at the interface between the topographic LIDAR data and hydrographic multi-beam data in order to avoid conflicting data and mesh the data together. The surface models were examined for invalid survey points, and any data that seemed incorrect and did not agree with adjacent survey data was removed.

The data was then combined into a single TIN surface model containing both the topographic LIDAR, Willow Creek topographic and hydrographic multibeam data for each grid area. Contours were generated at 2 foot intervals from these surface models. The surface models were then exported to DEM format on a 1 meter grid. AutoCAD Civil 3D 2009 was used for all surface modeling and contour generation.

3. RESULTS

3.1 SURVEY NETWORK – DATUMS

OPUS-derived elevations and positions were used for network control. A Precision Point Positioning (PPP) analysis was subsequently done on all RINEX files in order to give confidence to the OPUS-derived elevations and positions.

Results from the OPUS-based surveys and PPP analysis are shown in Table 4, and comparisons to NGS JT9623 and CENCH point DUNC are shown in Table 5.

Table 4 – Final Control Network Results (used for all surveys)

Mark	Date	State Plane Coordinates (Z2) (NAD 83)		Orthometric Height (NAVD 88 US Ft.)				Notes
		Northing (US Survey Ft.)	Easting (US Survey Ft.)	OPUS	OPUS-RS	PPP	RMS (one sigma)	
BM-1	10/2/2008	1,927,388.46	6,242,277.19	11.71*	11.71	11.73	0.008 m	Visitor Center
BM-2	10/2/2008	1,922,927.13	6,245,419.63	19.57	19.57	19.57	0.040 m	Hwys 1 & 116 intersection
BM-3	10/2/2008	1,928,686.98	6,260,357.65	35.22*	35.23	35.30	0.016 m	DUNC

Table 5 – Final Control Network Results (used for all surveys)

Mark	Survey Date	REPORTED State Plane Coordinates (Z2) (NAD 83)		Reported Elevation (NAVD 88 US Survey Ft.)	SURVEYED State Plane Coordinates (Z2) (NAD 83)		Surveyed Elevation (NAVD 88 US Survey Ft.)
		Northing (US Survey Ft.)	Easting (US Survey Ft.)		Northing (US Survey Ft.)	Easting (US Survey Ft.)	
JT9623	2/20/2009	1,932,686.439	6,232,582.451	206.59	1,932,685.859	6,232,582.846	206.34
DUNC	4/11/08	1,928,687.546	6,260,357.238	35.54	1794817.5697	6651803.9058	35.23

Since the project location is in a coastal area and is subject to tectonic movement, ocean loading, and the fact that the nearby HARN point JT9623 has an uncertain history of vertical stability (the recent height modernization elevation being listed as 0.81 Ft. lower than the previous elevation Table 6), we believe the best solution is to hold the OPUS-derived values from the 10/2/2008 survey, which were the coordinates used for the hydrographic and topographical surveys. Appendix E contains a stamped accuracy statement by Thomas Tucker (LS 4460, and ASPRS CP #1430).

Table 6 – History of NGS JT9623 (disregarding the Vertcon established elevation of 207.40 Ft)

Data Sheet Retrieval Date	NGS Epoch	Ellipsoid Ht. (Meters)	Geoid Ht. (Meters)	Geoid Model	NAVD 88 (Meters)	NAVD 88 (Feet)
7/16/2004	1991.35	31.16	-31.86	Geoid 03	63.020	206.76
2/17/2009	2007.00	31.015	-31.86	Geoid 03	62.875	206.28
6/9/2011	2007.00	31.015	-32.08	Geoid 09	63.095	207.00

3.2 TIDAL MONITORING

A total of five (5) water level recorders (gages) were deployed on September 19, 2008 and removed on 11/10/2008 (the downstream gage was left in place to 11/25/2008 – Figure 2). Two (2) closure events were captured during the deployment. The first closure event started on 10/2/08 at approximately 18:36 and ended when the mouth was mechanically breached on 10/9/08 at approximately 8:30 AM. The maximum water level during this time period reached an elevation of 7.87 Ft. NAVD 88 (Figure 13).

The second closure event to be captured by the time series started on 10/24/08 at approximately 20:00 and lasted until 11/3/2008 when the mouth was mechanically breached at approximately 13:00. The water surface elevation during this event reached a maximum level of 10.75 Ft. NAVD 88. Appendix F contains individual graphs showing the time series from each station.

The accuracy of each WSE time series is the result of the inherent accuracy of the recording pressure transducer (0.05% of the full scale range), the WSE calibration survey, and the accuracy of the primary benchmark. Given that each level loop from a primary project control point (BM-1, BM-2 and BM-3) to the tide gage location closed out with an average error of 0.05, and the activity of actually surveying the water surface contains an inherent error (is the rod actually exactly on the water surface? Is it slightly below the water surface? Does the WSE survey shot exactly correspond to the time that the gage measures and records a point?) the overall accuracy of each time series would be approximately ± 0.20 Ft.

Table 7 – Tide Time Series Period of Record

Gage Name	Period of Record	Location	Notes
Downstream	9/19/2008 12:18:00 PM - 11/25/2008 8:42:00 PM	38° 26' 59.269" 123° 07' 30.311"	Gage was not deployed low enough to capture all of the low waters.
Route 1 Bridge	9/19/2008 1:06:00 PM - 11/10/2008 4:54:00 PM	38° 26' 4.1707" 123° 06' 5.344"	Located on first pier from river left
Third Gage US	9/19/2008 1:48:00 PM - 11/10/2008 1:30:00 PM	38° 26' 51.053" 123° 05' 23.5531"	This gage was the subject of vandalism on 10/18/2008 @ 11:24. Time series before this time and after this time required two separate calibration numbers
Fourth Gage US	10/5/2008 11:00:00 PM - 11/10/2008 2:12:00 PM	38° 26' 48.7342" 123° 03' 36.2839"	This gage failed upon initial deployment. Logging started on 10/5/2008 (deployment was 9/19/2009)
Cassini Ranch	9/19/2008 3:06:00 PM - 11/10/2008 3:00:00 PM	38° 27' 46.614" 123° 03' 13.5537"	

3.2.1 Tidal Datum Analysis

There is a very close fit among the elevations of the high tides across all stations (Table 8). However, the data at each station covered a different period of record, which resulted in a variation among local values for MHW. The low tides did not match up very well across stations likely due to rising riverbed elevations (loss of low tide signature at monitoring location) and gage going dry at low tide. The low tides were completely cut off at the downstream location. This lack of data at many stations made it impossible to calculate an accurate value for MLW. The values given below should only be regarded as the average of the lowest water level observed (LWLO) at each station, and not MLW.

Table 8 – Tide Datum Analysis Results

Station	Elevation (US Ft. NAVD 88)		Elevation (ft NGVD 29)	
	MHW	LWLO	MHW	MLW
Downstream	4.24	3.01	2.502	1.272
Upstream 3	4.12	2.06	2.382	0.322
Upstream 4	4.04	2.43	2.302	0.692
Cassini Ranch	4.27	2.83	2.532	1.092

The high and low tide time series plots for all stations can be found in Figures 5 and 6, respectively.

As mentioned above, there was good agreement in the elevations of the high tides across all stations. Since the Downstream station had the longest period of record, the MHW value for this station should be used as the overall MHW value for the study area.

TIDAL DATUM DETERMINATION (Next Steps)

If requested, a tidal datum reckoning could be performed using the data from the Downstream monitoring station to determine the true value of MHW in the study area. The closest tidal reference station is at Point Reyes (Station ID: 941-5020). While this station is rather far from the study site, it presents the most reliable option for calculating tidal datums. It will not likely be possible to calculate a reliable value for MLW for the reasons described above.

3.3 BATHYMETRIC SURVEYS

Bathymetric data collection was timed to take advantage of the high water conditions present when the river mouth at Jenner closes. Mouth closures are the result of low river flows coinciding with a large swell event. Due to the shallow nature of the project reach, a mouth closure presented the only conditions that would be suitable to collect high-resolution bathymetric data.

The bathymetric survey started with patch tests commencing on October 3, 2008, with actual data collection starting the next day on October 4. The survey effort ended on October 9, 2008 without incident. The initial coverage maps (Figures 14, 15 and 16) showed gaps in the surveyed areas resulting from shallow areas such as various point bars, the shoal area upstream and downstream of Penny Island as well as areas containing snags or similar types of underwater hazards.

The singlebeam “gap-fill” survey effort occurred on December 2 - 3, 2008 and February 12, 18 and 19th 2009, during two mouth closure events. Table 9 outlines dates and other details describing each bathymetric survey effort. Appendix G contains the 2-ft bathymetric contours merged with the LiDAR-derived topographic data (generated in AutoCAD 2009 and printed to Adobe PDF format).

Table 9 – Bathymetric Survey Dates

<u>Date</u>	<u>Survey Activity</u>	<u>Survey Time Period</u>	<u>Water Surface Elevation (at the Route 1 Bridge)</u>
9/19/2008	Set survey network		
10/3/2008	Bathy survey day 1: patch test and calibration	8:00 – 17:00	4.68

Table 9 (continued)			
10/4/2008	Bathy survey day 2: patch test and calibration	8:00 – 17:00	5.47
10/5/2008	Survey day 3	8:00 – 17:00	6.61
10/6/2008	Survey day 4	8:00 – 17:00	7.08
10/7/2008	Survey day 5	8:00 – 17:00	7.50
10/8/2008	Survey day 6	8:00 – 18:00	7.87
10/9/2008	Survey day 7	8:00 – 18:00	8.08
12/2/2009	Singlebeam gap fill day 1		
12/3/2008	Singlebeam survey – Willow Creek	7:00 – 17:00	
1/14/2009	Willow Creek channel – total station	9:00 – 17:00	
1/15/2009	Willow Creek channel – total station	9:00 – 17:00	
2/12/2009	Single beam gap fill day 2	9:00 – 17:00	7.87
2/18/2009	Single beam gap fill day 3	9:00 – 17:00	
2/19/2009	Single beam gap fill day 4	9:00 – 17:00	

3.4 SIDESCAN SURVEY

The sidescan survey effort was comprised of two separate data acquisition periods. As stated in section 1.6 initial sidescan imagery was collected concurrently with the bathymetric data using the *C3D* (October 2008). Upon initial review of the imagery, it was determined that some sidescan data would have to be re-acquired.

Very slow moving, shallow rivers are notoriously difficult environments for sidescan data collection. Power settings need to be low enough so the receiving transducers are not “washed-out” with too much backscatter data. However, since high resolution data was a goal, power levels need to be of sufficient strength in order to “illuminate” details on the riverbed. This balance was sometimes difficult to achieve given the shallow nature of the survey (even with water depths maximized during mouth closure). Another difficult feature of the hydrology is the saline lenses which are present in the lower reaches of the river. The saline lens tends to refract the sonar swath resulting in “ghost images” present behind the actual shape of an object on the riverbed.

A second, intensive sidescan data re-acquisition effort commenced on September 14 and ended on September 16, 2009. An *Edgetech 4200* series dual-frequency sidescan sonar system was utilized for this second data collection effort. The second effort yielded robust, detailed imagery of the river bed. Once the images were placed in a mosaic, sediment type mapping could commence.

3.5 SEDIMENT TYPE MAPPING

The first step in building a sediment type map was to plot the sidescan sonar imagery in AutoCAD 2009 with the sediment sample locations. Certain areas of bottom roughness (as interpreted from the sidescan imagery) were associated with a sediment type. Once a catalogue of the association of image “roughness” to sediment type was established, the process of categorizing the river bed began.

While the sidescan imagery was of excellent quality and very fine detail, it was often difficult to delineate between the different types of fine sediments (coarse sand vs. fine sand and fine sand vs. mud). Boulders, bedrock, cobbles, and sand were fairly easy to see on the images. But noticing a difference between sand and mud and categories within sand was nearly impossible. As a result, some sediment type polygons are placed as a result of the closest sediment sample and not from the result of a sidescan signature.

Appendix H contains sheets showing the sidescan mosaic with the sediment classification polygons shown as a layer as well as the sediment analysis data presented in spreadsheet format.

4. REFERENCES

California Department of Transportation. 2006. *Survey Manual*, Sacramento, CA.

National Ocean Service. 2003. *NOS Hydrographic Surveys and Specifications and Deliverables*, Washington, D.C.

The International Hydrographic Organization's Standards for Hydrographic Surveys, Special Publication 48, Sixth Edition, April 1997

U.S. Army Corps of Engineers. 2002. *Hydrographic Survey Manual*, Engineering and Design Manual No. EM-1110-2-1003, Washington D.C.

U.S. Army Corps of Engineers. 2002. *Geodetic and Control Surveys*, Engineering and Design Manual No. EM-110-1-1004, Washington D.C.

Appendix A

Figures



 Survey Extent

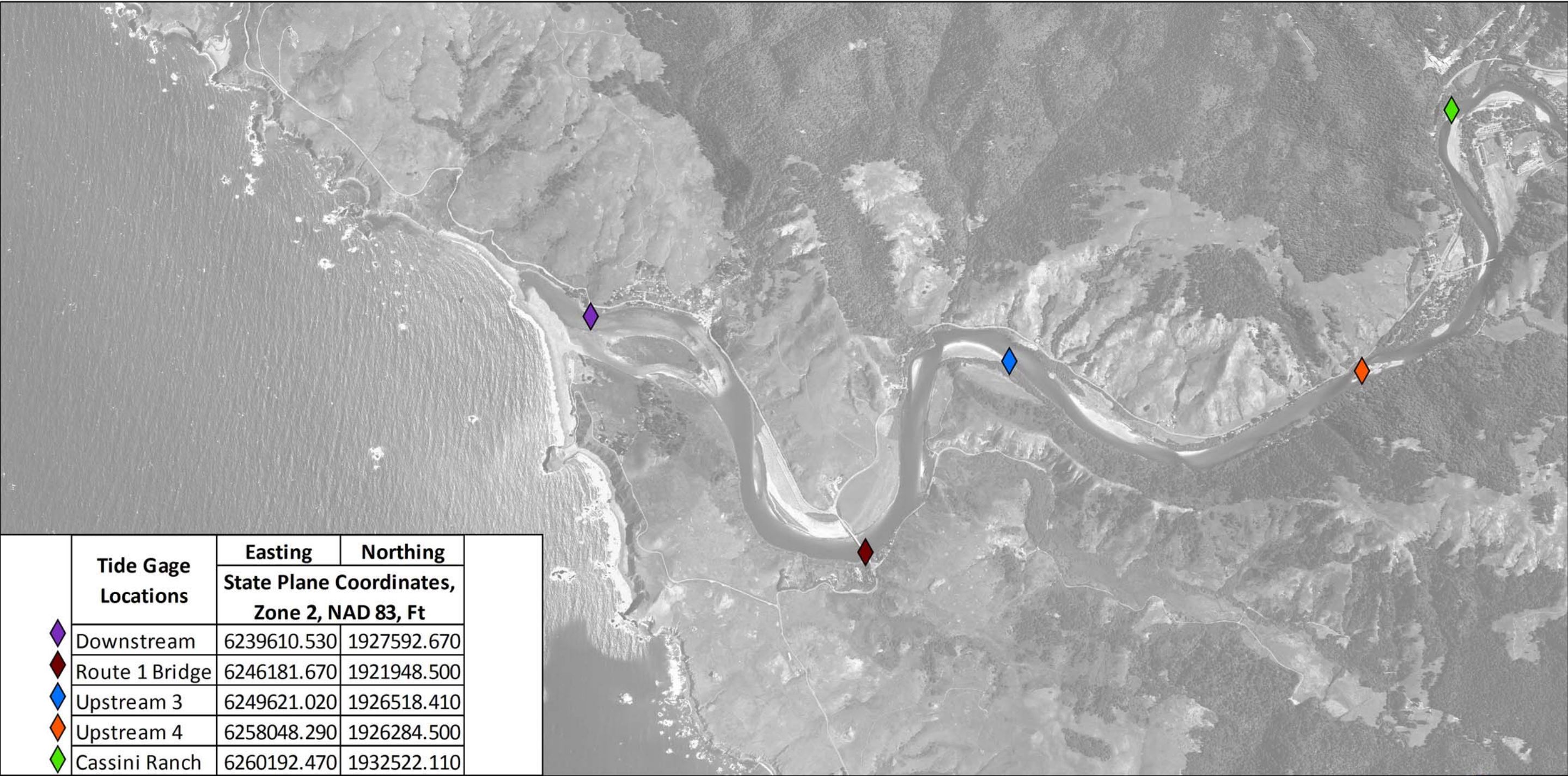


Data Sources: Sonoma County (NAIP 2005);
Survey Data (EDS 2008)
GIS/ Cartography by: Leigh Etheridge
map file: Russian-river-survey-extent_2009-1015lee.mxd

1:30,000 (1" = 2,500' at 11x17 layout)
0 0.25 0.5 Miles
0 0.5 1 Kilometers

**Russian River Estuary
Bathymetric Analysis**
Figure 1: Bathymetric / Sidescan Imagery Survey Extent
September-December 2008



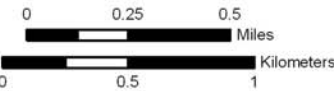


	Tide Gage Locations	Easting	Northing
		State Plane Coordinates, Zone 2, NAD 83, Ft	
	Downstream	6239610.530	1927592.670
	Route 1 Bridge	6246181.670	1921948.500
	Upstream 3	6249621.020	1926518.410
	Upstream 4	6258048.290	1926284.500
	Cassini Ranch	6260192.470	1932522.110



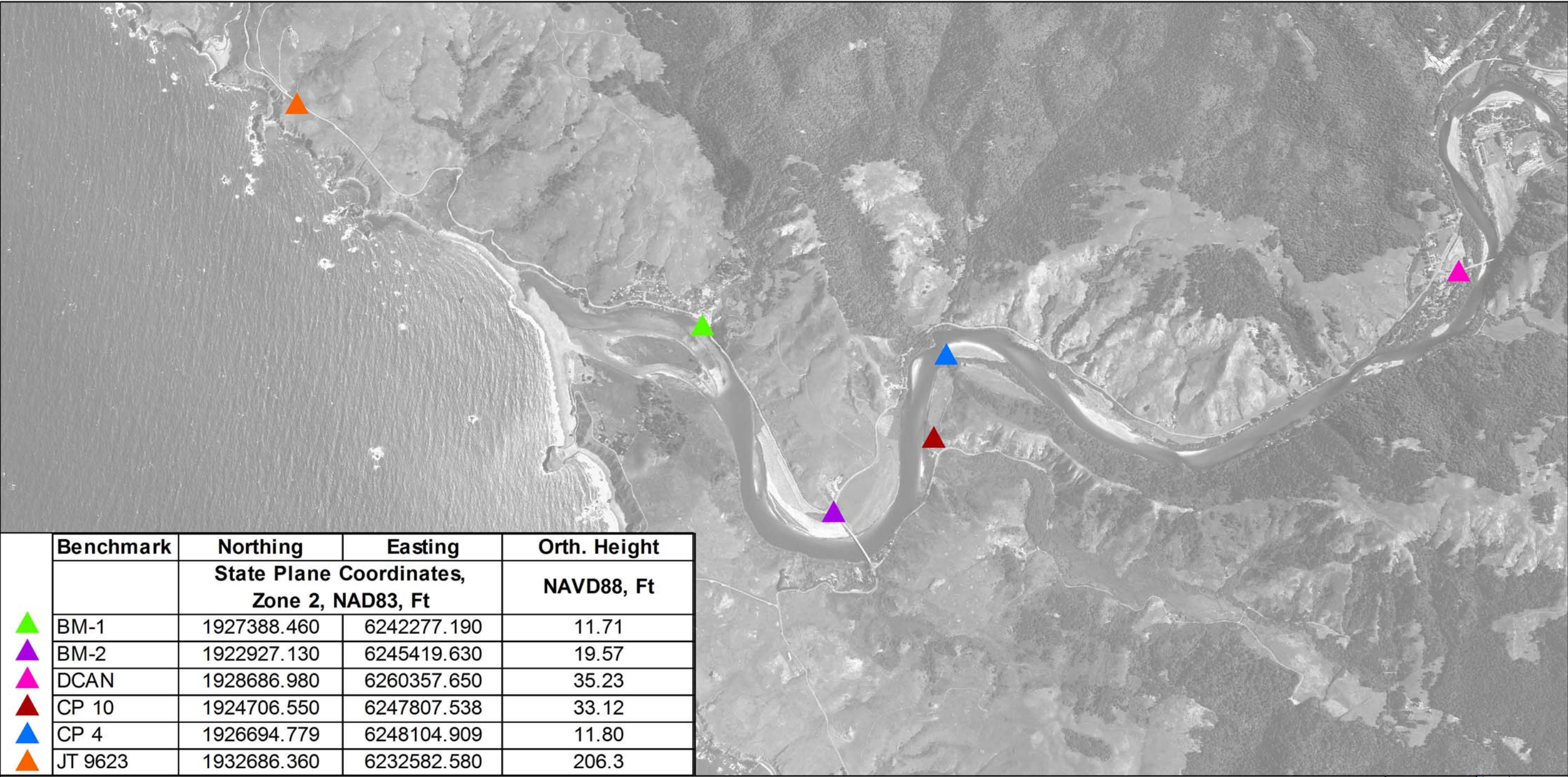
Data Sources: Sonoma County (NAIP 2005);
Survey Data (EDS 2008)
GIS/ Cartography by: Leigh Etheridge
map file: Russian-river-tide-gage-locations_2009-0802lee.mxd

1:30,000 (1" = 2,500' at 11x17 layout)



**Russian River Estuary
Hydrologic Monitoring
Figure 2: Tide Gage Locations
September 19 - December 9, 2008**



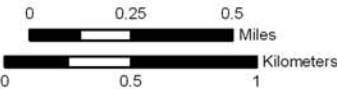


	Benchmark	Northing	Easting	Orth. Height
		State Plane Coordinates, Zone 2, NAD83, Ft		NAVD88, Ft
▲	BM-1	1927388.460	6242277.190	11.71
▲	BM-2	1922927.130	6245419.630	19.57
▲	DCAN	1928686.980	6260357.650	35.23
▲	CP 10	1924706.550	6247807.538	33.12
▲	CP 4	1926694.779	6248104.909	11.80
▲	JT 9623	1932686.360	6232582.580	206.3



Data Sources: Sonoma County (NAIP 2005);
Survey Network Control Points (EDS 2008)
GIS/ Cartography by: Leigh Etheridge
map file: Russian-river-survey-control-pts_2009-0801fee.mxd

1:30,000 (1" = 2,500' at 11x17 layout)



**Russian River Estuary
Bathymetric Analysis**
Figure 3: Survey Control Point Network
September-December 2008





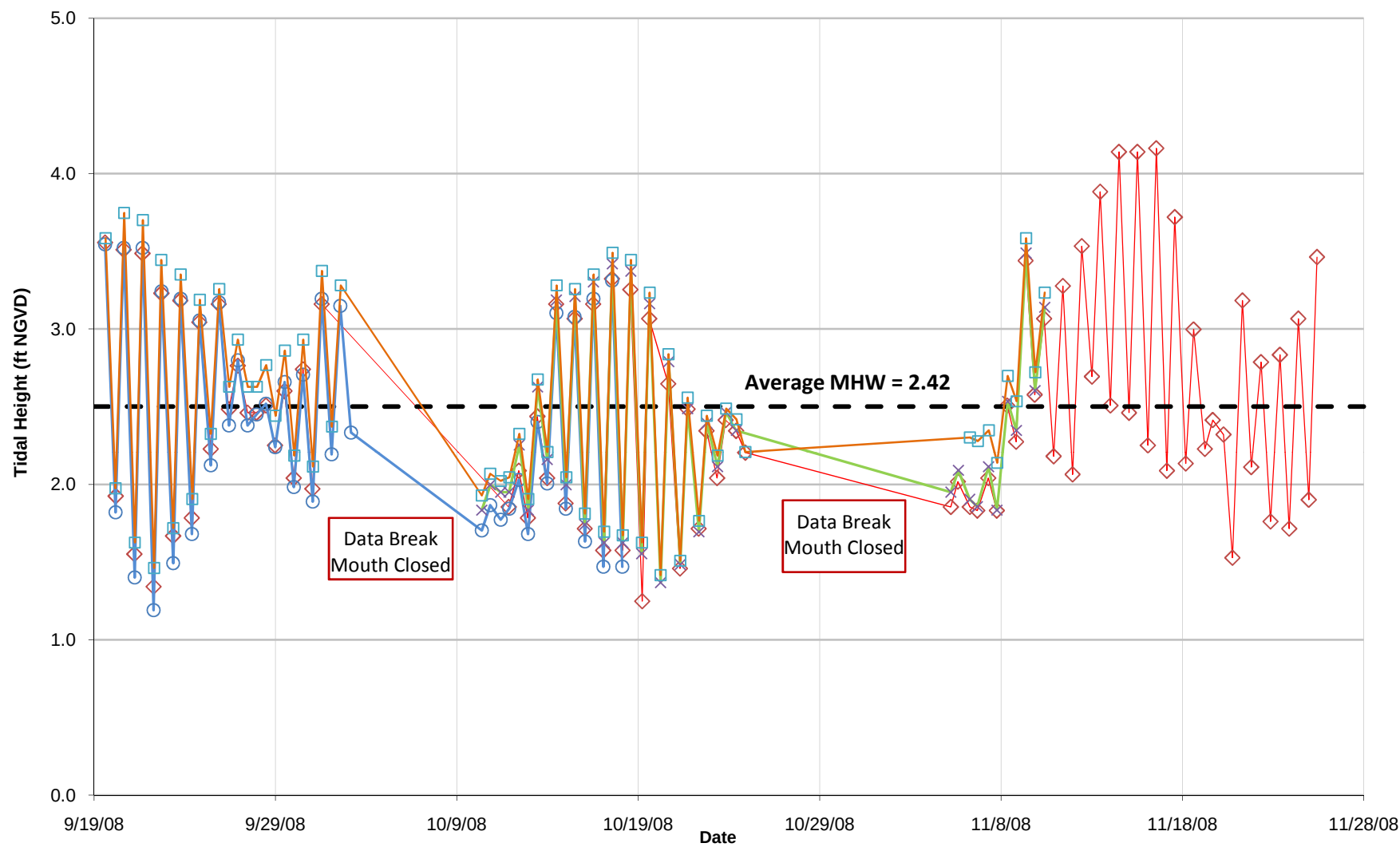
Environmental
Data Solutions

**Lower Russian River Bathymetric Study
Water Surface Elevation Monitoring
Gage Mooring
October 2008**

October 2009

Project No. 172

Figure 4

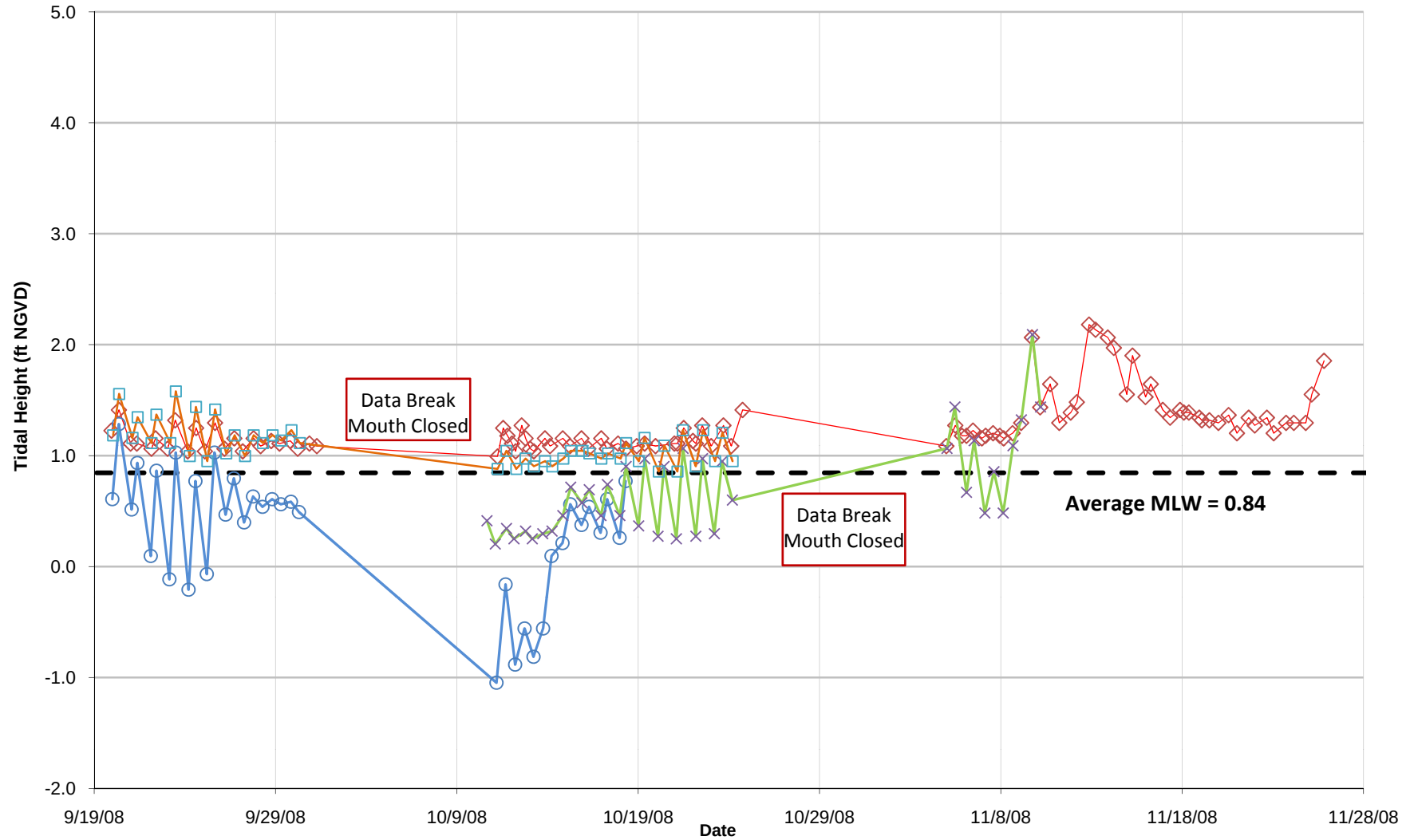


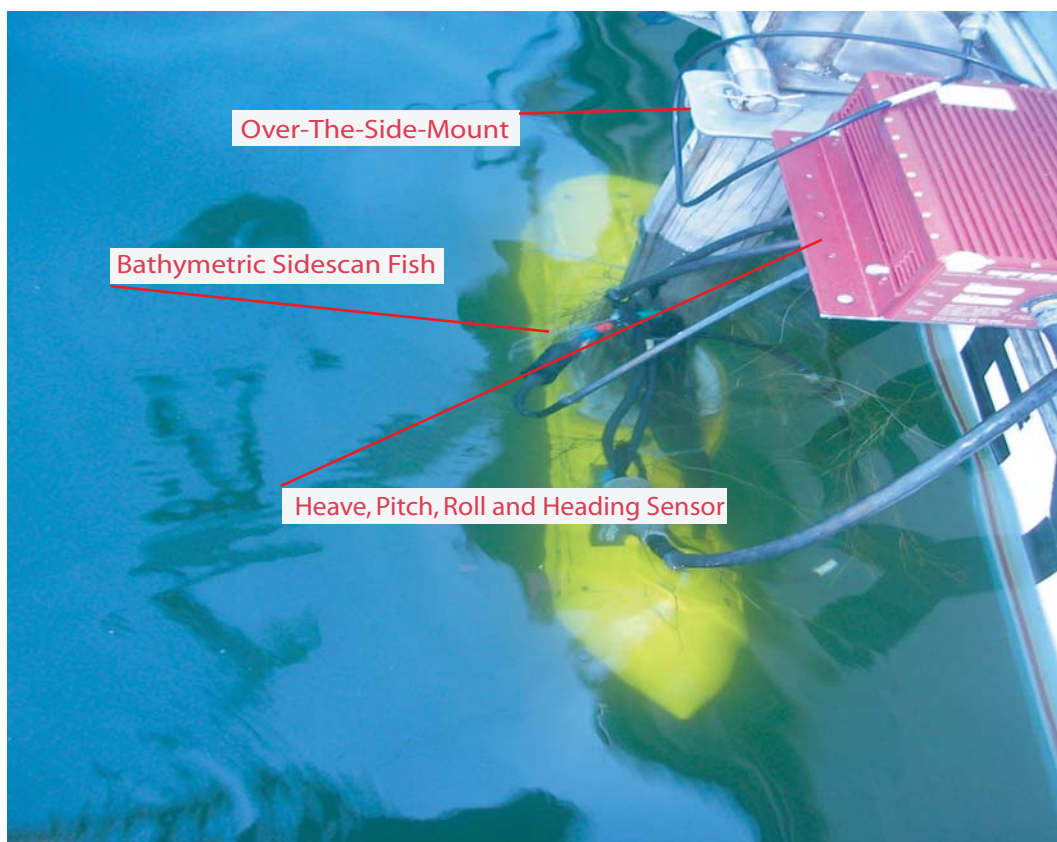
- ◇ Downstream Tides
- - - MHW
- Upstream #3 Station
- × Upstream #4 Station
- Cassini Ranch

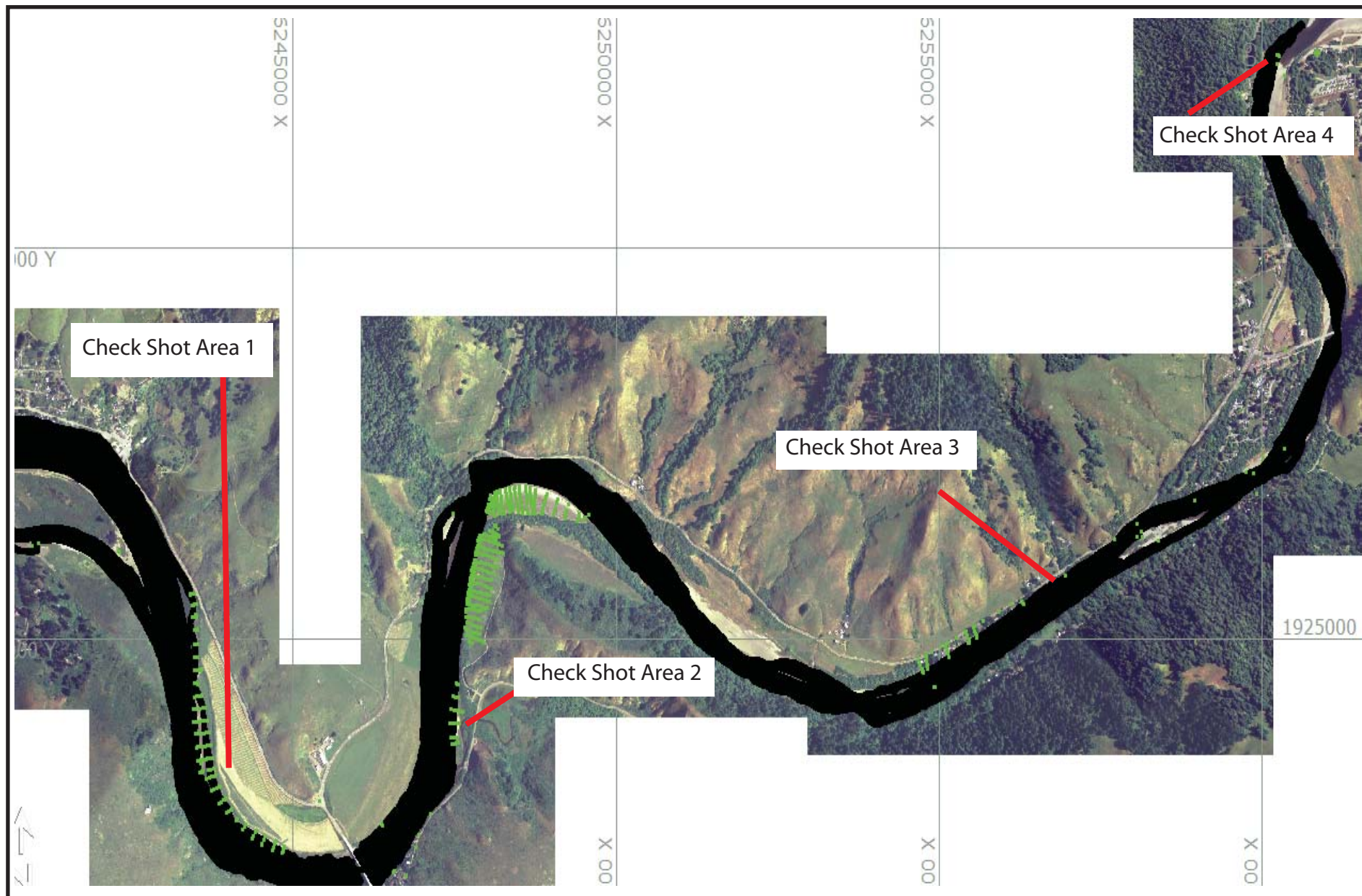
Lower Russian River Bathymetric Analysis

Tidal Time Series MHW Analysis: All Stations - High Tides
September 19 - November 28, 2008

Figure 5







Black Area: Bathymetric data
Green Shots: RTK-GPS survey shots



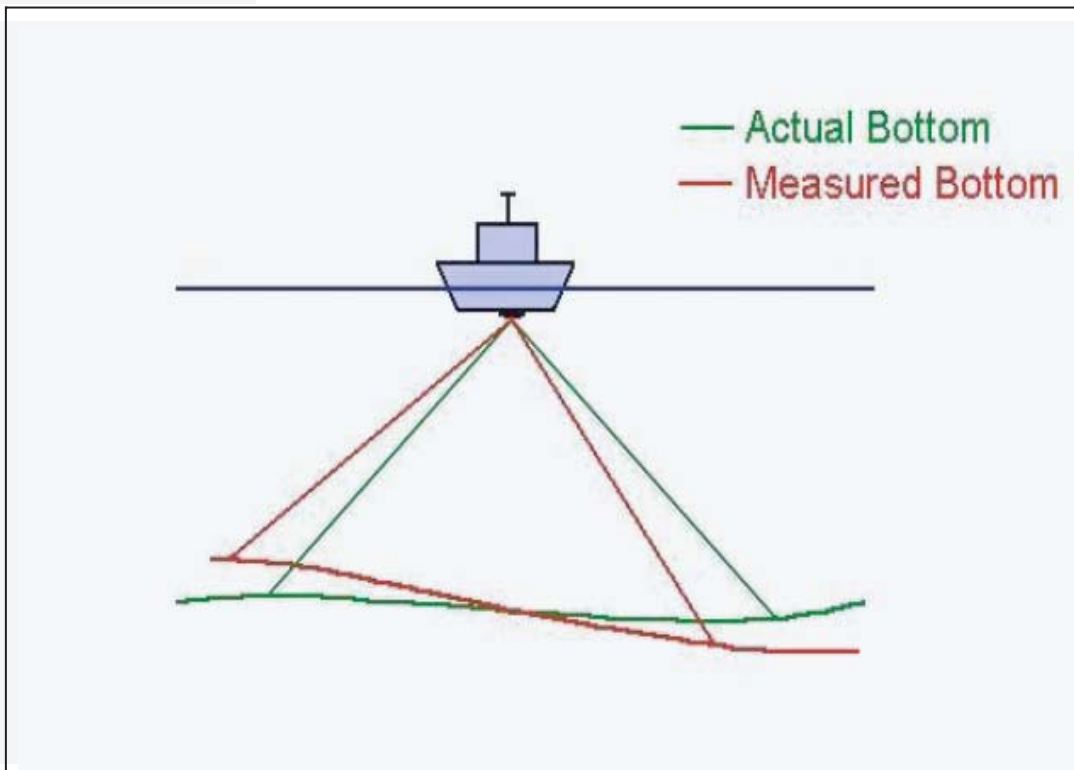
Lower Russian River Bathymetric Analysis Bathymetric and Topographic Check Shots

October 2008

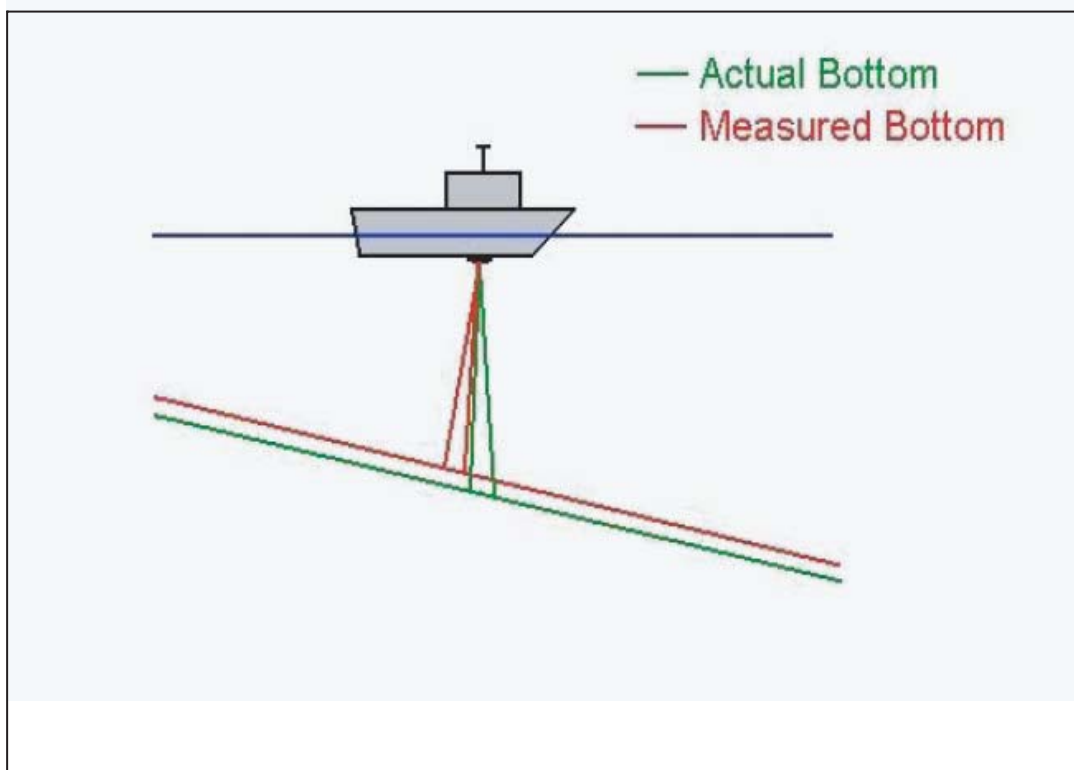
November 2009

Project No. 1742

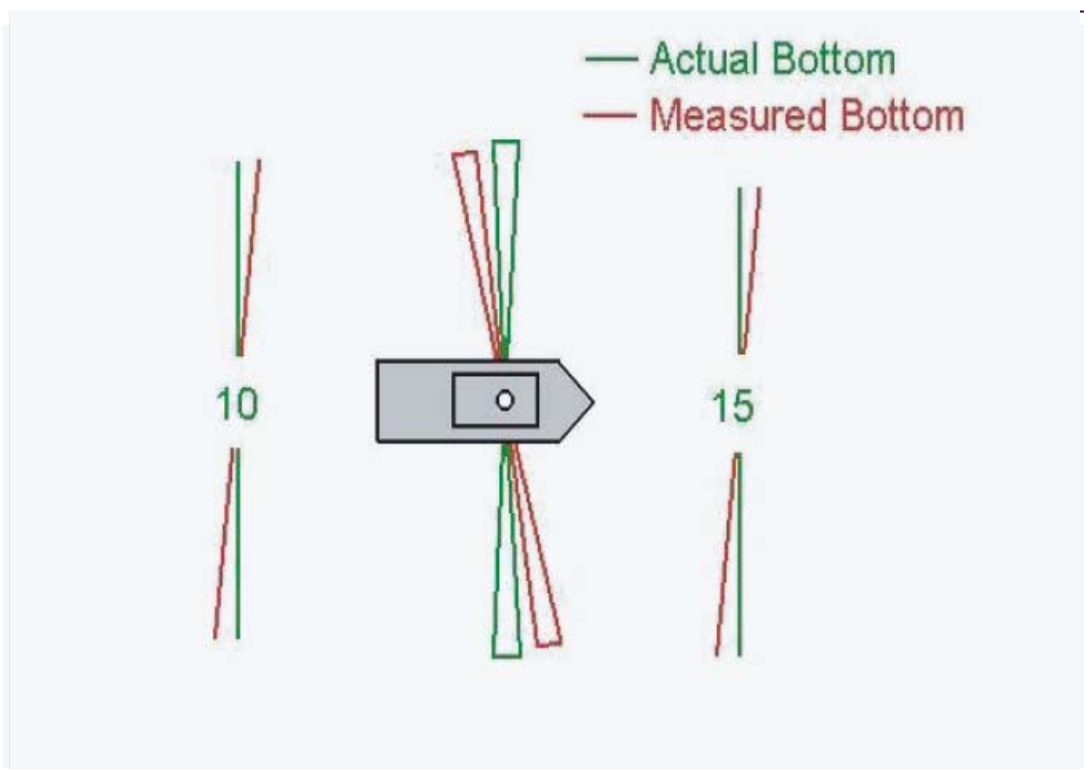
Figure 8



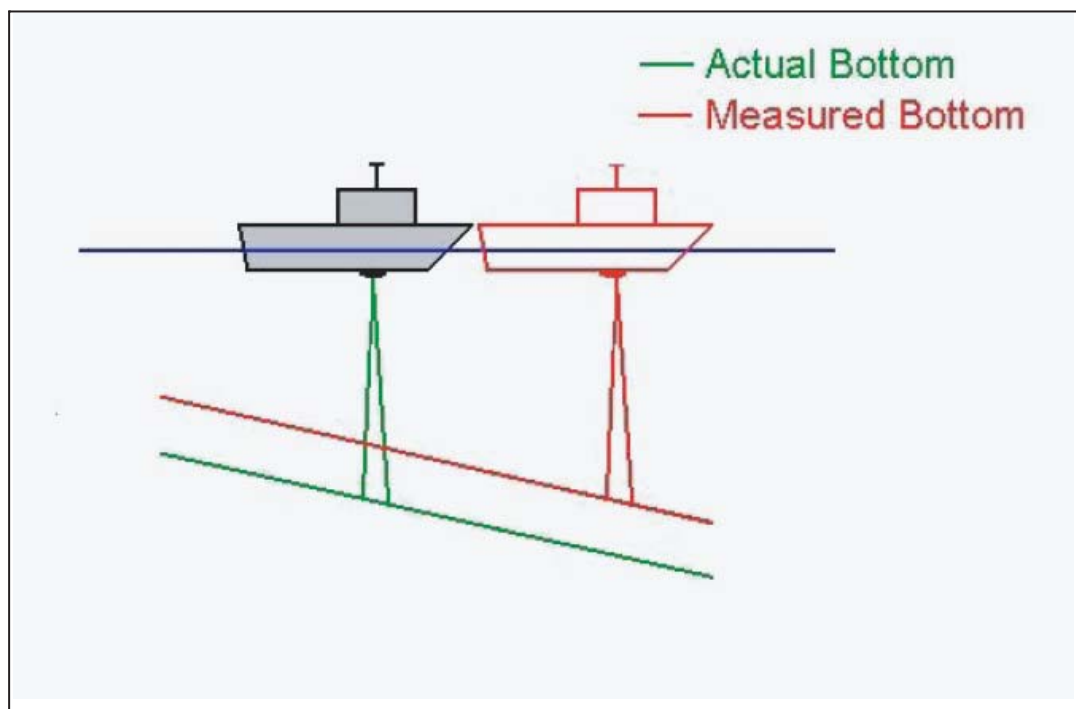
Roll Patch Test



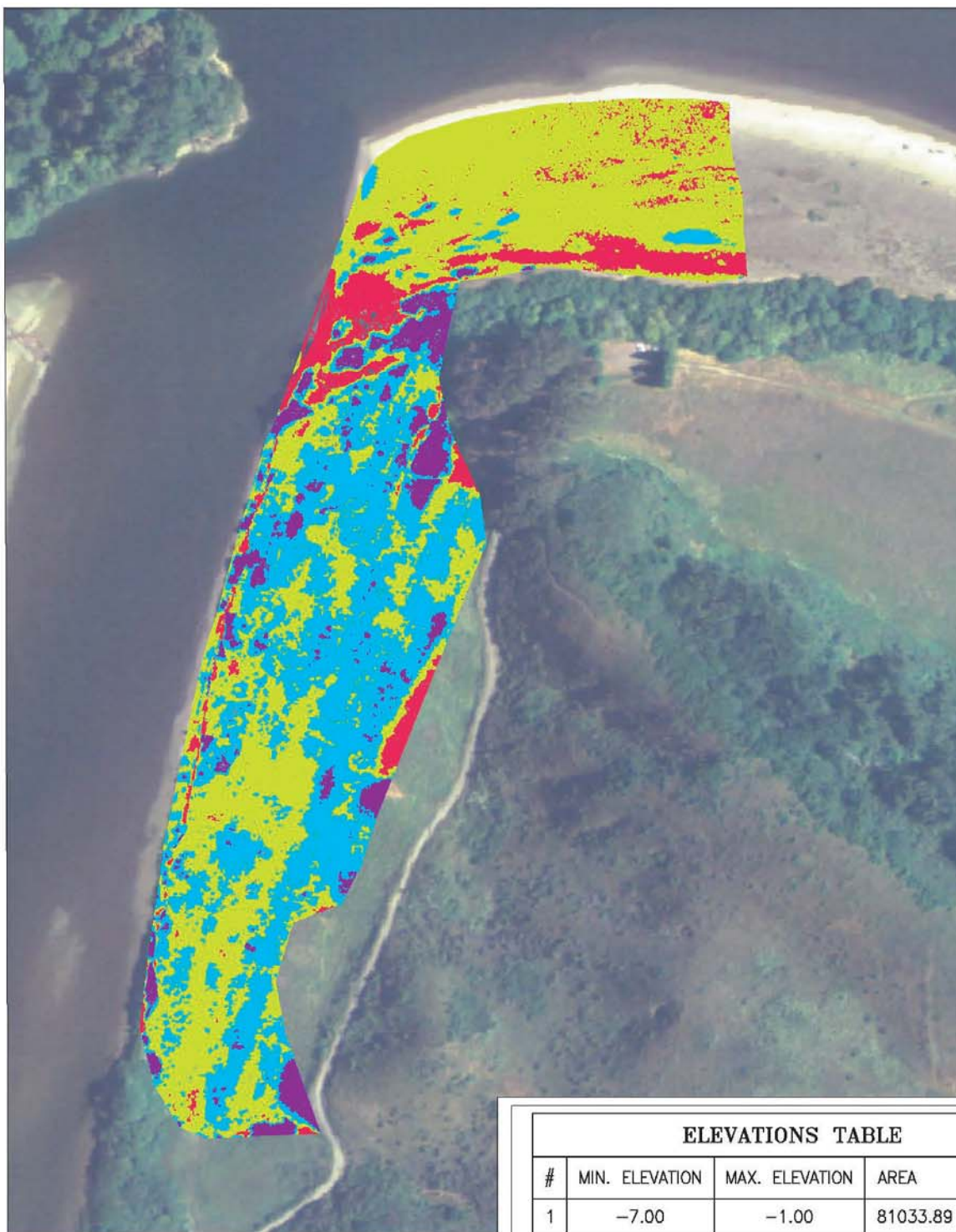
Pitch Patch Test



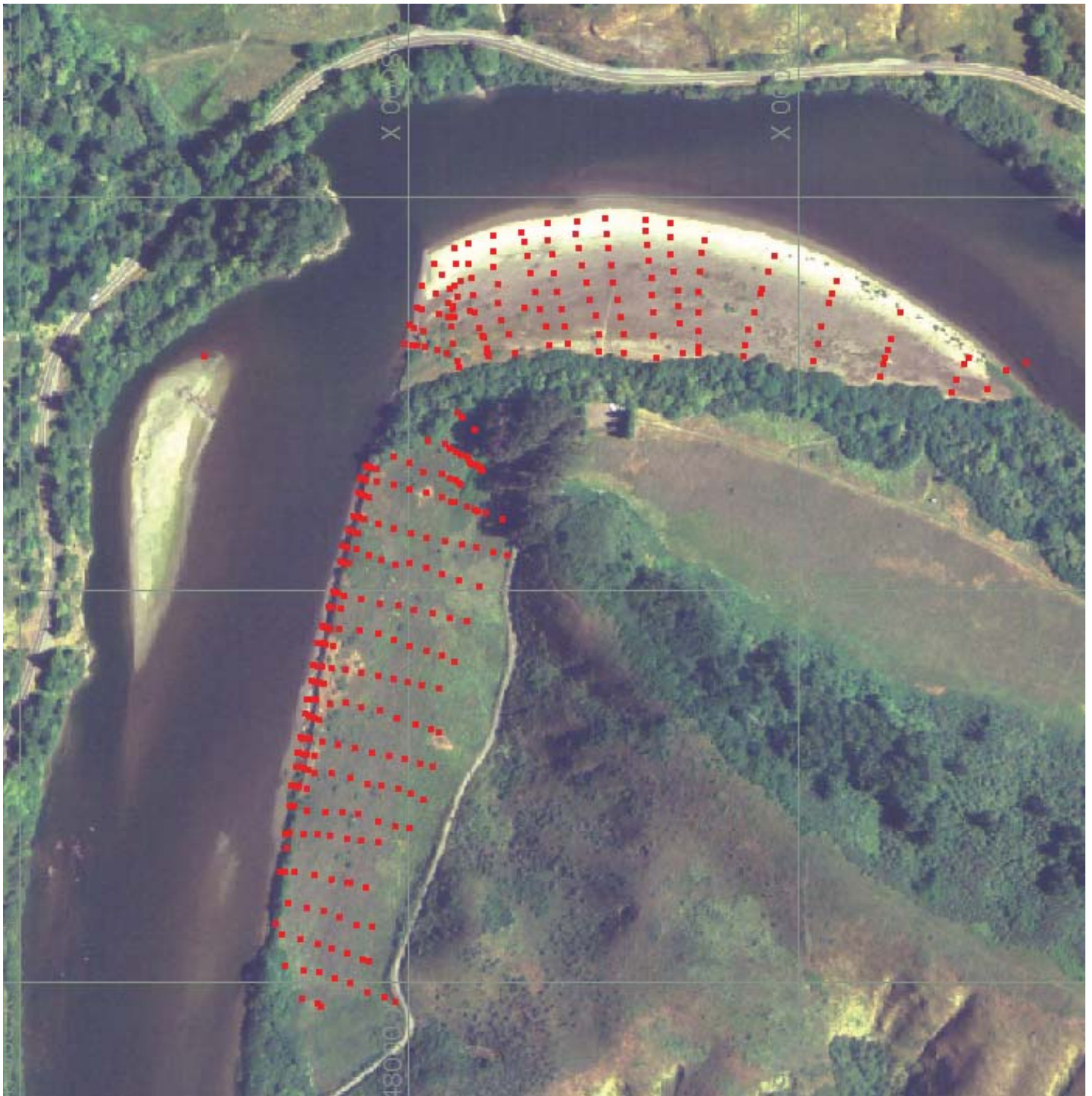
Yaw Patch Test



Latency Patch Test



ELEVATIONS TABLE				
#	MIN. ELEVATION	MAX. ELEVATION	AREA	COLOR
1	-7.00	-1.00	81033.89	Red
2	-1.00	0.00	376533.74	Yellow
3	0.00	1.00	232693.47	Blue
4	1.00	13.00	46799.82	Purple



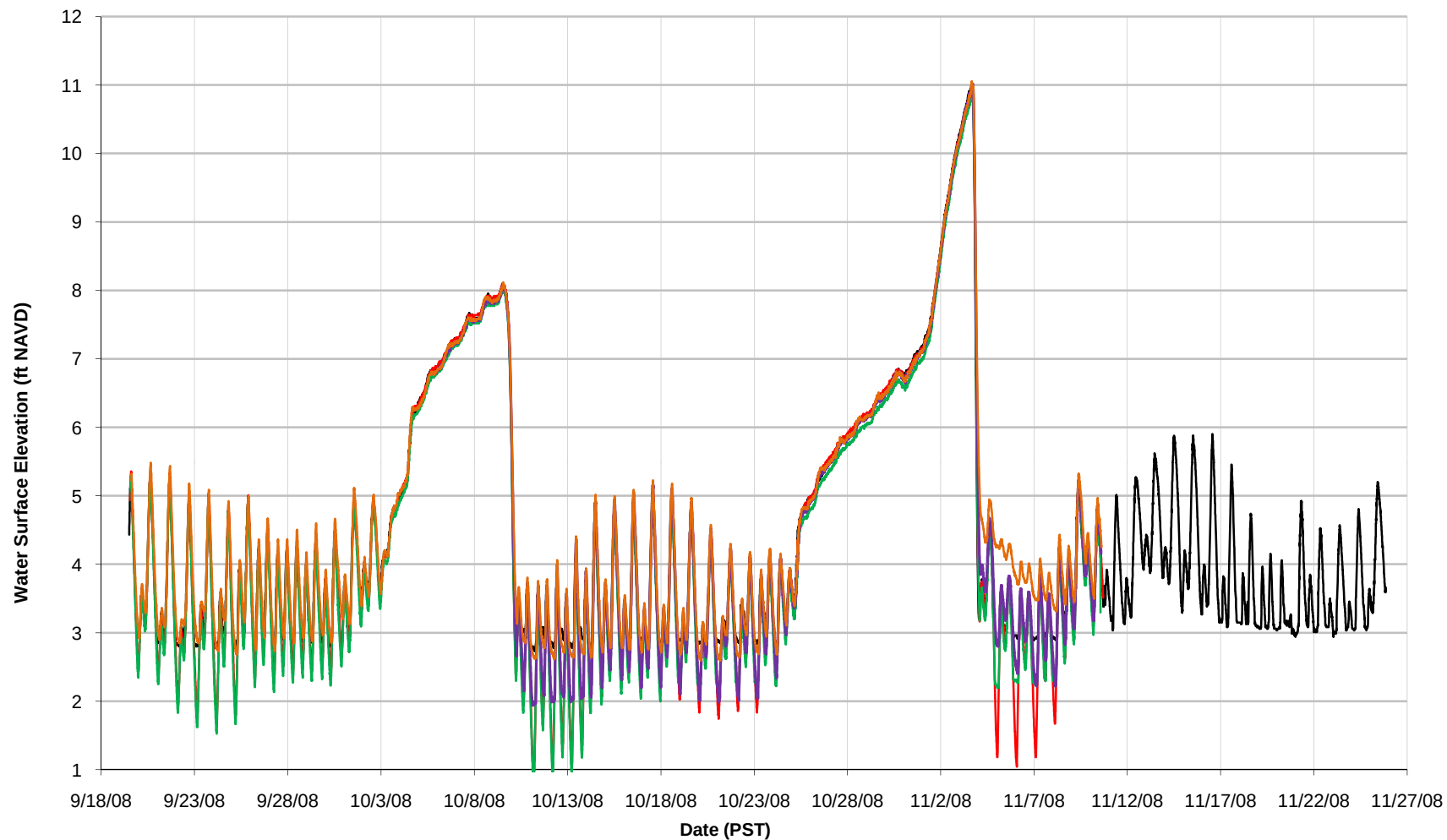
Environmental
Data Solutions

**Lower Russian River Bathymetric Study
LiDAR Derived Ground Surfaces vs.
RTK-GPS Derived Ground Surfaces
RTK-GPS Ground Shot Locations**

October 2009

Project No. 1742

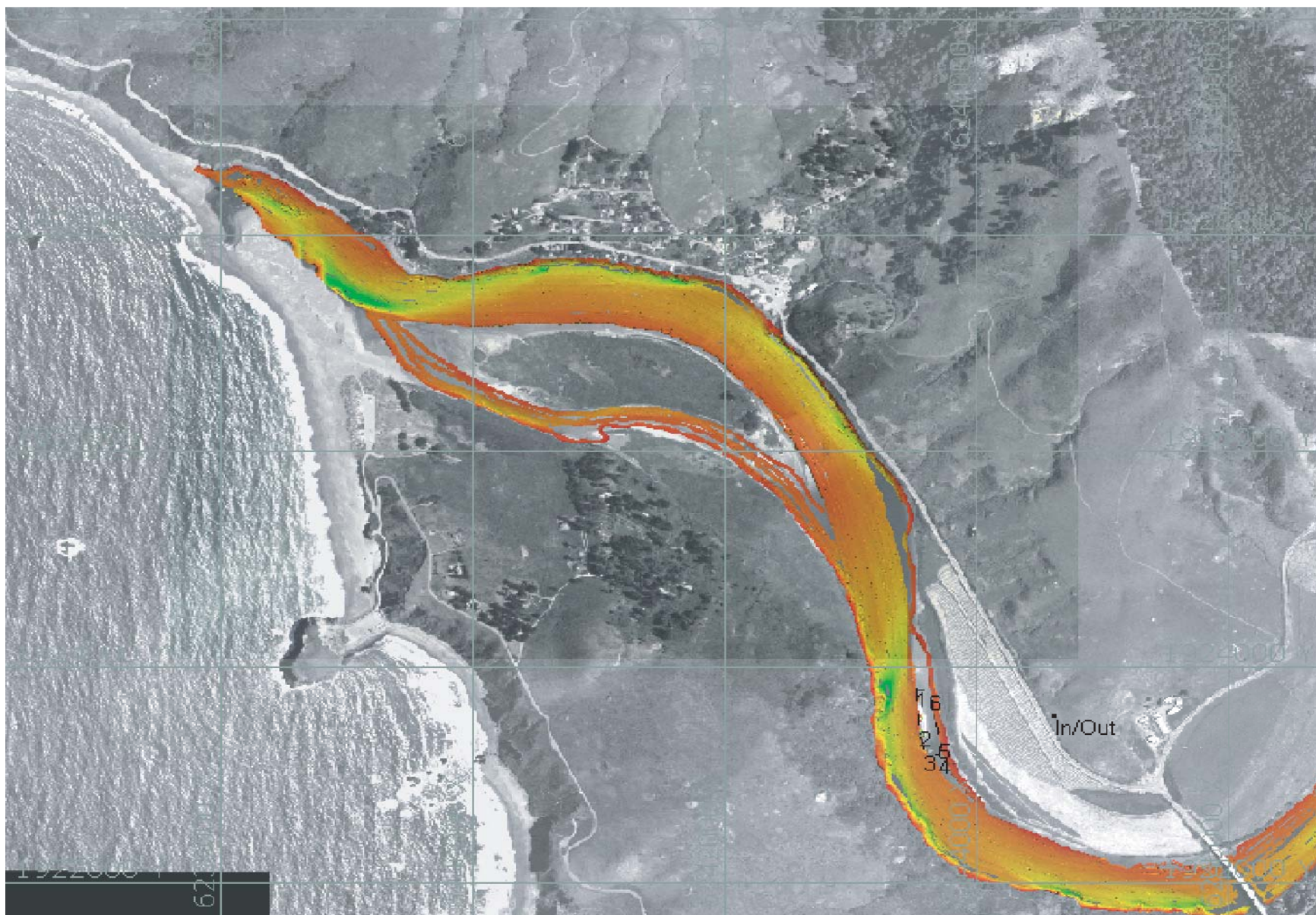
Figure 12a

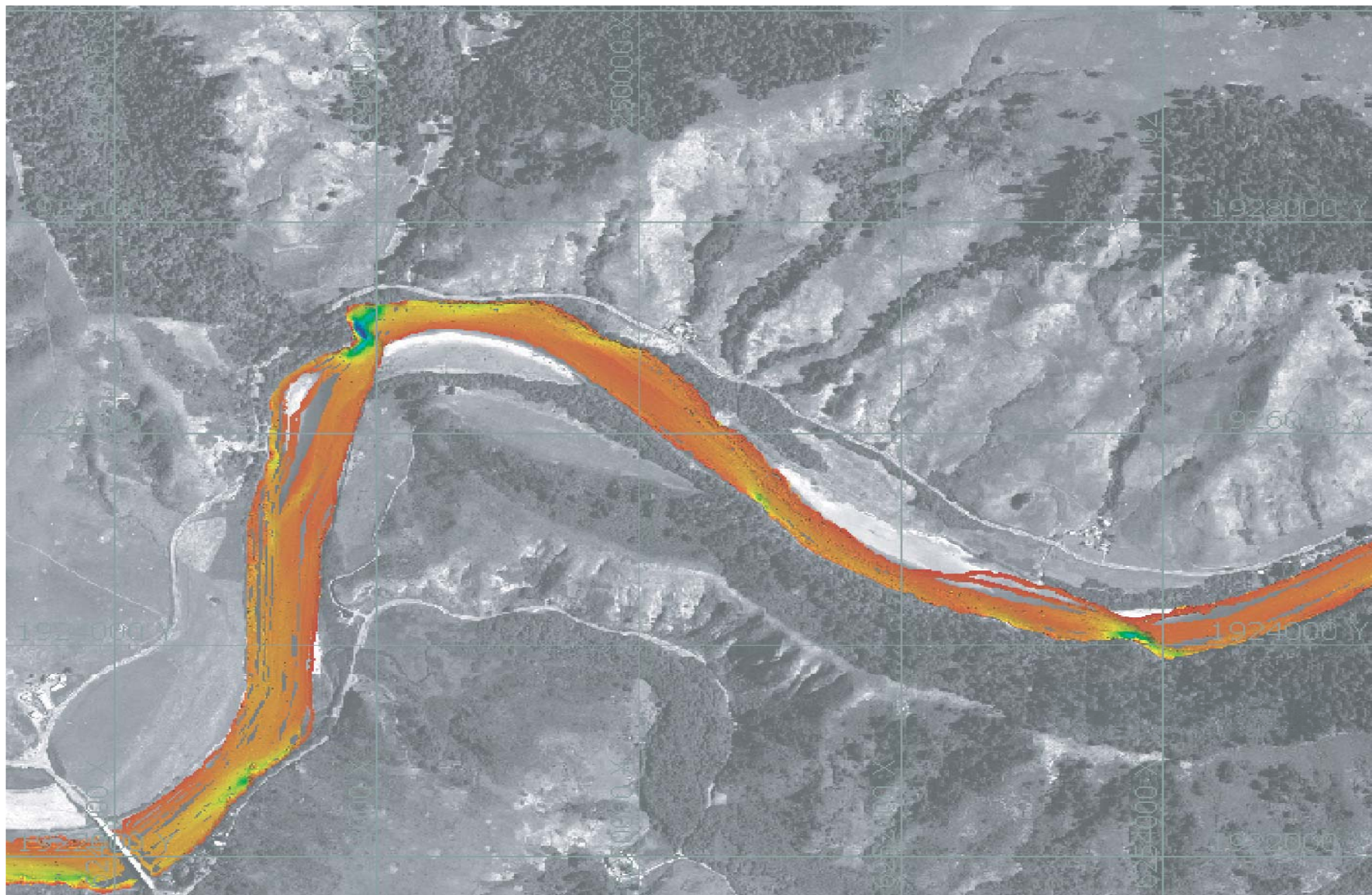


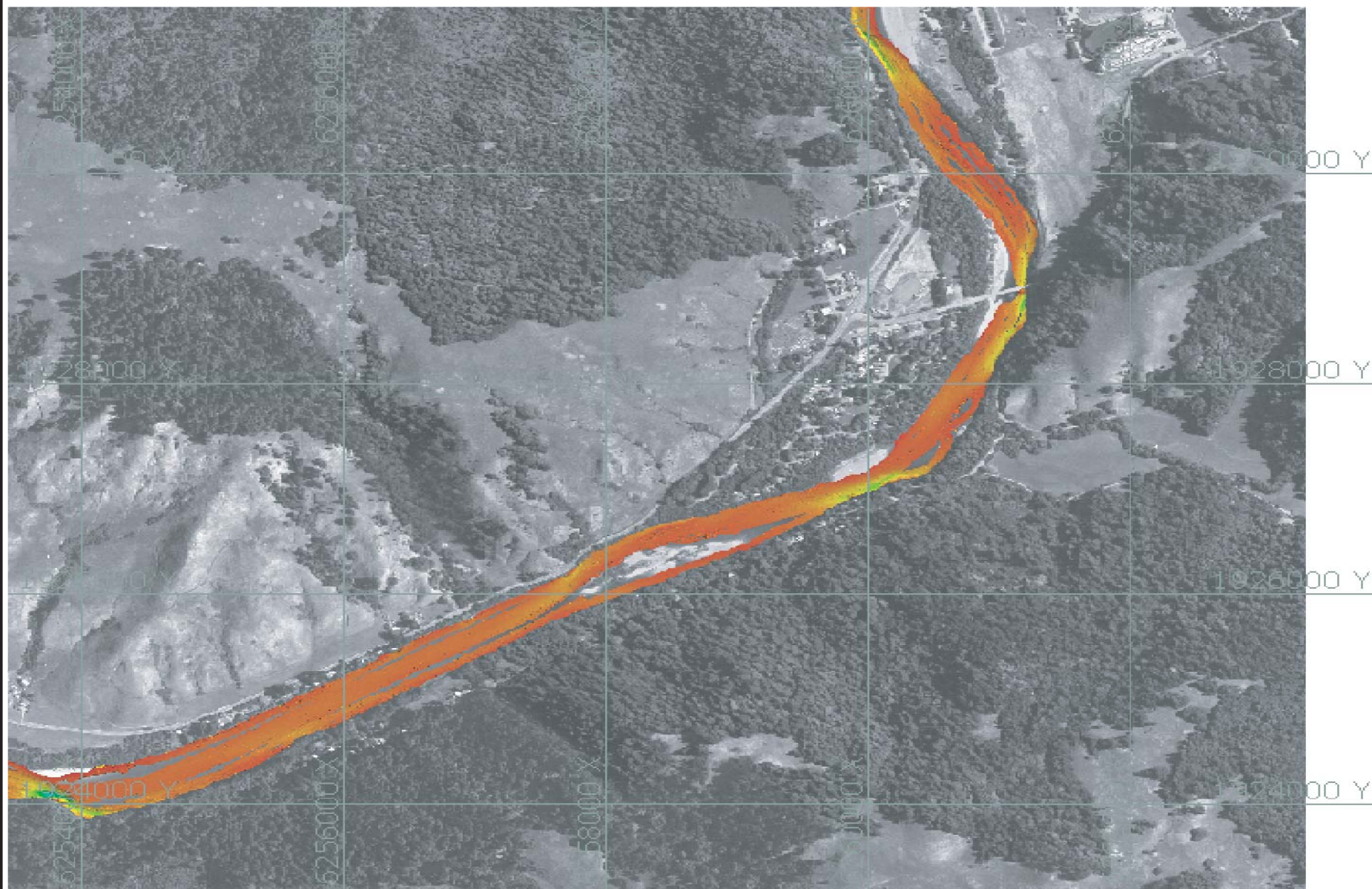
— Downstream
 — Route 1 Bridge
 — Third US Gage
 — Fourth US Gage
 — Cassini

Lower Russian River Bathymetric Analysis - Tidal Monitoring

*All Locations
 September 19, 2008 - November 10, 2008
 Figure 13*







Appendix B

OPUS Results and GPS Field Logs



OPUS solution : 96230510.dat 000523178

Monday, February 23, 2009 9:54 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 96230510.dat 000523178

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.
 For additional information: www.ngs.noaa.gov/OPUS/Using_OPUS.html#accuracy

USER: tasurv@sbcglobal.net

DATE: February 23, 2009

RINEX FILE: 9623051t.09o

TIME: 17:54:14 UTC

SOFTWARE: page5 0810.20 master11.pl 081023 START: 2009/02/20 19:28:00
 EPHEMERIS: igr15195.eph [rapid] STOP: 2009/02/20 21:30:30
 NAV FILE: brdc0510.09n OBS USED: 5328 / 5391 : 99%
 ANT NAME: TRM22020.00+GP NONE # FIXED AMB: 35 / 41 : 85%
 ARP HEIGHT: 1.540 OVERALL RMS: 0.011(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2009.1393)

X: -2734453.082(m)	0.021(m)	-2734453.904(m)	0.021(m)
Y: -4186757.856(m)	0.079(m)	-4186756.415(m)	0.079(m)
Z: 3945873.729(m)	0.040(m)	3945873.874(m)	0.040(m)

LAT: 38 27 48.46477	0.038(m)	38 27 48.48372	0.038(m)
E LON: 236 51 2.39280	0.034(m)	236 51 2.33191	0.034(m)
W LON: 123 8 57.60720	0.034(m)	123 8 57.66809	0.034(m)
EL HGT: 31.028(m)	0.069(m)	30.526(m)	0.069(m)
ORTHO HGT: 62.892(m)	0.126(m)	[NAVD88 (Computed using GEOID03)]	

206.34

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters]	4257249.735	589083.828
Easting (X) [meters]	486971.698	1899695.051
Convergence [degrees]	-0.09288901	-0.72461953
Point Scale	0.99960209	0.99997304
Combined Factor	0.99959722	0.99996817

US NATIONAL GRID DESIGNATOR: 10SDH8697257250(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMA	IRCN2004	CORS ARP	N381535.528 W1223626.762 52477.3
DK6396	P206 CRAZY	CREEKCN2006	CORS ARP	N384640.123 W1223432.798 60933.0
DG8210	P261 HUNTER	HILLCN2004	CORS ARP	N380910.640 W1221303.086 88488.1

NEAREST NGS PUBLISHED CONTROL POINT

JT9623	HPGN D CA 04 MC	N382748.469 W1230857.610	0.0
--------	-----------------	--------------------------	-----

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.

8002 The Opus solution for your submitted RINEX file appears to be
8002 quite close to an NGS published control point. This suggests that
8002 you may have set your GPS receiver up over an NGS control point.
8002 Furthermore, our files indicate that this control point has not
8002 been recovered in the last five years.
8002 If you did indeed recover an NGS control point, we would
8002 appreciate receiving this information through our web based
8002 Mark Recovery Form at
8002 http://www.ngs.noaa.gov/products_services.shtml#MarkRecoveryForm.
8002



OPUS-RS solution : 96230510.dat 000932797

Monday, February 23, 2009 10:33 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 96230510.dat 000932797

9623
OPUS-RS

6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.

Your station is 48.3 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net

DATE: February 23, 2009

RINEX FILE: 9623051t.09o

TIME: 18:33:21 UTC

SOFTWARE: rsgps 1.32 RS12.prl 1.53b START: 2009/02/20 19:28:50
EPHEMERIS: igr15195.eph [rapid] STOP: 2009/02/20 21:31:10
NAV FILE: brdc0510.09n OBS USED: 12546 / 15426 : 81%
ANT NAME: TRM22020.00+GP QUALITY IND. 51.66/142.41
ARP HEIGHT: 1.540 NORMALIZED RMS: 0.232

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2009.13933)

X: -2734453.080(m)	0.007(m)	-2734453.902(m)	0.007(m)
Y: -4186757.832(m)	0.004(m)	-4186756.391(m)	0.004(m)
Z: 3945873.714(m)	0.009(m)	3945873.859(m)	0.009(m)

LAT: 38 27 48.46482	0.004(m)	38 27 48.48377	0.004(m)
E LON: 236 51 2.39233	0.007(m)	236 51 2.33144	0.007(m)
W LON: 123 8 57.60767	0.007(m)	123 8 57.66856	0.007(m)
EL HGT: 31.002(m)	0.009(m)	30.500(m)	0.009(m)
ORTHO HGT: 62.866(m)	0.105(m)	[NAVD88 (Computed using GEOID03)]	

206.25

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters]	4257249.737	589083.830
Easting (X) [meters]	486971.686	1899695.040
Convergence [degrees]	-0.09288909	-0.72461962

Point Scale 0.99960209 0.99997304
Combined Factor 0.99959723 0.99996817

US NATIONAL GRID DESIGNATOR: 10SDH8697257250(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRC	N2004	CORS ARP	N381535.528 W1223626.762 52477.3
DK6396	P206 CRAZYCREEK	CN2006	CORS ARP	N384640.123 W1223432.798 60933.0
DG8210	P261 HUNTERHILL	CN2004	CORS ARP	N380910.640 W1221303.086 88488.0
DI1085	P181 MILLERKNOX	CN2005	CORS ARP	N375452.343 W1222236.261 91069.1
DH3879	P224 SIBLEYVOL	CCN2005	CORS ARP	N375150.009 W1221308.558 105244.5
DG8213	P271 WOODLAND1	_CA2004	CORS ARP	N383926.446 W1214252.323 126892.3
DK6402	P336 HUBBARDRD	GCN2007	CORS ARP	N393141.072 W1222549.684 133592.0
DH9021	P222 COYOTHILL	SCN2004	CORS ARP	N373221.244 W1220459.691 138889.7
DH3882	P230 MORGANTR	TYCN2005	CORS ARP	N374908.256 W1214710.981 139264.9

NEAREST NGS PUBLISHED CONTROL POINT

JT9623 HPGN D CA 04 MC N382748.469 W1230857.610 0.0

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.

8002 The Opus solution for your submitted RINEX file appears to be
8002 quite close to an NGS published control point. This suggests that
8002 you may have set your GPS receiver up over an NGS control point.
8002 Furthermore, our files indicate that this control point has not
8002 been recovered in the last five years.
8002 If you did indeed recover an NGS control point, we would
8002 appreciate receiving this information through our web based
8002 Mark Recovery Form at
8002 http://www.ngs.noaa.gov/products_services.shtml#MarkRecoveryForm.
8002

GPS LOG



Environmental Data Solutions

Project:

Russian River - MB Survey

Location:

Denner, CA

Job#

1024

[illegible]

GPS LOG



Environmental Data Solutions

Project:

R. River elev. check Bm1 to J+9623

Location:

Jenner, CA

Job#

1024

[illegible]

GPS LOG



Environmental Data Solutions

Project: Russian R.

Location: BM-2 / Jenner

Job # 1024

[illegible]

GPS LOG

Environmental
Data Solutions

Project:

RR

Location:

DUNCAN'S Mill - VERIFY DUNC FOR PPP

Job#

1024

Station Name	Start Stop	Ant.(m) Feet	Filename	Ser.#	Date	Description
DUNC	15:53	(1.7958)	20000150.117	9975	3/16/11	Re-occupy DUNC For PPP
BM-3	17:33	5.8917				
PDOP/#SV/Time						
PDOP/#SV/Time						
PDOP/#SV/Time						
PDOP/#SV/Time						
PDOP/#SV/Time						
PDOP/#SV/Time						
		()	- -			
PDOP/#SV/Time						
		()	- -			
PDOP/#SV/Time						
		()	- -			
PDOP/#SV/Time						
		()	- -			
PDOP/#SV/Time						
		()	- -			

James Kulpa

From: opus <opus@NGS.NOAA.GOV>
Sent: Tuesday, June 07, 2011 1:21 PM
To: jkulpa@felddata.biz
Subject: OPUS solution : 20000750.11o 000246976

DUNCAN
SECOND
OCCUPATION

FILE: 20000750.11o 000246976

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.

For additional information: <http://www.ngs.noaa.gov/OPUS/about.html#accuracy>

USER: jkulpa@felddata.biz DATE: June 07, 2011
RINEX FILE: 2000075x.11o TIME: 20:21:01 UTC

SOFTWARE: page5 1009.28 master2.pl 0607113 START: 2011/03/16 23:53:00
EPHEMERIS: igs16273.eph [precise] STOP: 2011/03/17 02:33:00
NAV FILE: brdc0750.11n OBS USED: 7317 / 7788 : 94%
ANT NAME: LEIAX1202 NONE # FIXED AMB: 51 / 53 : 96%
ARP HEIGHT: 1.79588 OVERALL RMS: 0.013(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2011.2056)

X:	-2727706.121(m)	0.014(m)	-2727706.985(m)	0.014(m)
Y:	-4191936.506(m)	0.012(m)	-4191935.027(m)	0.012(m)
Z:	3944967.367(m)	0.008(m)	3944967.519(m)	0.008(m)

LAT:	38 27 12.26388	0.011(m)	38 27 12.28324	0.011(m)
E LON:	236 56 52.15748	0.012(m)	236 56 52.09434	0.012(m)
W LON:	123 3 7.84252	0.012(m)	123 3 7.90566	0.012(m)
EL HGT:	-20.864(m)	0.011(m)	-21.371(m)	0.011(m)
ORTHO HGT:	10.753(m)	0.107(m)	[NAVD88 (Computed using GEOID03)]	

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters]	4256124.664	587864.983
Easting (X) [meters]	495447.218	1908160.818
Convergence [degrees]	-0.03244865	-0.66336521
Point Scale	0.99960026	0.99997494
Combined Factor	0.99960353	0.99997821

US NATIONAL GRID DESIGNATOR: 10SDH9544756124(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
-----	-------------	----------	-----------	-------------

DH7229 P198 PETALUMAIRCN2004 CORS ARP N381535.528 W1223626.762 44414.3
DK6396 P206 CRAZYCREEKCN2006 CORS ARP N384640.123 W1223432.798 54941.3
DM6171 P188 BURNTRIDGECN2006 CORS ARP N384004.263 W1231346.368 28385.5

NEAREST NGS PUBLISHED CONTROL POINT

JT2797 REDWOOD 2 N382410.710 W1230118.706 6205.3

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



Duncan 2007

opus Sol.

OPUS solution : Dcan2760.dat 000443323

Friday, October 3, 2008 8:25 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: Dcan2760.dat 000443323

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER

9999

9999 Your data file spans less than 2 hours. OPUS guidelines require a
9999 minimum data span of 2 hours and strongly recommend 4 hours to achieve
9999 best accuracies. However, due to popular demand by users, OPUS will
9999 accept data spans less than 2 hours, but does not provide any guarantee
9999 to the accuracies or results. The results may be seriously degraded.
9999 OPUS-RS may deliver better accuracy. OPUS-RS requires dense CORS
9999 coverage, so it will work mainly in the United States. See the OPUS-RS
9999 web site at <http://www.ngs.noaa.gov/OPUS/OPUS-RS.html>.

9999

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER

9999

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.

2005

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.

For additional information: www.ngs.noaa.gov/OPUS/Using_OPUS.html#accuracy

USER: tasurv@sbcglobal.net

DATE: October 03, 2008

RINEX FILE: dcan276u.08o

TIME: 15:25:33 UTC

SOFTWARE: page5 0612.06 master12.pl 080929 START: 2008/10/02 20:22:00

EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 21:43:00

NAV FILE: brdc2760.08n OBS USED: 2314 / 2729 : 85%

ANT NAME: TRM22020.00+GP NONE # FIXED AMB: 24 / 27 : 89%

ARP HEIGHT: 2.147 OVERALL RMS: 0.012(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.7538)

X: -2727706.120(m) 0.020(m) -2727706.931(m) 0.020(m)

Y: -4191936.500(m) 0.007(m) -4191935.085(m) 0.007(m)

Z: 3944967.345(m) 0.011(m) 3944967.470(m) 0.011(m)

LAT: 38 27 12.26343 0.004(m) 38 27 12.28160 0.004(m)

E LON: 236 56 52.15737 0.014(m) 236 56 52.09751 0.014(m)
W LON: 123 3 7.84263 0.014(m) 123 3 7.90249 0.014(m)
EL HGT: -20.882(m) 0.019(m) -21.387(m) 0.019(m)
ORTHO HGT: 10.735(m) 0.106(m) [NAVD88 (Computed using GEOID03)]

35.22

UTM COORDINATES STATE PLANE COORDINATES

	UTM (Zone 10)	SPC (0402 CA 2)
Northing (Y) [meters]	4256124.650	587864.970
Easting (X) [meters]	495447.216	1908160.815
Convergence [degrees]	-0.03244867	-0.66336523
Point Scale	0.99960026	0.99997494
Combined Factor	0.99960353	0.99997822

US NATIONAL GRID DESIGNATOR: 10SDH9544756125(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMA	IRCN2004	CORS ARP	N381535.528 W1223626.762 44414.3
DK6396	P206 CRAZY CREEK	IRCN2006	CORS ARP	N384640.123 W1223432.798 54941.3
DG8210	P261 HUNTER HILL	IRCN2004	CORS ARP	N380910.640 W1221303.086 80263.7

NEAREST NGS PUBLISHED CONTROL POINT

JT2892	KOERBER	N382553.	W1230042.	4299.3
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.

DUNCAN 2007



OPUS-RS Sol.

OPUS-RS solution : Dcan2760.dat 000889095

Friday, October 3, 2008 8:30 AM

From: "opus" <opus@ngs.noaa.gov>
To: tasurv@sbcglobal.net

—Inline Attachment Follows—

FILE: Dcan2760.dat 000889095

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.

2005

6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.

Your station is 39.8 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net
RINEX FILE: dcan276u.08o

DATE: October 03, 2008
TIME: 15:30:37 UTC

SOFTWARE: rsgps 1.28 RS11.prl 1.30 START: 2008/10/02 20:22:30
EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 21:43:00
NAV FILE: brdc2760.08n OBS USED: 5049 / 7542 : 67%
ANT NAME: TRM22020.00+GP QUALITY IND. 35.37/ 60.94
ARP HEIGHT: 2.147 NORMALIZED RMS: 0.286

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.75376)

X:	-2727706.112(m)	0.010(m)	-2727706.923(m)	0.010(m)
Y:	-4191936.511(m)	0.015(m)	-4191935.096(m)	0.015(m)
Z:	3944967.347(m)	0.016(m)	3944967.472(m)	0.016(m)

LAT:	38 27 12.26338	0.006(m)	38 27 12.28156	0.006(m)
E LON:	236 56 52.15790	0.007(m)	236 56 52.09804	0.007(m)
W LON:	123 3 7.84210	0.007(m)	123 3 7.90196	0.007(m)
EL HGT:	-20.877(m)	0.022(m)	-21.381(m)	0.022(m)
ORTHO HGT:	10.740(m)	0.107(m)	[NAVD88 (Computed using GEOID03)]	

35:23 14

UTM COORDINATES STATE PLANE COORDINATES

	UTM (Zone 10)	SPC (0402 CA 2)
Northing (Y) [meters]	4256124.649	587864.968
Easting (X) [meters]	495447.228	1908160.828
Convergence [degrees]	-0.03244858	-0.66336514
Point Scale	0.99960026	0.99997494
Combined Factor	0.99960353	0.99997822

1928686.98 ft.
6260357.65 ft.

US NATIONAL GRID DESIGNATOR: 10SDH9544756125(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRCN2004	CORS ARP	N381535.528 W1223626.762	44414.2
DK6396	P206 CRAZYCREEKCN2006	CORS ARP	N384640.123 W1223432.798	54941.3
DG8210	P261 HUNTERHILLCN2004	CORS ARP	N380910.640 W1221303.086	80263.6
DI1085	P181 MILLERKNOXCN2005	CORS ARP	N375452.343 W1222236.261	84139.9
AM7021	OHLN OHLONE PARK	CORS ARP	N380022.493 W1221622.704	84368.9
DH3879	P224 SIBLEYVOLCCN2005	CORS ARP	N375150.009 W1221308.558	98050.7
AF9690	CHAB CHAB_BARD_CN1991	CORS ARP	N374326.799 W1220709.461	115110.1
DG8213	P271 WOODLAND1_CA2004	CORS ARP	N383926.446 W1214252.323	118771.2
DH9024	P225 CULLCANYONCN2005	CORS ARP	N374249.899 W1220329.923	119753.1

NEAREST NGS PUBLISHED CONTROL POINT

JT2892	KOERBER	N382553. W1230042.	4299.3
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



BM-Z

OPUS-RS sol.

OPUS-RS solution : 0bm22760.dat 000889094

Friday, October 3, 2008 8:39 AM

From: "opus" <opus@ngs.noaa.gov>
To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 0bm22760.dat 000889094

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.

2005
6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.

Your station is 44.2 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net
RINEX FILE: 0bm2276s.08o

DATE: October 03, 2008
TIME: 15:39:15 UTC

SOFTWARE: rsgps 1.28 RS41.prl 1.30 START: 2008/10/02 18:32:40
EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 19:40:10
NAV FILE: brdc2760.08n OBS USED: 6363 / 7920 : 80%
ANT NAME: TRM22020.00+GP QUALITY IND. 13.82/101.76
ARP HEIGHT: 2.066 NORMALIZED RMS: 0.275

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.75354)

X:	-2732115.581(m)	0.009(m)	-2732116.392(m)	0.009(m)
Y:	-4190402.481(m)	0.014(m)	-4190401.064(m)	0.014(m)
Z:	3943547.016(m)	0.011(m)	3943547.144(m)	0.011(m)

LAT:	38 26 13.57816	0.004(m)	38 26 13.59641	0.004(m)
E LON:	236 53 45.26548	0.010(m)	236 53 45.20555	0.010(m)
W LON:	123 6 14.73452	0.010(m)	123 6 14.79445	0.010(m)
EL HGT:	-25.861(m)	0.016(m)	-26.364(m)	0.016(m)
ORTHO HGT:	5.966(m)	0.106(m)	[NAVD88 (Computed using GEOID03)]	

19.57 ft.

UTM COORDINATES STATE PLANE COORDINATES

	UTM (Zone 10)	SPC (0402 CA 2)	
Northing (Y) [meters]	4254319.672	586109.361	— 1922 927.13 ft
Easting (X) [meters]	490915.430	1903607.710	— 6245 419.63 ft
Convergence [degrees]	-0.06470995	-0.69609563	
Point Scale	0.99960102	0.99997809	
Combined Factor	0.99960507	0.99998215	

US NATIONAL GRID DESIGNATOR: 10SDH9091554320(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRC	N2004	CORS ARP	N381535.528 W1223626.762 47664.8
DK6396	P206 CRAZYCREEK	CN2006	CORS ARP	N384640.123 W1223432.798 59567.4
DG8210	P261 HUNTERHILL	CN2004	CORS ARP	N380910.640 W1221303.086 83724.6
DI1085	P181 MILLERKNOX	CN2005	CORS ARP	N375452.343 W1222236.261 86175.5
AM7021	OHLN OHLONE PARK		CORS ARP	N380022.493 W1221622.704 87085.0
DH3879	P224 SIBLEYVOLCC	CN2005	CORS ARP	N375150.009 W1221308.558 100333.9
AF9690	CHAB CHAB_BARD_CN	1991	CORS ARP	N374326.799 W1220709.461 117168.3
DH9024	P225 CULLCANYON	CN2005	CORS ARP	N374249.899 W1220329.923 121917.2
DG8213	P271 WOODLAND1_CA	2004	CORS ARP	N383926.446 W1214252.323 123574.7

NEAREST NGS PUBLISHED CONTROL POINT

JT1408	G 631	N382545.	W1230629.	948.9
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



BM-2

opus Sol.

OPUS solution : 0bm22760.dat 000443322

Friday, October 3, 2008 8:23 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 0bm22760.dat 000443322

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER
9999

9999 Your data file spans less than 2 hours. OPUS guidelines require a
9999 minimum data span of 2 hours and strongly recommend 4 hours to achieve
9999 best accuracies. However, due to popular demand by users, OPUS will
9999 accept data spans less than 2 hours, but does not provide any guarantee
9999 to the accuracies or results. The results may be seriously degraded.
9999 OPUS-RS may deliver better accuracy. OPUS-RS requires dense CORS
9999 coverage, so it will work mainly in the United States. See the OPUS-RS
9999 web site at <http://www.ngs.noaa.gov/OPUS/OPUS-RS.html> .
9999

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER
9999

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.
2005

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.
For additional information: www.ngs.noaa.gov/OPUS/Using_OPUS.html#accuracy

USER: tasurv@sbcglobal.net
RINEX FILE: 0bm2276s.08o

DATE: October 03, 2008
TIME: 15:23:40 UTC

SOFTWARE: page5 0612.06 master2.pl 0809291 START: 2008/10/02 18:32:00
EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 19:40:00
NAV FILE: brdc2760.08n OBS USED: 2355 / 2646 : 89%
ANT NAME: TRM22020.00+GP NONE # FIXED AMB: 22 / 23 : 96%
ARP HEIGHT: 2.066 OVERALL RMS: 0.011(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.7535)

X:	-2732115.595(m)	0.023(m)	-2732116.406(m)	0.023(m)
Y:	-4190402.491(m)	0.083(m)	-4190401.074(m)	0.083(m)
Z:	3943547.021(m)	0.058(m)	3943547.149(m)	0.058(m)

LAT: 38 26 13.57796 0.005(m) 38 26 13.59622 0.005(m)

E LON: 236 53 45.26522 0.027(m) 236 53 45.20529 0.027(m)
W LON: 123 6 14.73478 0.027(m) 123 6 14.79471 0.027(m)
EL HGT: -25.845(m) 0.100(m) -26.348(m) 0.100(m)
ORTHO HGT: 5.982(m) 0.145(m) [NAVD88 (Computed using GEOID03)]

19.63

UTM COORDINATES STATE PLANE COORDINATES

	UTM (Zone 10)	SPC (0402 CA 2)
Northing (Y) [meters]	4254319.666	586109.355
Easting (X) [meters]	490915.424	1903607.704
Convergence [degrees]	-0.06470999	-0.69609567
Point Scale	0.99960102	0.99997809
Combined Factor	0.99960507	0.99998214

US NATIONAL GRID DESIGNATOR: 10SDH9091554320(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMA	IRCN2004	CORS ARP	N381535.528 W1223626.762 47664.8
DK6396	P206 CRAZY CREEK	IRCN2006	CORS ARP	N384640.123 W1223432.798 59567.4
DG8210	P261 HUNTER HILL	IRCN2004	CORS ARP	N380910.640 W1221303.086 83724.6

NEAREST NGS PUBLISHED CONTROL POINT

JT1408	G 631	N382545.	W1230629.	948.8
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.

James Kulpa

From: opus <opus@NGS.NOAA.GOV>
Sent: Monday, June 06, 2011 7:27 PM
To: jkulpa@fielddata.biz
Subject: OPUS-RS solution : 20003650.08o 000262902

BM2
SECOND
OCCUPATION

FILE: 20003650.08o 000262902

6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.
Your station is 28.9 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: jkulpa@fielddata.biz DATE: June 07, 2011
RINEX FILE: 2000365t.08o TIME: 02:26:36 UTC

SOFTWARE: rsgps 1.35.1 RS52.prl 1.70 START: 2008/12/30 19:36:55
EPHEMERIS: igs15122.eph [precise] STOP: 2008/12/30 21:32:10
NAV FILE: brdc3650.08n OBS USED: 12564 / 13275 : 95%
ANT NAME: LEIAX1202 NONE QUALITY IND. 90.60/145.41
ARP HEIGHT: 1.78999 NORMALIZED RMS: 0.243

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.99688)

X: -2732115.583(m) 0.008(m) -2732116.399(m) 0.008(m)
Y: -4190402.493(m) 0.012(m) -4190401.065(m) 0.012(m)
Z: 3943547.024(m) 0.007(m) 3943547.158(m) 0.007(m)

LAT: 38 26 13.57814 0.005(m) 38 26 13.59667 0.005(m)
E LON: 236 53 45.26568 0.009(m) 236 53 45.20533 0.009(m)
W LON: 123 6 14.73432 0.009(m) 123 6 14.79467 0.009(m)
EL HGT: -25.847(m) 0.013(m) -26.352(m) 0.013(m)
ORTHO HGT: 5.980(m) 0.013(m) [NAVD88 (Computed using GEOID03)]

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters] 4254319.672 586109.360
Easting (X) [meters] 490915.435 1903607.715
Convergence [degrees] -0.06470991 -0.69609559
Point Scale 0.99960102 0.99997809
Combined Factor 0.99960507 0.99998214

US NATIONAL GRID DESIGNATOR: 10SDH9091554319(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229 P198	PETALUMAIRCN2004 CORS ARP	N381535.528	W1223626.762	47664.8
DK6396 P206	CRAZYCREEKCN2006 CORS ARP	N384640.123	W1223432.798	59567.4
DG8210 P261	HUNTERHILLCN2004 CORS ARP	N380910.640	W1221303.086	83724.6
AM7021 OHLN	OHLONE PARK CORS ARP	N380022.493	W1221622.704	87085.0
DH3879 P224	SIBLEYVOLCCN2005 CORS ARP	N375150.009	W1221308.558	100333.9
AF9690 CHAB	CHAB_BARD_CN1991 CORS ARP	N374326.799	W1220709.461	117168.3
DH9024 P225	CULLCANYONCN2005 CORS ARP	N374249.899	W1220329.923	121917.2
DH9021 P222	COYOTHILLSCN2004 CORS ARP	N373221.244	W1220459.691	134064.0
DH5855 P314	CAHTOPEAK_CN2004 CORS ARP	N394108.372	W1233454.593	144662.0

NEAREST NGS PUBLISHED CONTROL POINT

JT2794	PEAKED HILL RESET	N382552.262	W1230707.182	1431.2
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



BM-1

OPUS-RS Sol.

OPUS-RS solution : 0bm12760.dat 000889093

Friday, October 3, 2008 8:43 AM

From: "opus" <opus@ngs.noaa.gov>
To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 0bm12760.dat 000889093

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.

2005

6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.

Your station is 45.2 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net
RINEX FILE: 0bm1276s.08o

DATE: October 03, 2008
TIME: 15:43:52 UTC

SOFTWARE: rsgps 1.28 RS12.prl 1.30 START: 2008/10/02 18:11:40
EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 19:45:50
NAV FILE: brdc2760.08n OBS USED: 11016 / 11475 : 96%
ANT NAME: TRM22020.00+GP QUALITY IND. 21.71/ 74.33
ARP HEIGHT: 2.101 NORMALIZED RMS: 0.309

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.75353)

X:	-2732472.895(m)	0.007(m)	-2732473.706(m)	0.007(m)
Y:	-4189166.637(m)	0.013(m)	-4189165.220(m)	0.013(m)
Z:	3944601.361(m)	0.008(m)	3944601.489(m)	0.008(m)

LAT:	38 26 57.29686	0.004(m)	38 26 57.31511	0.004(m)
E LON:	236 53 5.08579	0.009(m)	236 53 5.02585	0.009(m)
W LON:	123 6 54.91421	0.009(m)	123 6 54.97415	0.009(m)
EL HGT:	-28.240(m)	0.013(m)	-28.743(m)	0.013(m)
ORTHO HGT:	3.570(m)	0.106(m)	[NAVD88 (Computed using GEOID03)]	

11.71 ft

UTM COORDINATES STATE PLANE COORDINATES

	UTM (Zone 10)	SPC (0402 CA 2)	
Northing (Y) [meters]	4255668.345	587469.179	→ 1927 328.44 46
Easting (X) [meters]	489943.052	1902649.893	→ 6242 277.19 4
Convergence [degrees]	-0.07166741	-0.70313230	
Point Scale	0.99960125	0.99997574	
Combined Factor	0.99960567	0.99998017	

US NATIONAL GRID DESIGNATOR: 10SDH8994355668(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRC	N2004	CORS ARP	N381535.528 W1223626.762 49113.5
DK6396	P206 CRAZYCREEK	CN2006	CORS ARP	N384640.123 W1223432.798 59482.5
DG8210	P261 HUNTERHILL	CN2004	CORS ARP	N380910.640 W1221303.086 85135.4
DI1085	P181 MILLERKNOX	CN2005	CORS ARP	N375452.343 W1222236.261 87802.7
AM7021	OHLN OHLONE PARK		CORS ARP	N380022.493 W1221622.704 88638.9
DH3879	P224 SIBLEYVOLCC	CN2005	CORS ARP	N375150.009 W1221308.558 101941.0
AF9690	CHAB CHAB_BARD_CN	1991	CORS ARP	N374326.799 W1220709.461 118796.0
DH9024	P225 CULLCANYON	CN2005	CORS ARP	N374249.899 W1220329.923 123536.5
DG8213	P271 WOODLAND1_CA	2004	CORS ARP	N383926.446 W1214252.323 124260.8

NEAREST NGS PUBLISHED CONTROL POINT

JT1405	64 M	N382706.	W1230729.	868.2
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



BM-1

opus sol.

OPUS solution : 0bm12760.dat 000443319

Friday, October 3, 2008 8:22 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

10/2/08

-----Inline Attachment Follows-----

FILE: 0bm12760.dat 000443319

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER
9999

9999 Your data file spans less than 2 hours. OPUS guidelines require a
9999 minimum data span of 2 hours and strongly recommend 4 hours to achieve
9999 best accuracies. However, due to popular demand by users, OPUS will
9999 accept data spans less than 2 hours, but does not provide any guarantee
9999 to the accuracies or results. The results may be seriously degraded.
9999 OPUS-RS may deliver better accuracy. OPUS-RS requires dense CORS
9999 coverage, so it will work mainly in the United States. See the OPUS-RS
9999 web site at <http://www.ngs.noaa.gov/OPUS/OPUS-RS.html> .
9999

9999 OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER OPUS DISCLAIMER
9999

2005 NOTE: The IGS precise and IGS rapid orbits were not available
2005 at processing time. The IGS ultra-rapid orbit was/will be used to
2005 process the data.
2005

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.
For additional information: www.ngs.noaa.gov/OPUS/Using_OPUS.html#accuracy

USER: tasurv@sbcglobal.net
RINEX FILE: 0bm1276s.08o

DATE: October 03, 2008
TIME: 15:22:24 UTC

SOFTWARE: page5 0612.06 master.pl 0809291 START: 2008/10/02 18:11:00
EPHEMERIS: igu14994.eph [ultra-rapid] STOP: 2008/10/02 19:44:30
NAV FILE: brdc2760.08n OBS USED: 3756 / 3895 : 96%
ANT NAME: TRM22020.00+GP NONE # FIXED AMB: 27 / 30 : 90%
ARP HEIGHT: 2.101 OVERALL RMS: 0.010(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.7535)

X:	-2732472.902(m)	0.019(m)	-2732473.713(m)	0.019(m)
Y:	-4189166.631(m)	0.020(m)	-4189165.214(m)	0.020(m)
Z:	3944601.358(m)	0.017(m)	3944601.486(m)	0.017(m)

LAT:	38 26 57.29680	0.002(m)	38 26 57.31506	0.002(m)
------	----------------	----------	----------------	----------

E LON: 236 53 5.08541 0.012(m) 236 53 5.02547 0.012(m)
 W LON: 123 6 54.91459 0.012(m) 123 6 54.97453 0.012(m)
 EL HGT: -28.242(m) 0.029(m) -28.745(m) 0.029(m)
 ORTHO HGT: 11.71 3.568(m) 0.109(m) [NAVD88 (Computed using GEOID03)]

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters] 4255668.343 587469.177
 Easting (X) [meters] 489943.042 1902649.884
 Convergence [degrees] -0.07166747 -0.70313237
 Point Scale 0.99960125 0.99997574
 Combined Factor 0.99960568 0.99998017

US NATIONAL GRID DESIGNATOR: 10SDH8994355668(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMA	IRCN2004	CORS ARP	N381535.528 W1223626.762 49113.5
DK6396	P206 CRAZY CREEK	IRCN2006	CORS ARP	N384640.123 W1223432.798 59482.5
DG8210	P261 HUNTER HILL	IRCN2004	CORS ARP	N380910.640 W1221303.086 85135.4

NEAREST NGS PUBLISHED CONTROL POINT

JT1405 64 M N382706. W1230729. 868.2

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



20

OPUS-RS solution : 0bm12760.dat 000930411

Tuesday, February 17, 2009 11:59 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: 0bm12760.dat 000930411

6011 Warning - OPUS-RS was able to find a set of reference stations
 6011 with data suitable for use with your dataset. However, your
 6011 position does not fall within the polygon enclosing these reference
 6011 stations. This means that the geographic interpolation algorithms
 6011 performed within OPUS-RS must instead perform extrapolation.
 6011 Extrapolation, especially if your position is far from the
 6011 reference stations, is prone to error. Use this solution with
 6011 caution.

Your station is 45.2 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net
 RINEX FILE: 0bm1276s.08o

DATE: February 17, 2009
 TIME: 19:59:07 UTC

SOFTWARE: rsgps 1.32 RS29.prl 1.53b START: 2008/10/02 18:11:40
 EPHEMERIS: igs14994.eph [precise] STOP: 2008/10/02 19:45:50
 NAV FILE: brdc2760.08n OBS USED: 11016 / 11475 : 96%
 ANT NAME: TRM22020.00+GP QUALITY IND. 22.41/ 73.96
 ARP HEIGHT: 2.101 NORMALIZED RMS: 0.308

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2008.75353)

X: -2732472.895(m)	0.007(m)	-2732473.706(m)	0.007(m)
Y: -4189166.637(m)	0.012(m)	-4189165.220(m)	0.012(m)
Z: 3944601.361(m)	0.007(m)	3944601.489(m)	0.007(m)

LAT: 38 26 57.29686	0.004(m)	38 26 57.31511	0.004(m)
E LON: 236 53 5.08579	0.009(m)	236 53 5.02585	0.009(m)
W LON: 123 6 54.91421	0.009(m)	123 6 54.97415	0.009(m)
EL HGT: -28.240(m)	0.012(m)	-28.743(m)	0.012(m)
ORTHO HGT: 11.71	3.570(m)	0.105(m)	[NAVD88 (Computed using GEOID03)]

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters]	4255668.345	587469.179
Easting (X) [meters]	489943.052	1902649.893
Convergence [degrees]	-0.07166741	-0.70313230

2/17/09 precise ephemeris
Same ortho ht as previous

Point Scale 0.99960125 0.99997574
Combined Factor 0.99960567 0.99998017

US NATIONAL GRID DESIGNATOR: 10SDH8994355668(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRC	N2004	CORS ARP	N381535.528 W1223626.762 49113.5
DK6396	P206 CRAZYCREEK	CN2006	CORS ARP	N384640.123 W1223432.798 59482.5
DG8210	P261 HUNTERHILL	CN2004	CORS ARP	N380910.640 W1221303.086 85135.4
DI1085	P181 MILLERKNOX	CN2005	CORS ARP	N375452.343 W1222236.261 87802.7
AM7021	OHLN OHLONE PARK		CORS ARP	N380022.493 W1221622.704 88638.9
DH3879	P224 SIBLEYVOLCC	CN2005	CORS ARP	N375150.009 W1221308.558 101941.0
AF9690	CHAB CHAB_BARD_	CN1991	CORS ARP	N374326.799 W1220709.461 118796.0
DH9024	P225 CULLCANYON	CN2005	CORS ARP	N374249.899 W1220329.923 123536.5
DG8213	P271 WOODLAND1_	CA2004	CORS ARP	N383926.446 W1214252.323 124260.8

NEAREST NGS PUBLISHED CONTROL POINT

JT1405 64 M N382706. W1230729. 868.2

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



OPUS-RS solution : Bm1x0510.dat 000932802

Monday, February 23, 2009 10:58 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: Bm1x0510.dat 000932802

BM 14
OPUS-RS

6011 Warning - OPUS-RS was able to find a set of reference stations
6011 with data suitable for use with your dataset. However, your
6011 position does not fall within the polygon enclosing these reference
6011 stations. This means that the geographic interpolation algorithms
6011 performed within OPUS-RS must instead perform extrapolation.
6011 Extrapolation, especially if your position is far from the
6011 reference stations, is prone to error. Use this solution with
6011 caution.

Your station is 45.2 KM outside the polygon enclosing the reference stations.

NGS OPUS-RS SOLUTION REPORT

=====

USER: tasurv@sbcglobal.net

DATE: February 23, 2009

RINEX FILE: bm1x051t.09o

TIME: 18:58:14 UTC

SOFTWARE: rsgps 1.32 RS11.prl 1.53b START: 2009/02/20 19:06:00
EPHEMERIS: igr15195.eph [rapid] STOP: 2009/02/20 21:52:40
NAV FILE: brdc0510.09n OBS USED: 20079 / 21501 : 93%
ANT NAME: TRM22020.00+GP QUALITY IND. 73.23/144.11
ARP HEIGHT: 2.085 NORMALIZED RMS: 0.240

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2009.13933)

X: -2732473.392(m) 0.007(m) -2732474.211(m) 0.007(m)
Y: -4189165.388(m) 0.005(m) -4189163.962(m) 0.005(m)
Z: 3944602.205(m) 0.008(m) 3944602.336(m) 0.008(m)

LAT: 38 26 57.33391 0.004(m) 38 26 57.35230 0.004(m)
E LON: 236 53 5.04048 0.006(m) 236 53 4.98006 0.006(m)
W LON: 123 6 54.95952 0.006(m) 123 6 55.01994 0.006(m)
EL HGT: -28.321(m) 0.009(m) -28.825(m) 0.009(m)
ORTHO HGT: 3.489(m) 0.105(m) [NAVD88 (Computed using GEOID03)]

PLDS

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters] 4255669.489 587470.335
Easting (X) [meters] 489941.955 1902648.809
Convergence [degrees] -0.07167525 -0.70314023

Point Scale 0.99960125 0.99997573
Combined Factor 0.99960569 0.99998018

US NATIONAL GRID DESIGNATOR: 10SDH8994255669(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRCN2004	CORS ARP	N381535.528 W1223626.762	49114.9
DK6396	P206 CRAZYCREEKCN2006	CORS ARP	N384640.123 W1223432.798	59482.7
DG8210	P261 HUNTERHILLCN2004	CORS ARP	N380910.640 W1221303.086	85136.8
DI1085	P181 MILLERKNOXCN2005	CORS ARP	N375452.343 W1222236.261	87804.3
DH3879	P224 SIBLEYVOLCCN2005	CORS ARP	N375150.009 W1221308.558	101942.5
DG8213	P271 WOODLAND1_CA2004	CORS ARP	N383926.446 W1214252.323	124261.7
DK6402	P336 HUBBARDRDGCN2007	CORS ARP	N393141.072 W1222549.684	133655.1
DH9021	P222 COYOTHILLSCN2004	CORS ARP	N373221.244 W1220459.691	135718.4
DH3882	P230 MORGANTRTYCN2005	CORS ARP	N374908.256 W1214710.981	135902.2

NEAREST NGS PUBLISHED CONTROL POINT

JT1405 64 M N382706. W1230729. 866.8

This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.



OPUS solution : Bm1x0510.dat 000523179

Monday, February 23, 2009 9:55 AM

From: "opus" <opus@ngs.noaa.gov>

To: tasurv@sbcglobal.net

-----Inline Attachment Follows-----

FILE: Bm1x0510.dat 000523179

NGS OPUS SOLUTION REPORT

=====

All computed coordinate accuracies are listed as peak-to-peak values.
 For additional information: www.ngs.noaa.gov/OPUS/Using_OPUS.html#accuracy

USER: tasurv@sbcglobal.net

DATE: February 23, 2009

RINEX FILE: bm1x051t.09o

TIME: 17:55:49 UTC

SOFTWARE: page5 0810.20 master29.pl 081023 START: 2009/02/20 19:06:00
 EPHEMERIS: igr15195.eph [rapid] STOP: 2009/02/20 21:52:00
 NAV FILE: brdc0510.09n OBS USED: 7306 / 7320 : 100%
 ANT NAME: TRM22020.00+GP NONE # FIXED AMB: 34 / 36 : 94%
 ARP HEIGHT: 2.085 OVERALL RMS: 0.009(m)

REF FRAME: NAD_83(CORS96)(EPOCH:2002.0000) ITRF00 (EPOCH:2009.1393)

X: -2732473.383(m)	0.005(m)	-2732474.202(m)	0.005(m)
Y: -4189165.393(m)	0.027(m)	-4189163.967(m)	0.027(m)
Z: 3944602.205(m)	0.039(m)	3944602.336(m)	0.039(m)

LAT: 38 26 57.33393	0.023(m)	38 26 57.35232	0.023(m)
E LON: 236 53 5.04090	0.014(m)	236 53 4.98049	0.014(m)
W LON: 123 6 54.95910	0.014(m)	123 6 55.01951	0.014(m)
EL HGT: -28.322(m)	0.034(m)	-28.826(m)	0.034(m)
ORTHO HGT: 3.488(m)	0.110(m)	[NAVD88 (Computed using GEOID03)]	

11.44

UTM COORDINATES STATE PLANE COORDINATES

UTM (Zone 10) SPC (0402 CA 2)

Northing (Y) [meters]	4255669.489	587470.335
Easting (X) [meters]	489941.965	1902648.819
Convergence [degrees]	-0.07167518	-0.70314016
Point Scale	0.99960125	0.99997573
Combined Factor	0.99960569	0.99998018

US NATIONAL GRID DESIGNATOR: 10SDH8994255669(NAD 83)

BASE STATIONS USED

PID	DESIGNATION	LATITUDE	LONGITUDE	DISTANCE(m)
DH7229	P198 PETALUMAIRCN2004	CORS ARP	N381535.528 W1223626.762	49114.9
DK6396	P206 CRAZYCREEKCN2006	CORS ARP	N384640.123 W1223432.798	59482.7
DG8210	P261 HUNTERHILLCN2004	CORS ARP	N380910.640 W1221303.086	85136.8

NEAREST NGS PUBLISHED CONTROL POINT

JT1405	64 M	N382706.	W1230729.	866.8
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This position and the above vector components were computed without any knowledge by the National Geodetic Survey regarding the equipment or field operating procedures used.

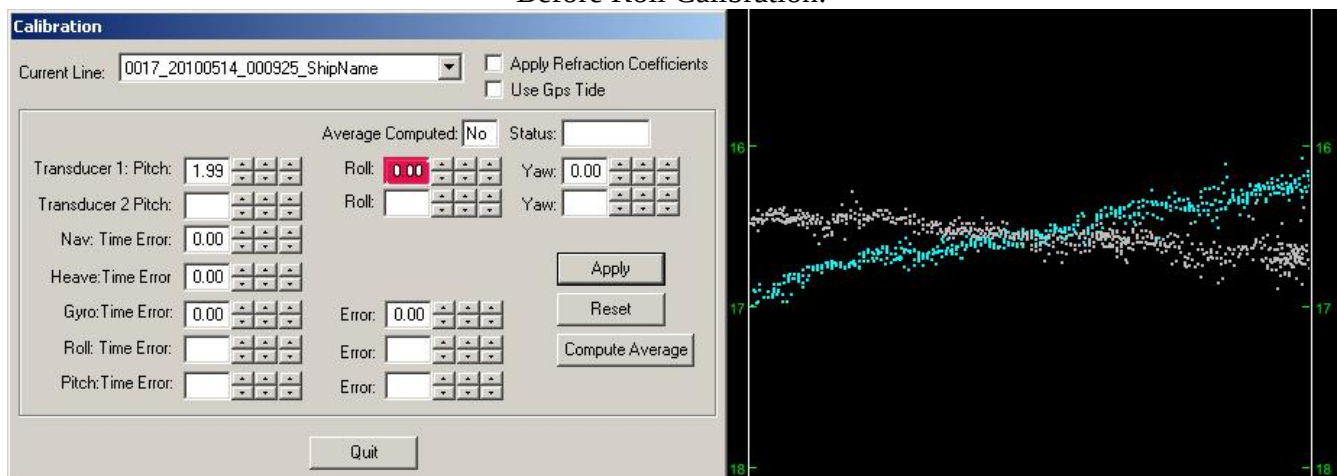
Appendix C
Patch Test Report

System Calibration Report

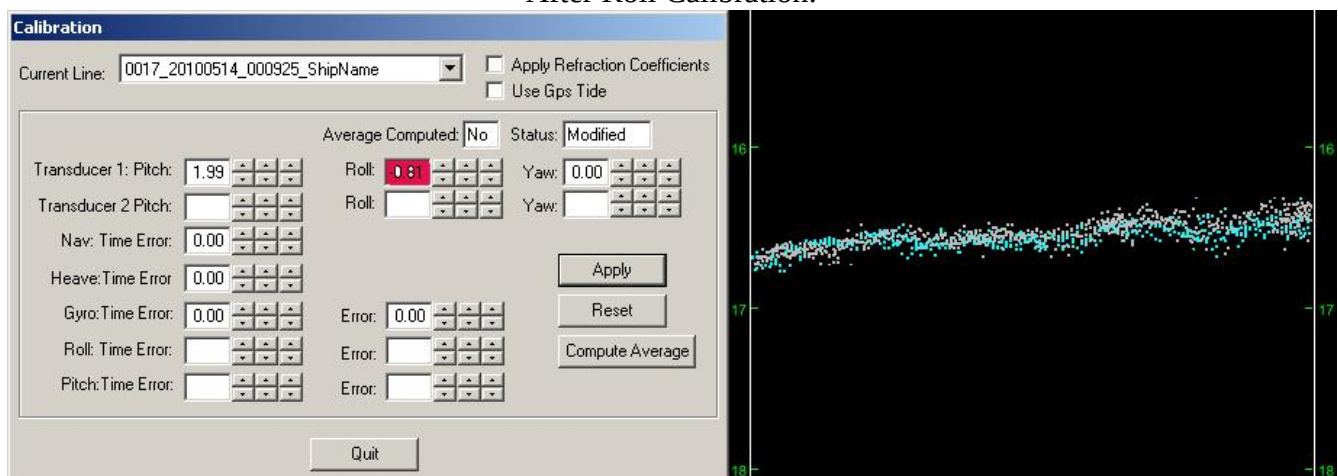
A Patch Test calibration routine is performed to measure and account for installation offsets in angular relations of the Motion Sensor, Sonar Head, and GPS Antenna.

The Roll Calibration is conducted by running a single line twice, in opposite directions over flat seabed. A short, wide, across-track swath of the overlapping data is used for a comparative analysis and software-applied adjustment.

Before Roll Calibration:

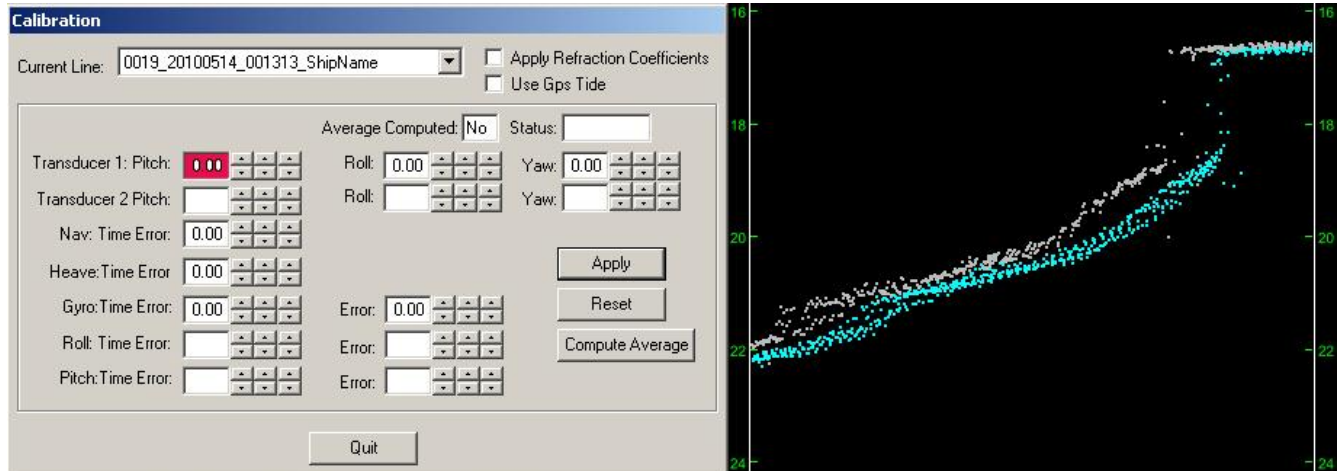


After Roll Calibration:

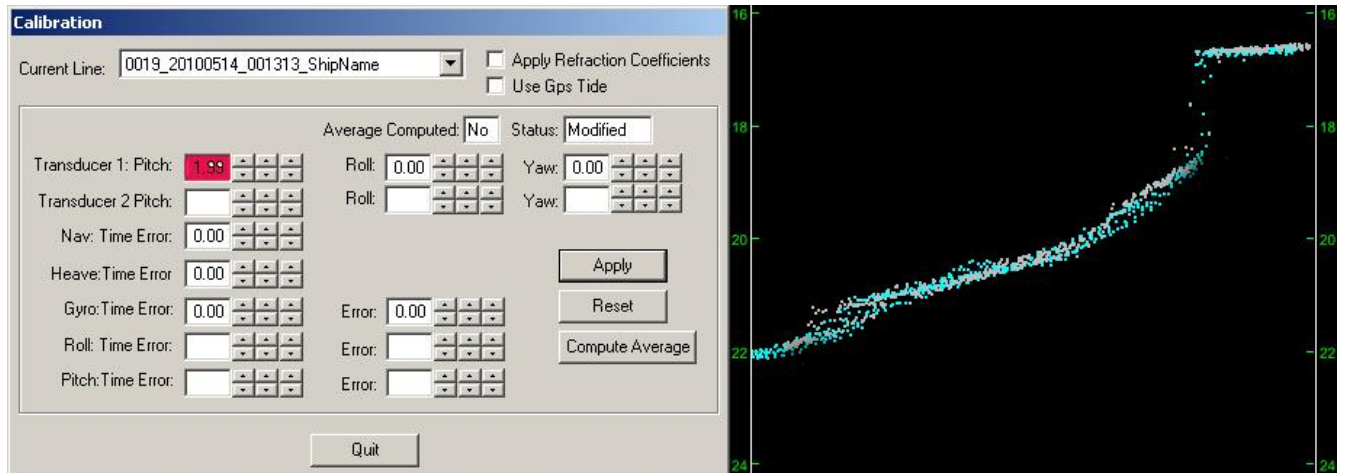


The Pitch Calibration is conducted by running a single line twice, in opposite directions, over sloped seafloor or a target that has noticeable relief above flat seabed. A long, narrow, along-track swath of overlapping data from under nadir is used for comparative analysis and software-applied adjustment.

Before Pitch Calibration:

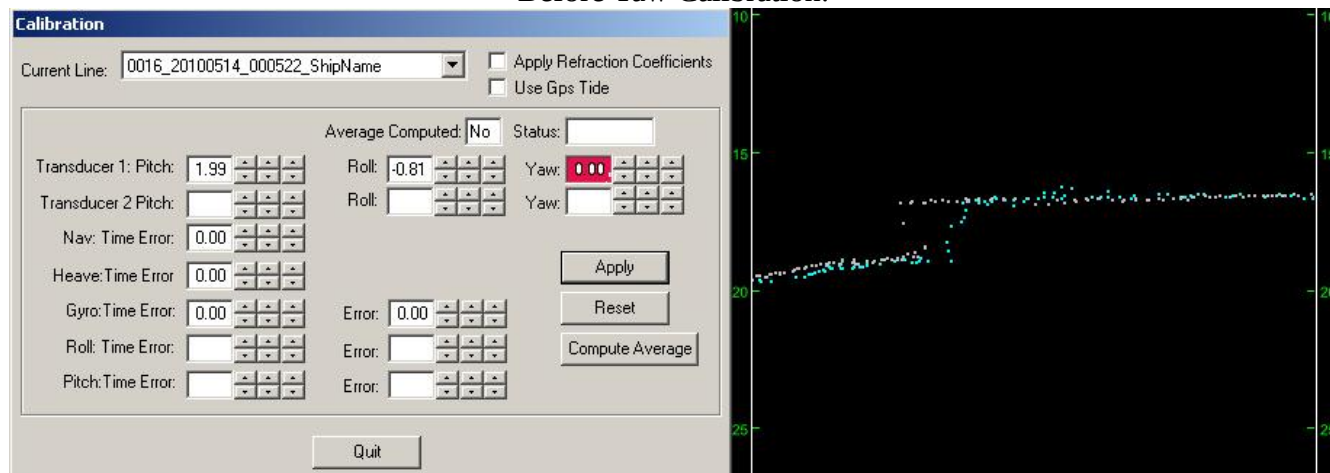


After Pitch Calibration:

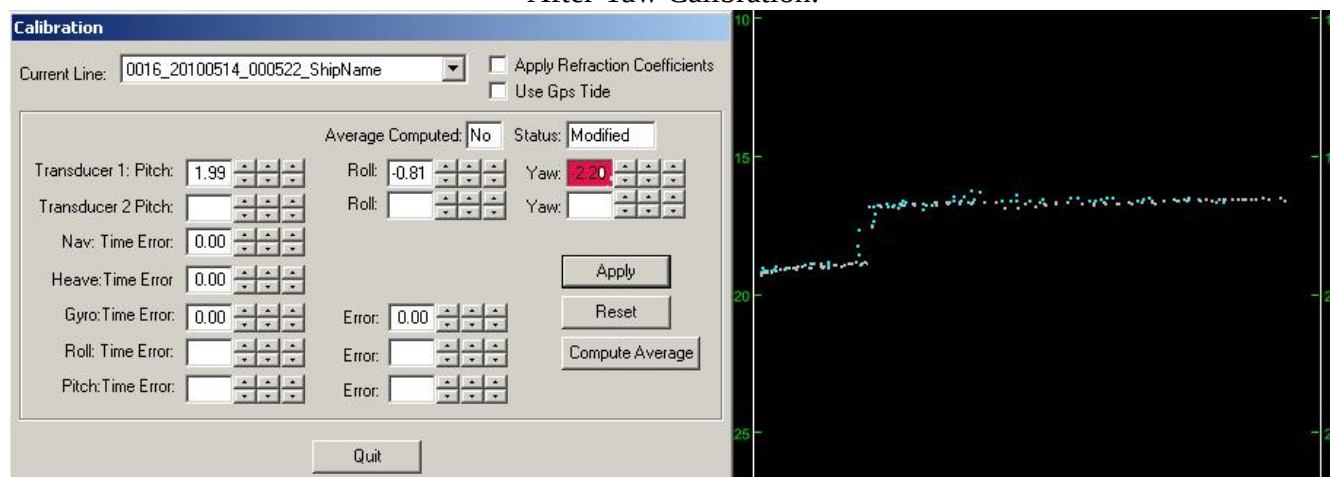


The Yaw Calibration is conducted by running parallel lines, in the same direction, over sloped seafloor or a target that has noticeable relief above flat seabed. A long, narrow, along-track swath of overlapping data from between the lines is used for comparative analysis and software-applied adjustment.

Before Yaw Calibration:



After Yaw Calibration:

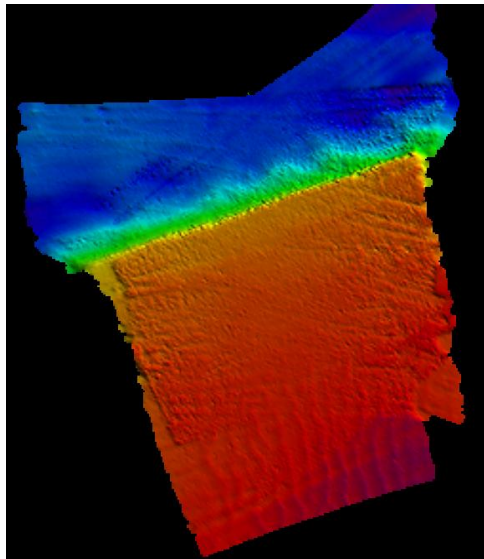


The following offsets were calculated and applied in HYPACK for processing:

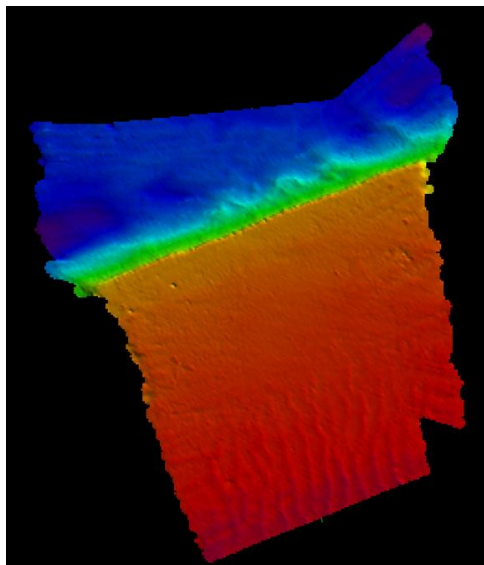
Roll Offset:	-0.81 Degrees
Pitch Offset:	1.99 Degrees
Yaw Offset:	-2.20 Degrees

The impact of this calibration routine can be seen in the results of the surfaces below:

Before Calibration:



After Calibration



Appendix D

Delta Geomatics Reporting

LiDAR DEM Quality Control Report

Methodology

The vertical accuracy of a LiDAR DEM is estimated by determining the root mean square error ($RMSE_{(z)}$). $RMSE_{(z)}$ is the square root of the average of the set of squared differences between dataset co-ordinate values and co-ordinate values from an independent source of higher accuracy for identical points. If those differences are normally distributed and average zero, 95 percent of any sufficiently large sample should be less than 1.96 times the $RMSE_{(z)}$. Therefore 0.6' $RMSE_{(z)}$ is often referred to as "1.2' accuracy and 1.2' $RMSE_{(z)}$ is often referred to as "2.4' accuracy at the 95-percent confidence level". Following that convention, the vertical accuracy of any DEM is defined as 1.96 times the $RMSE_{(z)}$ of linearly interpolated elevations in the DEM, as compared with known elevations from high-accuracy test points.

The Quality Control check was intended to ensure that data would meet NMAS Standards for 2' contours for the Russian River (Priority 1) area. Field verification of the vertical accuracy of this DEM was performed to ensure that the 0.61' $RMSE_{(z)}$ requirements for the NSSDA equivalent to the NMAS standard were satisfied for all major vegetation categories predominate within the priority project area. The categories may include:

- a) Bare-earth and low grass (plowed fields, lawns, golf courses);
- b) High grass and crops (hay fields, cornfields, wheat fields);
- c) Brush lands and low trees (chaparrals, mesquite, mangrove swamps);
- d) Fully covered by trees (hardwoods, evergreens, mixed forests);
- e) Urban areas (high, dense man-made structures);
- f) Sawgrass

The $RMSE_{(z)}$ calculated from a sample of test points is not the $RMSE_{(z)}$ of the DEM. The calculated value may be higher or it may be lower than that of the DEM. Confidence in the calculated value increases with the number of test points. If the errors (lack of accuracy) associated with the DEM are normally distributed and unbiased, the confidence in the calculated $RMSE_{(z)}$ can be determined as a function of sample size.

The test points were to be selected in areas to evaluate DEM accuracy under trees and in vegetation representative of the study area. The Precise Dilution of Position (PDOP) during the LiDAR data collection was consistently less than 3.0 and was determined to be of no issue. Test points on sloping or irregular terrain would be unreasonably affected by the linear interpolation of test points from surrounding DEM points and, therefore, were not selected.

Optimal Geomatics collected test points using RTK (Real-Time Kinematic) GPS techniques. Over one hundred sixty points were collected in total. All RMSE calculations were performed on the bare-earth, orthometric surface.

The Willow Creek, priority area#2, was validated qualitatively against the Russia River, priority area#1, to ensure a fit between the two datasets.

Results

The comparisons between each validation point and the LiDAR DEM are shown in Appendix A. The $RMSE_{(z)}$ was determined for the project area.

Results of the validation points:

Russian River – Priority Area#1 (2' Contour): Fundamental

	US Survey Ft	Meters
Average dz	-0.09	-0.03
Minimum dz	-0.58	-0.18
Maximum dz	0.26	0.08
Average magnitude	0.15	0.04
Root mean square	0.18	0.06
Std deviation	0.16	0.05

Consolidated

	US Survey Ft	Meters
Average dz	0.01	0.00
Minimum dz	-0.58	-0.18
Maximum dz	1.808	0.55
Average magnitude	0.20	0.06
Root mean square	0.30	0.09
Std deviation	0.30	0.09

The favorable result of the DEM comparison to the validation points provides an overall confidence that the LiDAR system was operating properly during data collection.

Those validation points marked as outside are such as they fall outside of a predetermined maximum triangle size. Therefore, there are an insufficient number of LiDAR points hitting the ground in the immediate vicinity of these test points. Any attempt to assign a value from the triangulated surface will result in erroneous values and so these points are excluded from the $RMSE_{(z)}$ calculation.

Due to the nature of the area and in-definite spot of each individual LiDAR point, an $RMSE_h$ value was not reported. Any particular point cannot be tested. However, accuracy statements can be made about the performance of the ABGPS, IMU and LiDAR sensors. The ABGPS data are quality controlled by comparing multiple solutions from multiple base stations. On this project, these solutions all agreed to better than 5 cm horizontally. The IMU sensor combines the post-processed GPS data with the raw inertial data to produce a best estimate of trajectory. Automated quality control checks will not allow the IMU solution to be of a lower accuracy than the provided input from the GPS solution. The altitude of the sensor on this project was 1500 meters AGL providing a spot size of 42 cm in diameter. Each return is located somewhere within the spot on the ground, meaning the location of the point is located within 21 cm of the center of the spot. The stated horizontal accuracy of the system is 1/2000 of the altitude. On this project, the combination of all the errors from all the components of the sensor is much less than the stated accuracy.

Accuracy Statements

The DEMs for these survey areas are believed to be accurate.

The following statements are derived in accordance with the ASPRS Guidelines for Vertical Accuracy Reporting for LiDAR Data (Flood, M., 2004).

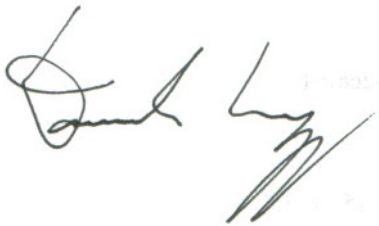
The Russian River, Priority Area#1, tested 0.36 US Survey Feet fundamental vertical accuracy at ninety-five percent confidence level in open terrain using $RMSE_{(z)} \times 1.9600$.

The Russian River, Priority Area#1, tested 0.60 US Survey Feet consolidated vertical accuracy at ninety-five percent confidence level in open terrain, high grasses, low and high trees using $RMSE_{(z)} \times 1.9600$.

The Willow Creek, Priority Area#2, compiled to meet 2.38 US Survey Feet fundamental vertical accuracy at ninety-five percent confidence level in open terrain.

Expected horizontal accuracy of elevation products as determined from system studies and other methods is $1/2000^{\text{th}}$ of the flight height, which, in the instance of this particular project, was 1500m (4921.25 US survey feet) AGL, giving a horizontal tolerance of less than 0.75m (2.46 US survey feet).

Respectfully Submitted,
Optimal Geomatics, Inc.

A handwritten signature in black ink, appearing to read 'Darrick L. Wagg', with a stylized flourish at the end.

Darrick L. Wagg, P.Geo.
02Feb2009

APPENDIX A – Validation Points vs LiDAR DEM

The following table is created from the sets of points collected from different control points. All positions as shown in the tables are in State Plane Co-ordinate System (SPCS) California Zone II, NAD83, US Survey Feet and the elevations are in NAVD88, US Survey Feet.

CODE	DESCRIPTION	VEGETATION CATEGORY
	Base Station used for RTK collection	Various
BASECHEC	RTK processing check	Various
STRIPE	Paint stripe on pavement	A
BARE	Bare open ground	A
HI_GRASS	High Grasses or crops	B
LO_TREES	Low trees or bushes	C
HI_TREES	High trees	D

Number	Easting	Northing	Known Z	Laser Z	Dz	Surface
1	6232584	1932685	207.117	outside	*	BASECHEC
2	6239707	1927831	70.807	70.63	-0.177	STRIPE
3	6239696	1927834	71.051	70.8	-0.251	STRIPE
4	6239685	1927838	71.18	70.96	-0.22	STRIPE
5	6239671	1927845	71.447	71.3	-0.147	STRIPE
6	6239659	1927851	71.593	71.3	-0.293	STRIPE
7	6239640	1927863	71.773	71.37	-0.403	STRIPE
8	6239630	1927870	71.754	71.7	-0.054	STRIPE
9	6239622	1927877	71.661	71.5	-0.161	STRIPE
10	6239614	1927884	71.519	71.57	0.052	STRIPE
11	6239606	1927892	71.449	71.35	-0.099	STRIPE
12	6239599	1927899	71.357	71.1	-0.257	STRIPE
13	6239592	1927906	71.259	71.19	-0.069	STRIPE
14	6239585	1927915	71.103	70.84	-0.263	STRIPE
15	6239576	1927927	70.871	70.6	-0.271	STRIPE
16	6239566	1927941	70.468	70.01	-0.457	STRIPE
17	6239713	1927829	70.689	70.46	-0.229	STRIPE
18	6239726	1927826	70.641	70.33	-0.311	STRIPE
19	6239748	1927822	69.837	69.75	-0.087	STRIPE
20	6239762	1927820	69.495	69.34	-0.155	STRIPE
21	6239774	1927818	69.2	68.83	-0.37	STRIPE
22	6239786	1927817	69.017	68.82	-0.197	STRIPE
23	6239796	1927816	68.799	68.59	-0.209	STRIPE
24	6239808	1927814	68.571	68.41	-0.161	STRIPE
25	6239799	1927807	66.728	slope	*	LO_TREES
26	6239781	1927810	67.127	68.11	0.983	LO_TREES
27	6239765	1927810	67.388	68.1	0.712	LO_TREES
28	6239748	1927812	67.965	68.6	0.636	LO_TREES
29	6239735	1927812	68.456	slope	*	LO_TREES

APPENDIX A – Validation Points vs LiDAR DEM

Number	Easting	Northing	Known Z	Laser Z	Dz	Surface
77	6239241	1926388	38.892	39.03	0.138	BARE
78	6239248	1926379	37.908	38.15	0.242	BARE
79	6239256	1926382	38.463	38.69	0.227	BARE
80	6239286	1926420	37.868	38.13	0.262	BARE
81	6239269	1926462	36.597	36.2	-0.397	BARE
82	6239281	1926463	36.244	36.32	0.076	BARE
83	6239294	1926468	36.581	36.57	-0.011	BARE
84	6239303	1926473	37.678	37.49	-0.188	BARE
85	6239304	1926484	39.848	39.67	-0.178	BARE
86	6239306	1926493	41.27	41.12	-0.15	BARE
87	6239302	1926504	41.841	41.87	0.029	BARE
88	6239289	1926500	40.533	40.39	-0.143	BARE
89	6239277	1926497	39.929	39.85	-0.079	BARE
90	6239264	1926495	38.178	38.14	-0.038	BARE
91	6239252	1926498	36.123	36.14	0.017	BARE
92	6239248	1926510	35.127	35.15	0.023	BARE
93	6239238	1926517	35.006	34.78	-0.226	BARE
94	6239226	1926510	34.027	33.84	-0.187	BARE
95	6239212	1926508	30.798	30.83	0.032	BARE
96	6239208	1926525	30.909	30.83	-0.079	BARE
97	6239205	1926535	31.189	31.08	-0.109	BARE
98	6239193	1926550	30.693	30.72	0.027	BARE
99	6239191	1926562	32.08	32.22	0.14	BARE
100	6239195	1926572	33.34	33.28	-0.06	BARE
101	6239195	1926583	33.607	33.38	-0.227	BARE
102	6239199	1926595	34.033	34.1	0.067	BARE
103	6239189	1926599	34.103	34.17	0.067	BARE
104	6239183	1926606	34.018	34.17	0.152	BARE
105	6239179	1926612	34.001	34.05	0.049	BARE
106	6239184	1926621	35.697	35.69	-0.007	BARE
107	6239189	1926629	37.145	36.56	-0.584	BARE
108	6239194	1926637	37.361	37.57	0.209	BARE
109	6239196	1926645	37.025	36.67	-0.355	BARE
110	6239206	1926650	34.414	34.67	0.256	BARE
111	6239212	1926643	34.17	34.23	0.06	BARE
112	6239216	1926632	34.687	34.8	0.113	BARE
113	6239213	1926620	34.887	35.02	0.133	BARE
114	6239217	1926606	35.042	34.86	-0.182	BARE
115	6239212	1926593	34.728	34.7	-0.028	BARE
116	6239206	1926582	33.94	33.94	0	BARE
117	6239201	1926567	32.898	32.94	0.042	BARE
118	6239207	1926559	33.74	33.73	-0.01	BARE
119	6239214	1926549	33.059	33.07	0.011	BARE
120	6244261	1924106	16.083	16.01	-0.073	STRIPE
121	6244268	1924091	16.13	15.97	-0.16	STRIPE
122	6244274	1924079	16.111	16.08	-0.031	STRIPE
123	6244275	1924077	16.106	16.19	0.084	STRIPE

APPENDIX A – Validation Points vs LiDAR DEM

Number	Easting	Northing	Known Z	Laser Z	Dz	Surface
124	6244280	1924066	16.136	15.77	-0.366	STRIPE
125	6244285	1924056	16.213	16.01	-0.203	STRIPE
126	6244291	1924045	16.343	16.09	-0.253	STRIPE
127	6244296	1924033	16.501	16.28	-0.221	STRIPE
128	6244302	1924021	16.614	16.33	-0.284	STRIPE
129	6244307	1924010	16.666	16.54	-0.127	STRIPE
130	6244313	1923998	16.788	16.69	-0.098	STRIPE
131	6244319	1923987	16.878	16.7	-0.178	STRIPE
132	6244324	1923975	16.833	16.54	-0.293	STRIPE
133	6244330	1923964	16.9	16.59	-0.31	STRIPE
134	6244335	1923953	16.894	16.57	-0.324	STRIPE
135	6244341	1923942	16.854	16.63	-0.224	STRIPE
136	6244346	1923930	16.846	16.72	-0.126	STRIPE
137	6244351	1923919	16.792	16.53	-0.262	STRIPE
138	6244357	1923908	16.901	16.64	-0.261	STRIPE
139	6244377	1923789	16.489	16.22	-0.269	BARE
140	6244380	1923796	16.755	16.79	0.035	BARE
141	6244371	1923803	16.478	16.39	-0.088	BARE
142	6244372	1923814	16.793	16.68	-0.113	BARE
143	6244364	1923820	16.475	16.55	0.075	BARE
144	6244363	1923833	16.713	16.68	-0.033	BARE
145	6244354	1923841	16.37	16.41	0.04	BARE
146	6244354	1923852	16.75	16.69	-0.06	BARE
147	6244344	1923860	16.628	16.6	-0.028	BARE
148	6244345	1923871	16.857	16.77	-0.087	BARE
149	6244335	1923879	16.424	16.27	-0.154	BARE
150	6244335	1923891	16.592	16.54	-0.052	BARE
151	6244326	1923897	16.228	16.15	-0.078	BARE
152	6244326	1923909	16.553	16.61	0.057	BARE
153	6244317	1923916	16.207	16.29	0.083	BARE
154	6244318	1923927	16.567	16.58	0.013	BARE
155	6244308	1923934	16.096	16.14	0.044	BARE
156	6244310	1923944	16.393	16.42	0.027	BARE
157	6244301	1923951	15.917	15.73	-0.187	BARE
158	6244301	1923960	16.252	16.22	-0.032	BARE
159	6244284	1924120	17.199	17.36	0.161	LO_TREES
160	6244285	1924114	16.531	17.18	0.649	LO_TREES
161	6244272	1924143	16.82	16.85	0.03	LO_TREES
162	6244276	1924141	17.332	17.64	0.308	LO_TREES
CA04MC	6232583	1932686	206.959	outside	*	



Delta Geomatics Corporation

**Willow Creek, Russian River
CA**

FLIGHT REPORT

**LiDAR Data Acquisition flown in January 2009 in support
of:**

Topographic LIDAR Data Services



Map the present. Manage the future.

Flight Report

A Cessna 208B Grand Caravan, N704MD, based out of Folsom Field, Vinemont, AL was utilized on this project. This aircraft was outfitted with an Optech ALTM LIDAR system (S/N 06SEN192). This aircraft was mobilized to Sonoma County Airport (STS) in Santa Rosa, California for this project.

Mission planning for the project determined that eleven flightlines would be needed to successfully cover the specified area and would take approximately 1.5 hours to complete. These lines would be flown using the following settings:

Altitude	: 1500 m
Overlap	: 40%
Speed	: 120 kts
System PRF	: 70 kHz
Scan Freq	: 34 Hz
Scan Half Angle	: 18°
Cross Track Resolution	: 0.95m
Down Track Resolution	: 0.91m

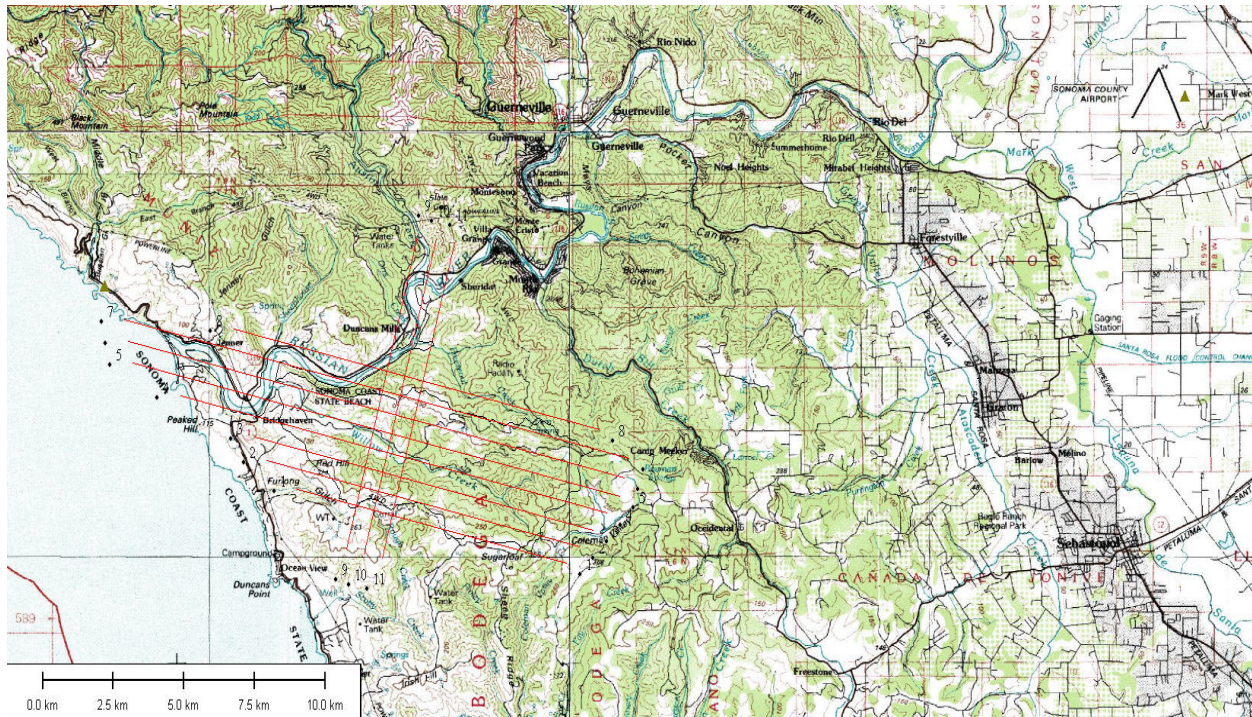


Figure 1-LiDAR Flight Plan with ABGPS control locations

Two GPS base stations located within the project area were used to support the precise positioning of the ALTM's sensor head during the entire duration of flight. The base stations were Leica System 500 units operated by Optimal Geomatics, Inc. See appendix A for the base station coordinates.

The aerial LiDAR survey was completed in one day. Data collection on the flight started around 21h45 UTC and ended around 23h59 UTC on 01Jan2009. All lines for the project were completed in a single flight. Line one was repeated twice and all the data were used in the delivered dataset.

Appendix A: Base Station Locations

Base Station	Latitude	Longitude	Ellipsoid Height (m)
CA04MC	38 27 48.46969 N	123 08 57.61059 W	31.218
STS108	38 30 30.23719 N	122 48 18.67518 W	4.202

Appendix B: Flight Logs



Flight Log

Project Number: 20417
S/N : 06SEN192
Operator : T. ROBBINS
Pilot(s) : A. SCHOOLING
Aircraft : N704MD
Airport : STS
Mission : DELTA GEOMATICS

Weather

Date : January 01, 2009
Julian Day : 001
Temperature : 13 C
Visibility : 10
Clouds : FEW 1000 SCT 25000
Precipitation : NONE
Wind Dir : VAR
Wind Speed : 04
Pressure : 3014

Statistics

Laser Time : 00:54:22

START	STOP	LINE#	ALT	PRF	FREQ	ANGLE MP	DIV	RC	HDG	Plan File
21:49:50.119	21:53:41.118	1	1621	70	34.00	18.00	NAR	OFF	OFF	284.52 09-027_Full_1500-16-40.PLN
21:57:16.817	22:00:09.917	1	1735	70	34.00	18.00	NAR	OFF	OFF	104.52 09-027_Full_1500-16-40.PLN
22:04:59.916	22:09:04.315	2	1655	70	34.00	18.00	NAR	OFF	OFF	284.56 09-027_Full_1500-16-40.PLN
22:12:07.115	22:15:25.614	3	1690	70	34.00	18.00	NAR	OFF	OFF	104.53 09-027_Full_1500-16-40.PLN
22:18:16.814	22:23:41.313	4	1515	70	34.00	18.00	NAR	OFF	OFF	284.54 09-027_Full_1500-16-40.PLN
22:26:19.113	22:30:54.112	5	1669	70	34.00	18.00	NAR	OFF	OFF	104.50 09-027_Full_1500-16-40.PLN
22:34:35.312	22:39:57.011	6	1449	70	34.00	18.00	NAR	OFF	OFF	284.52 09-027_Full_1500-16-40.PLN
22:43:35.811	22:48:42.51	7	1773	70	34.00	18.00	NAR	OFF	OFF	104.55 09-027_Full_1500-16-40.PLN
22:52:44.21	22:56:59.209	8	1560	70	34.00	18.00	NAR	OFF	OFF	284.47 09-027_Full_1500-16-40.PLN



Delta Geomatics Corporation
Willow Creek, Russian River, CA LiDAR Survey
Contract No. RV12259-000-00

22:59:54.809	23:02:52.609	9	1557	70	34.00	18.00	NAR	OFF	OFF	193.52 09-027_Full_1500-16-40.PLN
23:06:09.808	23:09:14.908	10	1529	70	34.00	18.00	NAR	OFF	OFF	13.52 09-027_Full_1500-16-40.PLN
23:13:05.508	23:16:18.107	11	1638	70	34.00	18.00	NAR	OFF	OFF	193.53 09-027_Full_1500-16-40.PLN
23:19:39.607	23:22:24.107	1	1701	70	34.00	18.00	NAR	OFF	OFF	104.52 09-027_Full_1500-16-40.PLN
23:40:03.506	23:41:48.706	1	1536	70	34.00	18.00	NAR	OFF	OFF	29.06 STS_09-027_Full_1500-16-40_CAL.PLN
23:44:43.305	23:46:28.805	1	1501	70	34.00	18.00	NAR	OFF	OFF	209.06 STS_09-027_Full_1500-16-40_CAL.PLN
23:49:21.505	23:50:09.505	3	1500	70	34.00	18.00	NAR	OFF	OFF	119.22 STS_09-027_Full_1500-16-40_CAL.PLN

LiDAR Data Processing Procedures Report

Data collection of the survey area resulted in a total of thirteen flightlines covering the project area, including two lines that were reflights. The range files, flight logs, raw air and ground GPS files were then taken to the office for data processing using DASHMap, a LiDAR processing software package from Optech.

For redundancy and accuracy purposes, the airborne GPS data were processed from two base stations using POSGPS from Applanix, Inc. The agreement between a minimum of two solutions checked or combined between a minimum of two stations was better than 10 cm in each of X, Y, and Z. These trajectories were used in the processing of the inertial data. The inertial data were processed using POSPac MMS from Applanix, Inc. This software produces an SBET (“smooth best estimate of trajectory”) using the GPS trajectory from POSGNSS and the roll, pitch and heading information recorded by the POS (Position Orientation System).

DASHMap uses the SBET to generate a set of data points for each laser return in the LAS file format. Each data point is assigned an echo value so it can be segregated based on the first and last pulse information. This project’s data were processed in strip form, meaning each flight line was processed independently. Processing the lines individually provides the data analyst with the ability to QC the overlap between lines.

The sensor calibration parameters derived for the flight were as follows:

JD001F01:

AltmSerialNo=06SEN192
ImuType=LN200A1
ImuRate=200
ScannerScale=1.020020
ScannerOffset=0.004800
IMURoll=-0.035300
IMUPitch=-0.084800
IMUHeading=0.015200
UserToImuEx=-0.012000
UserToImuEy=-0.007000
UserToImuEz=0.170000
UserToImuDx=-0.090000
UserToImuDy=-0.008000
UserToImuDz=-0.096000
UserToRefDx=-0.051000
UserToRefDy=-0.030000
UserToRefDz=-0.488000
TimeLag=0.00001200
IntensityGainFor3070=20.000000
UseLeftDroopCorrection=15.000000
UseRightDroopCorrection=15.000000
Temperature=9.470000
Pressure=932.010000

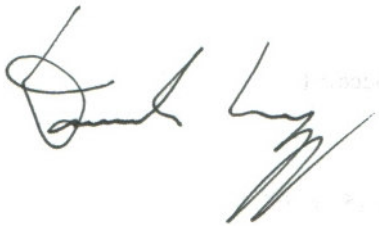
Each strip was then imported into a project using TerraScan (Terrasolid, Ltd.) and the project management tool GeoCue (GeoCue Corp.). By creating a project the various flightlines are combined while breaking the dataset as a whole into manageable pieces. This process also converts the dataset from geographic coordinates to the State Plane Coordinate System (NAD83), California Zone II, US Survey Feet, utilizing NADCON. The ellipsoid height values were converted to NAVD88, US Survey Feet, orthometric values using Geoid03, provided by NGS.

Individual lines were then checked against adjacent lines to ensure a cohesive dataset. The data from each line were then combined and a classification routine was then run to determine the initial surface model. This initial surface model was then reduced using Optimal Geomatics' proprietary methods to create the final bare-earth dataset.

The bareearth data were then checked against the validation points across the project area. The results of these checks showed the DEM fitting the validation points well (see LiDAR DEM Quality Control Report for results).

Stereo pairs were generated from the LiDAR intensity data using Geocue and LiDAR1CuePac (Geocue Corp.). LiDARgrammetry was then utilized to collect polygons around hydro features for the sole purpose of hydro point reclassification. These features were collected as a 3D element in the MicroStation (Bentley Systems, Inc.) environment utilizing ISSD (Z/I Imaging). The LiDAR point data were delivered in LAS format and ASCII XYZ text format.

Respectfully Submitted,
Optimal Geomatics, Inc.

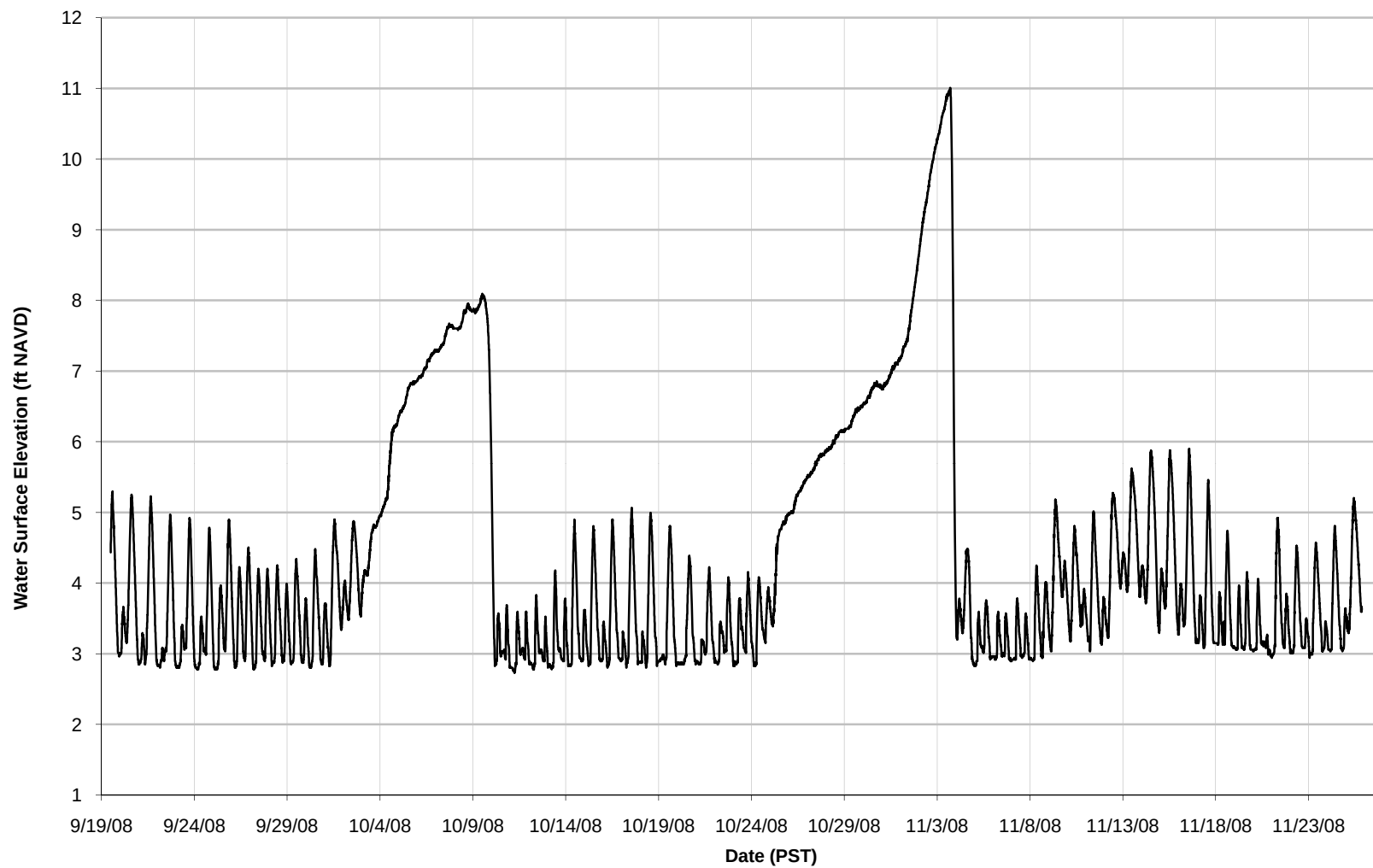
A handwritten signature in black ink, appearing to read 'Darrick L. Wagg', with a stylized flourish at the end.

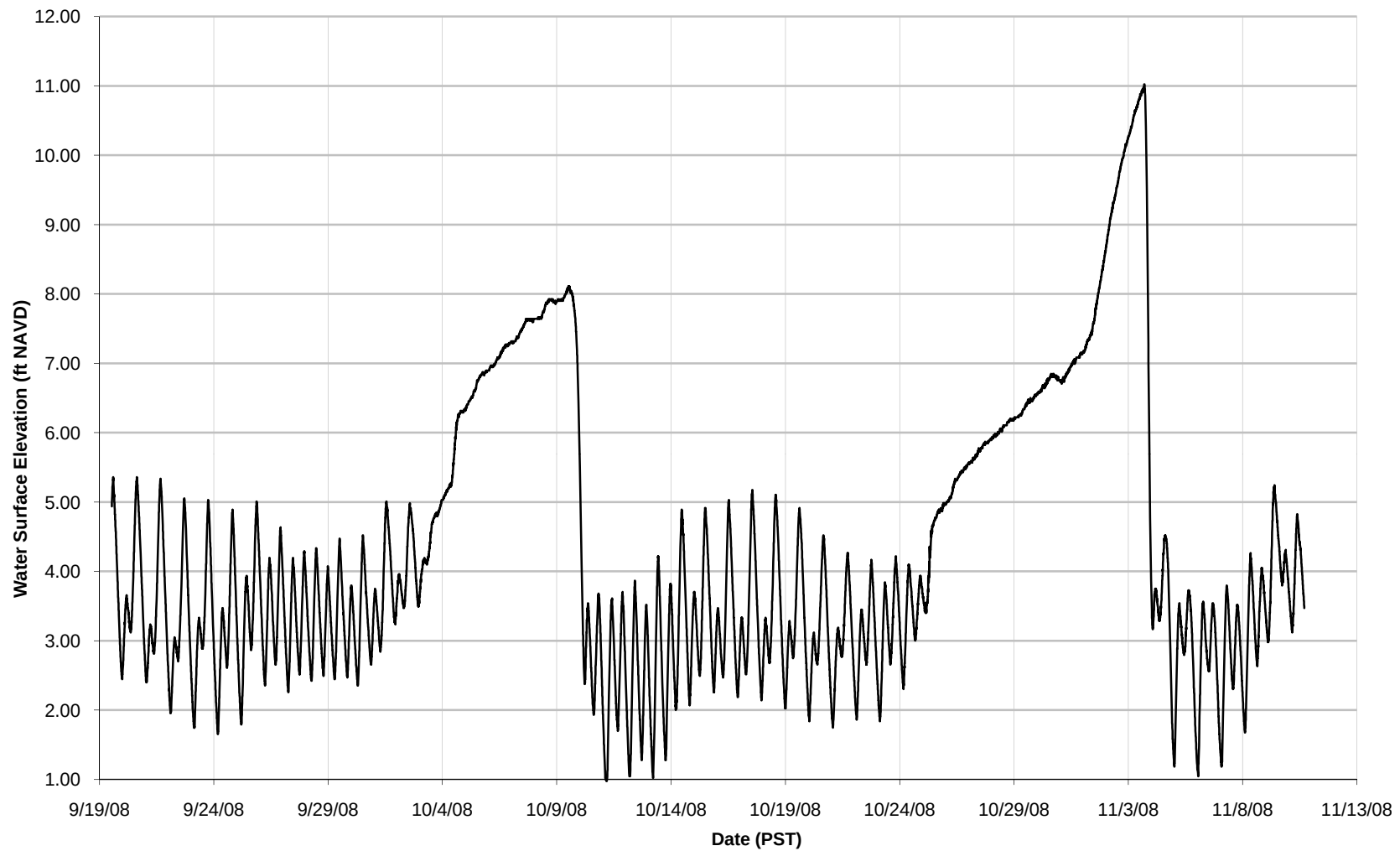
Darrick L. Wagg, P.Geo.
02Feb2009

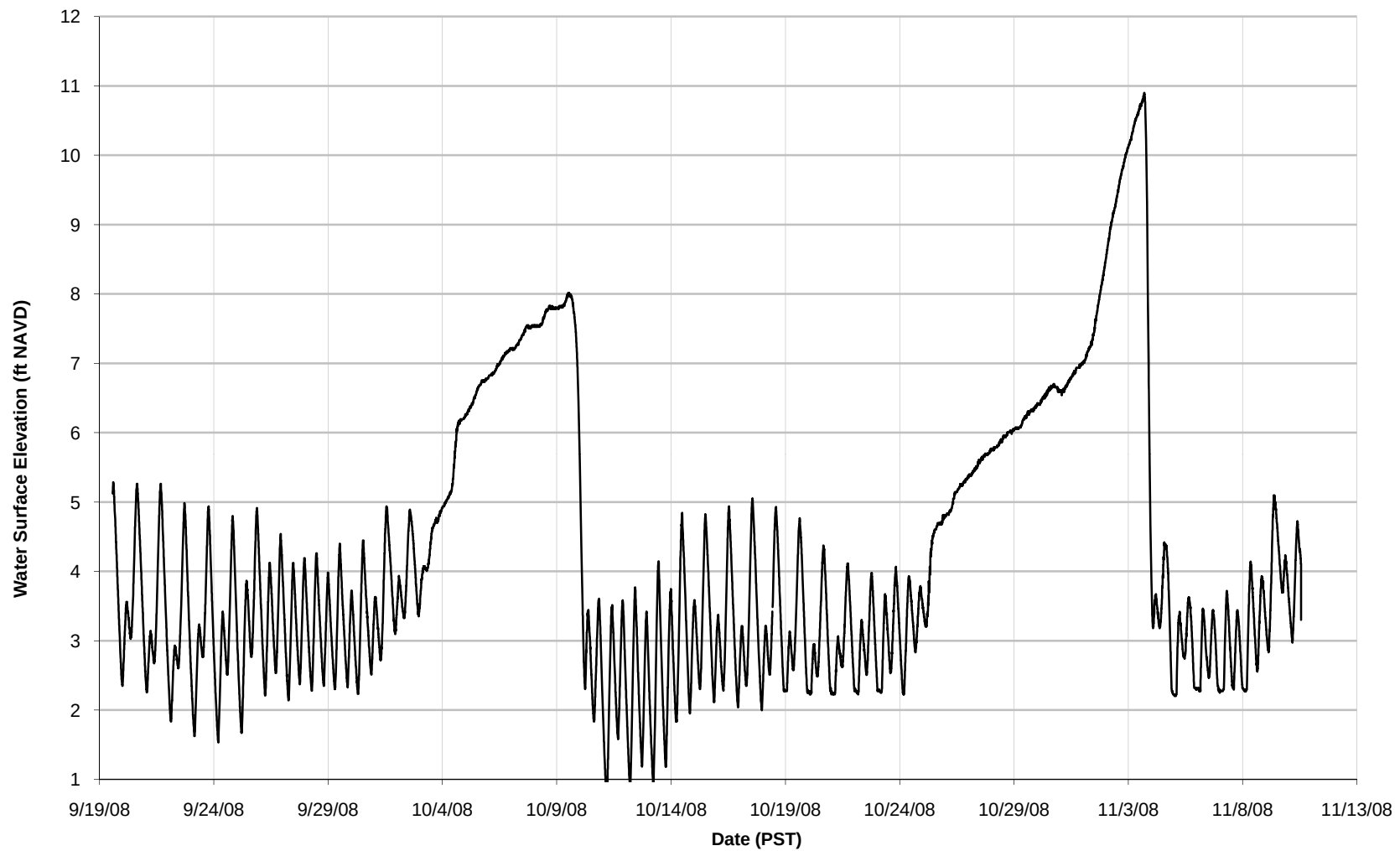
Appendix E
Accuracy Statement

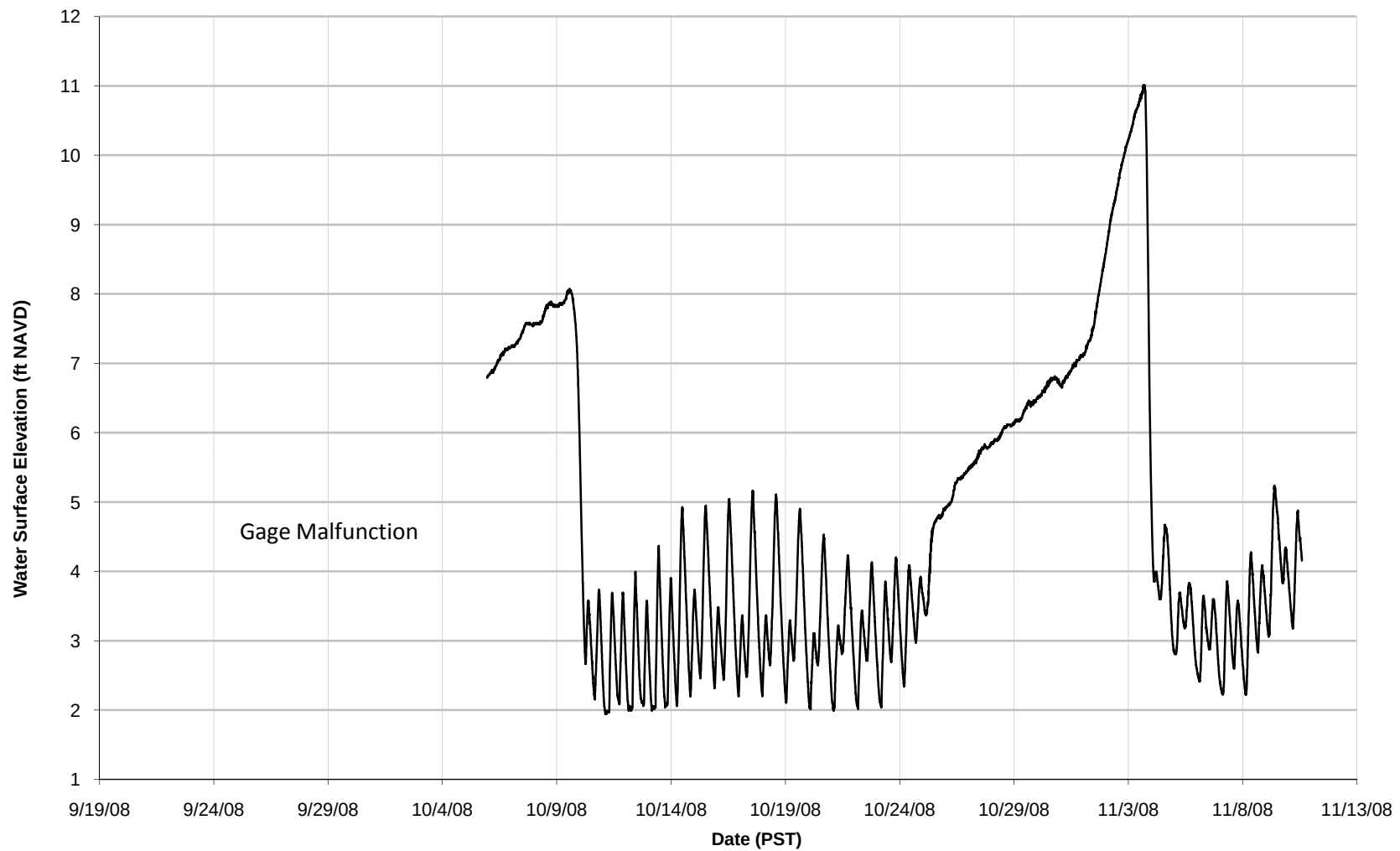
Appendix F

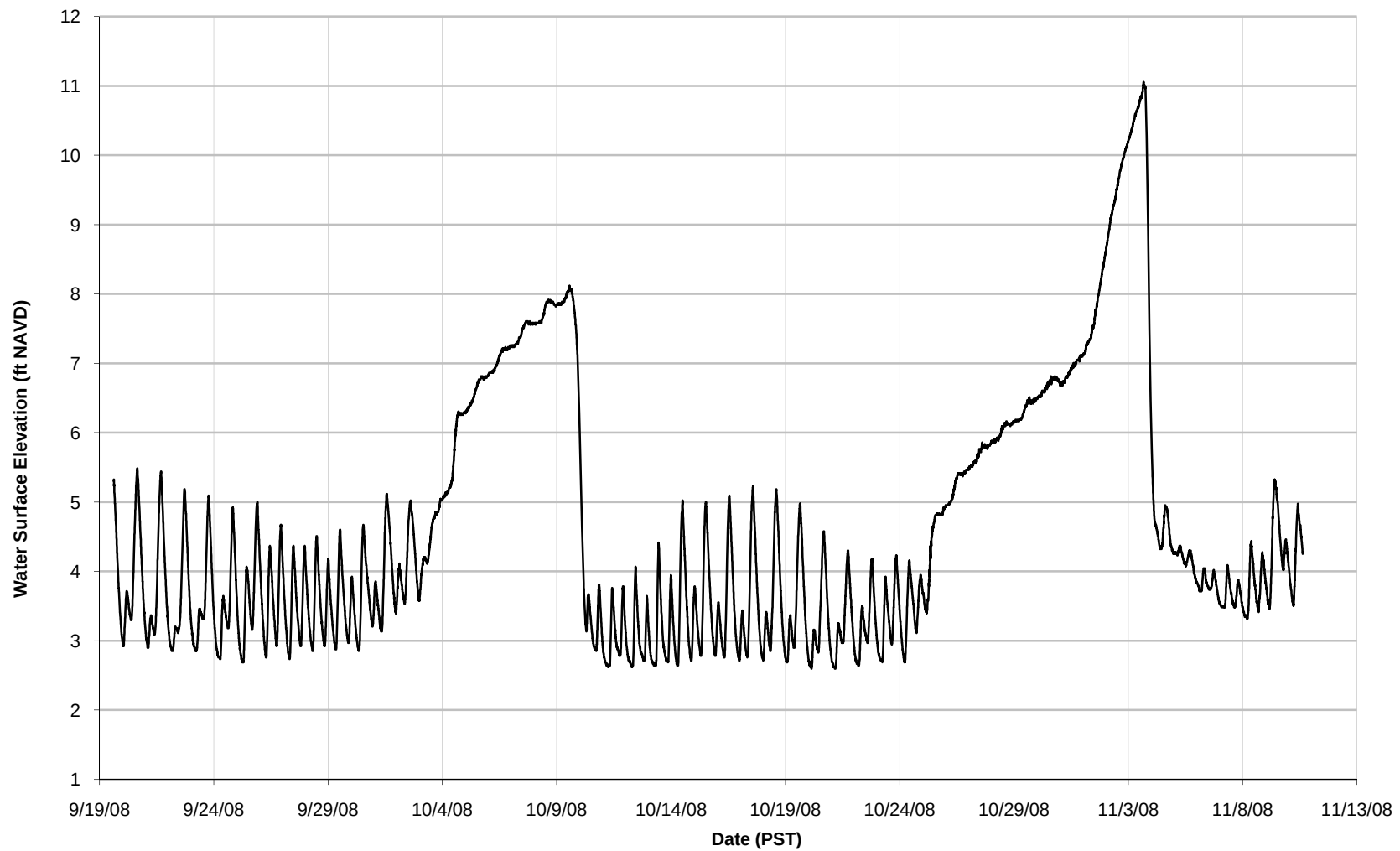
Water Surface Elevation Graphs











Appendix G
Contour Maps

Appendix H
Sidescan Sonar Maps